



Industry Report on Logistics in India

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Prepared for

Skyways Air Services Limited



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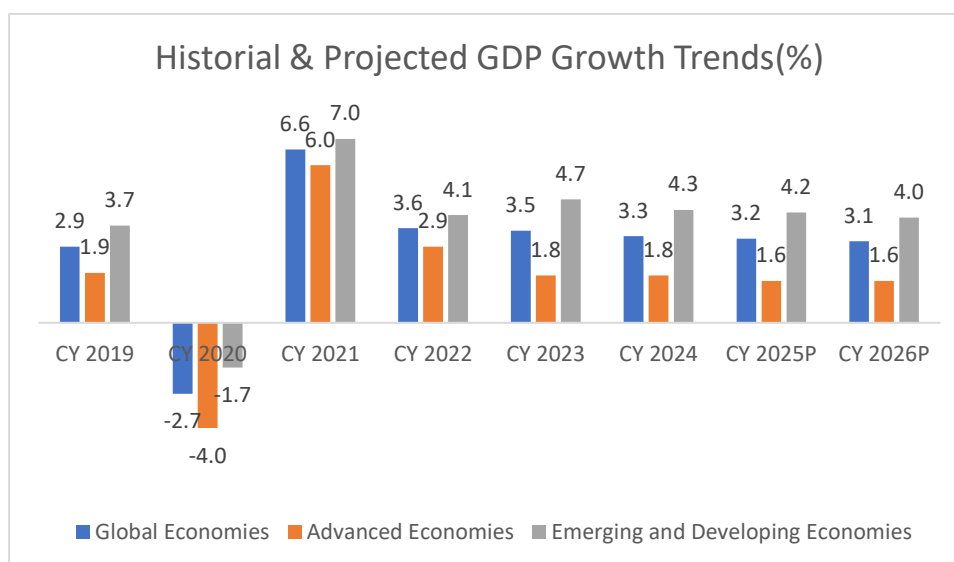
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Global Economic Overview

The global economy, which recorded GDP growth at 3.3% in CY 2024, is expected to show resilience at 3.2% in CY 2025. This marks the slowest expansion since 2020 and reflects a -0.1%point downgrade from the January 2025 forecast. Moreover, the projection for CY 2026 has also reduced to 3.1%. This slowdown is mainly attributed to numerous factors such as high inflation in many economies despite central bank efforts to curb inflation, continuing energy market volatility driven by geopolitical tensions, particularly in Ukraine and the Middle East, and the re-election of Donald Trump as US President, which extended uncertainty around the trade policies as well as overall global economic growth. High inflation and rising borrowing costs affected the private consumption on one hand, while fiscal consolidation impacted the government consumption on the other hand. As a result, global GDP growth is projected to slow down from 3.3% in CY 2024 to 3.2% in CY 2025.



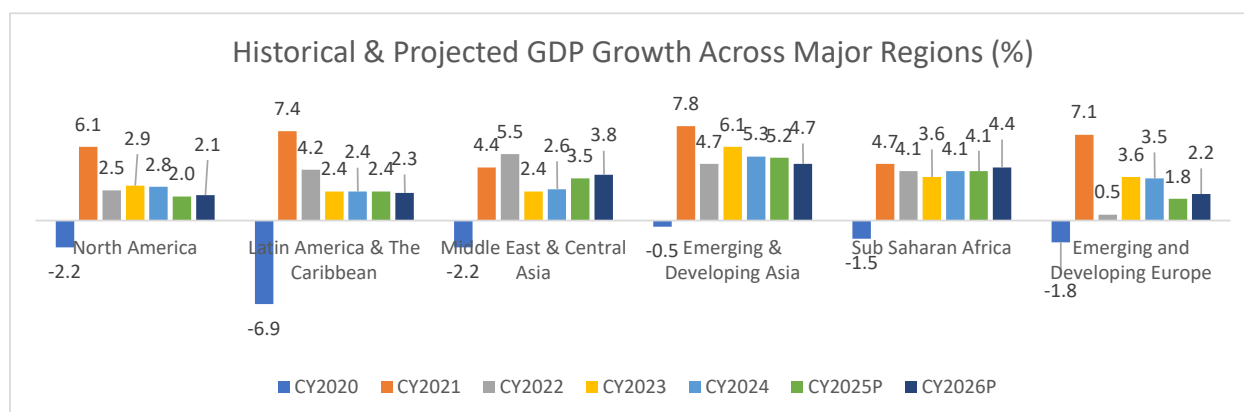
Source – IMF Global GDP Forecast Release October 2025

Note: Advanced Economies and Emerging & Developing Economies are as per the classification of the World Economic Outlook (WEO). This classification is not based on strict criteria, economic or otherwise, and it has evolved over time. It comprises 40 countries under the Advanced Economies, including the G7 (the United States, Japan, Germany, France, Italy, the United Kingdom, and Canada) and selected countries from the Euro Zone (Germany, Italy, France, etc.). The group of emerging market and developing economies (156) includes all those that are not classified as Advanced Economies (India, China, Brazil, Malaysia etc.)

Historical and Projected GDP Growth

GDP growth across major regions exhibited a mixed trend between 2022-23, with GDP growth in many regions, including North America, Emerging and Developing Asia, and Emerging and Developing Europe, slowing further in 2024. In 2025, the GDP growth rate in Emerging and Developing Asia (India, China, Indonesia, Malaysia, etc.) is expected to moderate further to 5.2% from 5.3% in the previous year, while in North America, it is expected to moderate to 2.0% in CY 2025 from 2.8% in CY 2024. Similarly, in Emerging and Developing Europe is expected to moderate further to 1.8% from 3.5% in the previous year.

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Source-IMF World Economic Outlook, October 2025 update.

Except Middle East & Central Asia, all other regions, like Emerging and Developing Asia, Emerging and Developing Europe, Latin America & The Caribbean, Sub-Saharan Africa and North America, are expected to record a moderation in GDP growth rate in CY 2025 as compared to CY 2024. Further, growth in the United States is expected to come down at 2.0% in CY 2025 from 2.8% in CY 2024 due to lagged effects of monetary policy tightening, gradual fiscal tightening, and a softening in labour markets, slowing aggregate demand.

Global Economic Outlook

The global macroeconomic environment remains shaped by divergent regional trends and continued geopolitical and policymaking uncertainties. A wave of new U.S. tariffs, mostly effective from August 7, has shaken markets and raised costs for global trade. On August 1, the U.S. announced higher tariff rates for countries from which it imports goods, with most of the rates effective from August 7. A 15% rate will act as a baseline floor for countries with which the U.S. has a trade deficit; a 10% rate applies for those with which the U.S. has a trade surplus. However, there are some countries that are subject to higher U.S. tariffs.

In North America, the United States continues to engage in trade negotiations with multiple countries and has announced plans to introduce sector-specific tariffs, targeting industries such as copper and pharmaceuticals. However, talks with Canada have stalled, despite Canada's decision to withdraw its Digital Services Tax in an effort to ease tensions. As a result, the U.S. imposed a 35% tariff on Canadian goods that do not meet USMCA compliance standards, effective August 1. This move has further strained bilateral relations and added complexity to the regional trade landscape.

By August 7, the U.S. had announced increased tariffs of 15-50% on Asian economies, with most rates around 20%. Although these tariffs are lower than the levels announced in April, they remain higher than those applied to most Western counterparts, impacting exporters such as Taiwan Region (20%) and India (25%, with the U.S. saying this could rise to 50% at the end of August). Moreover, on July 28, the US imposed a



15% tariff on most EU imports under a new trade agreement, impacting Nordic countries such as Denmark, Finland, and Sweden. Key exemptions include aircraft parts and semiconductor equipment, while steel and aluminium continue to face 50% tariffs.

Tariffs and their unpredictable application have weighed on consumer and business sentiment, sunk global stock markets, raised recession risks, and made a global slowdown more likely. Our latest Global Business Optimism Insights report indicates a further decline in business optimism as firms continue to grapple with trade-related policy uncertainty and its broader economic implications. Export-driven sectors reported sharp declines in optimism. Financial risk perceptions remain elevated as businesses contend with high borrowing costs and persistent inflation expectations. More broadly, the uncertainty is reflected in delayed capital expenditure and a pullback in hiring.

Tariffs have begun to exert pressure on central banks by contributing to inflationary pressures and increasing financial market volatility. Central banks are adjusting forward guidance and policy frameworks and may begin to consider the likelihood of softer growth being a bigger priority than high inflation by starting to cut interest rates to support economies. For businesses, this uncertainty translates into unpredictable cost structures, fluctuating credit availability, and the management of operational costs through diversified supply networks.

The latest Dun & Bradstreet Global Business Optimism Insights report reveals a further decline in business optimism, though at a more moderate pace than in the prior quarter, as businesses continued to grapple with trade-related policy uncertainty and its broader economic implications. Export-driven sectors such as automotives, electricals, and metals saw sharp declines in optimism, particularly in the U.S., Mexico, South Korea, and Japan, where rising tariffs and shifting trade policies have fueled cost pressures and demand volatility. Financial risk perceptions remain elevated.

Global Growth Projection

At broader level, the global economy is expected to experience a slowdown in 2025, with GDP growth projected to decline to 3.2%, down from 3.3% in 2024. This deceleration reflects persistent inflationary pressure, geopolitical uncertainties and tightened monetary policies. However, a slight recovery is anticipated in 2026, with growth projected to improve to 3.1%. In the United Kingdom, headline inflation, which began rising in 2024, is expected to continue increasing in 2025, partly due to changes in regulated prices. This rise is projected to be temporary, with a loosening labour market and moderating wage growth helping inflation return to target by the end of 2026. In the United States, inflation is expected to rise in the second half of 2025, as the impact of tariffs is no longer absorbed within supply chains and is instead passed on to consumers. Inflation is then expected to return to the Federal Reserve's 2 per cent target in 2027. This forecast assumes modest second-round effects, implying upside risks to U.S. inflation and downside risks to employment.



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Among emerging markets and developing economies, inflation forecasts for Brazil and Mexico are revised upward. For Brazil, the revision is more pronounced and partly reflects the stabilization of inflation expectations above target, due to fiscal policy credibility challenges in the previous year, although currency appreciation is expected to provide relief in late 2025 and 2026. For Mexico, the upward revision is driven by volatile categories such as food and more persistent-than-expected services inflation. For several other economies, inflation forecasts are revised downward compared with the October 2024 WEO. In much of emerging and developing Asia, this is the case. The revision largely reflects lower-than-expected outturns, with food, energy, and administrative prices playing a significant role—particularly in China, India, and Thailand.

In the United States, growth is projected to slow to 2.0 percent in 2025 and remain steady at 2.1 percent in 2026, broadly consistent with July projections and improved from April due to lower effective tariff rates, a fiscal boost from the OBBBA, and easing financial conditions. This reflects a significant slowdown from 2024 and a cumulative downward revision of 0.1 percentage point from the October 2024 WEO and 0.7 percentage point from the January 2025 WEO Update. The revision is primarily driven by greater policy uncertainty, higher trade barriers, and slower labour force and employment growth.

Growth in the euro area is expected to increase modestly to 1.2 percent in 2025 and to 1.1 percent in 2026. While this marks an improvement from April and July, it represents a cumulative downward revision of 0.4 percentage points compared to the October 2024 WEO. The main contributing factors are elevated uncertainty and higher tariffs. Recovering private consumption from higher real wages and fiscal easing in Germany in 2026 provides only a partial offset, while a strong performance in Ireland supports growth in 2025. The euro area economy is expected to grow at its potential in 2026.

For emerging market and developing economies, growth is projected to moderate from 4.3 percent in 2024 to 4.2 percent in 2025, and further to 4.0 percent in 2026. This is virtually unchanged from the July WEO Update and reflects a cumulative upward revision of 0.6 percentage points from the April 2025 WEO, but remains 0.2 percentage points lower than the October 2024 forecast, with low-income developing countries facing a larger downward revision than middle-income economies.

Growth in emerging and developing Asia is expected to decline from 5.3 percent in 2024 to 5.2 percent in 2025, and further to 4.7 percent in 2026. In several countries—particularly in ASEAN, among the most affected—growth forecasts closely followed changes in effective tariff rates. In China, the 2025 GDP growth forecast was revised downward by 0.6 percentage points in the April 2025 WEO due to escalating trade tensions with the United States and then revised upward by 0.8 percentage points in the July WEO Update following the pause on higher tariffs in May.

In Latin America and the Caribbean, growth is projected to remain stable at 2.4 percent in 2025 and decline slightly to 2.3 percent in 2026. The 2025 forecast is revised upward by 0.4 percentage points relative to April, driven by lower tariff rates for most countries in the region and stronger-than-expected incoming data. The



revision is largely attributed to Mexico, which is expected to grow at 1.0 percent in 2025, 1.3 percentage points higher than forecast in the April 2025 WEO. For Brazil, the 2025 projection is revised upward, while the 2026 forecast is revised downward, partly due to the higher tariff rate on exports to the United States. For the region overall, the 2025–2026 forecast is cumulatively 0.5 percentage point lower than the October 2024 WEO, reflecting trade policy changes and uncertainty.

In emerging and developing Europe, growth is projected to decline significantly from 3.5 percent in 2024 to 1.8 percent in 2025, followed by a modest recovery to 2.2 percent in 2026. This decline is primarily driven by a sharp drop in Russia's growth forecast, from 4.3 percent in 2024 to 0.6 percent in 2025 and 1.0 percent in 2026. The 2025 growth forecast is 0.9 percentage points lower than in the April 2025 WEO, largely due to recent data showing a concentration of fiscal expenditures in Q4 2024, which raised the 2024 GDP estimate from 4.1 percent to 4.3 percent. The payback effect is reflected in the 2025 projection.



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India Macroeconomic Analysis

The International Monetary Fund (IMF), in its latest World Economic Outlook, has projected India's economy to grow at 6.6% in 2025-26, marking a 20-basis-point upward revision from its previous estimate. This boost is largely credited to a strong first-quarter performance in FY26, which helped offset the negative impact of increased U.S. tariffs on Indian exports. With this projection, India is set to remain one of the fastest-growing emerging markets and developing economies, outpacing China's expected growth of 4.8%. Despite global trade policy shifts and economic uncertainties, India's growth continues to be driven by resilient domestic demand and strong economic fundamentals. However, the IMF slightly lowered its forecast for 2026-27 to 6.2%, anticipating a natural moderation as the early momentum fades

Country	CY 2020	CY 2021	CY 2022	CY 2023	CY 2024	CY 2025 P	CY 2026 P
India	-5.8%	9.7%	7.6%	9.2%	6.5%	6.6%	6.2%
China	2.3%	8.6%	3.1%	5.4%	5.0%	4.8%	4.2%
United States	-2.2%	6.1%	2.5%	2.9%	2.8%	2.0%	2.1%
Japan	-4.2%	2.7%	0.9%	1.4%	0.1%	1.1%	0.6%
United Kingdom	-10.3%	8.6%	4.8%	0.4%	1.1%	1.3%	1.3%
Russia	-2.7%	5.9%	-1.4%	4.1%	4.3%	0.6%	1.0%

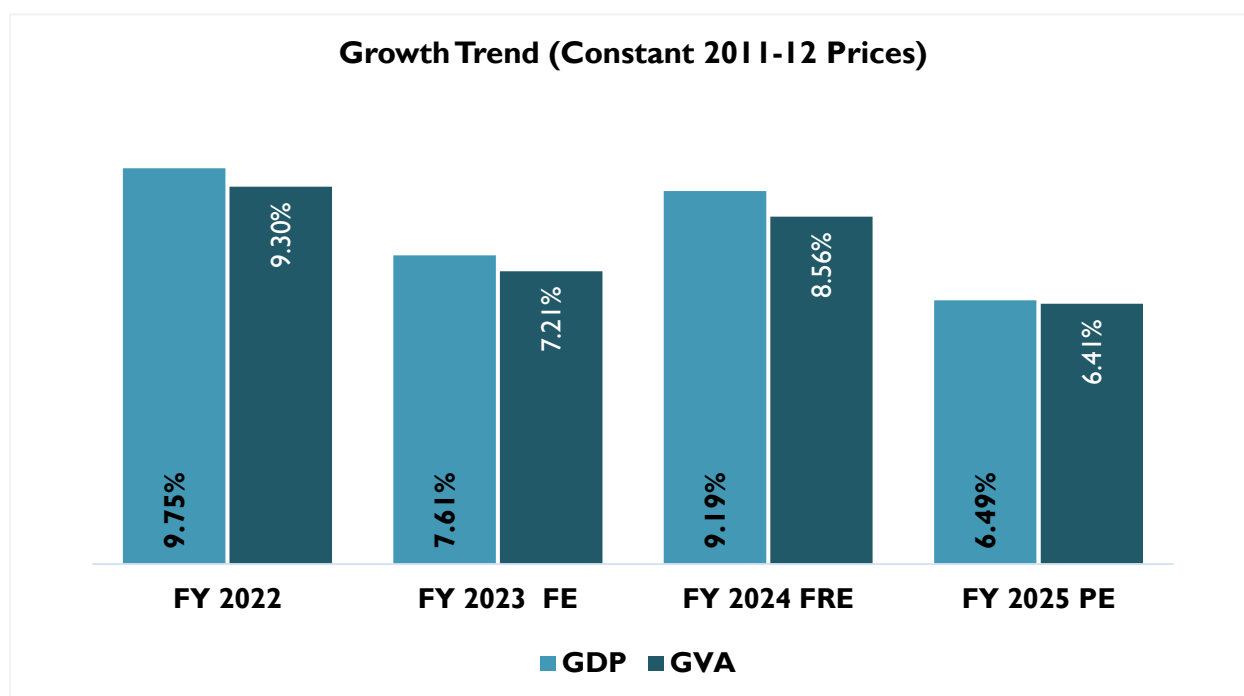
Source: World Economic Outlook, October 2025

In the Union Budget 2025-2026, the government announced INR 11.21 billion capex on infrastructure (10.12% higher than the previous year's revised estimates), coupled with INR 1.5 trillion in interest-free loans to states. This has provided much-needed confidence to the private sector, and in turn, is expected to attract private investment.

Historical GDP and GVA Growth trend

As per the latest estimates, India's GDP at constant prices is estimated to grow to INR 187.96 trillion in FY 2025 (Provisional Estimates), with the real GDP growth rates estimated to be 6.5% for FY 2025. Similarly, real Gross Value Added (GVA) growth is estimated to have moderated to 6.4% in FY 2025. Even amidst global economic uncertainties, India's economy exhibited resilience supported by robust consumption and government spending.

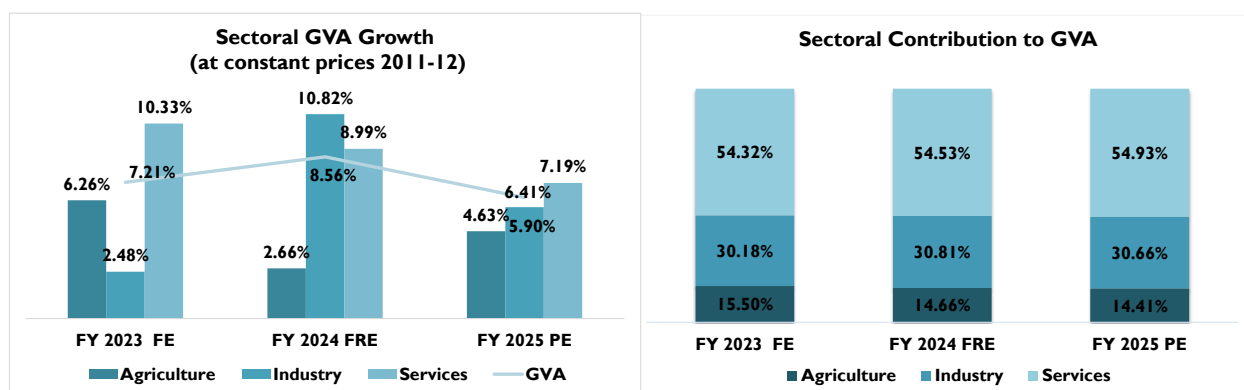




Source: Ministry of Statistics & Programme Implementation (MOSPI), National Account Statistics: FY2025.

FE is Final Estimates, FRE is First Revised Estimate and PE is Provisional Estimates

Sectoral Contribution to GVA and annual growth trend



Source: Ministry of Statistics & Programme Implementation (MOSPI)

FE is Final Estimates, FRE is First Revised Estimate, and PE is Provisional Estimates

Sectoral analysis of GVA reveals that the industrial sector experienced a moderation in FY 2025, recording a 5.90% year-over-year growth against 10.82% year-on-year growth in FY 2024. Within the industrial sector, growth moderated across sub-sectors with mining, manufacturing, and construction activities growing by 2.69%, 4.52%, and 9.35% respectively in FY 2025, compared to 3.21%, 12.30%, and 10.41% in FY 2024. Growth in the utilities sector too moderated to 6.03% in FY 2025 from 8.64% in the previous year. The industrial sector's contribution to GVA moderated marginally from 30.81% in FY 2024 to 30.66% in FY 2025.

The services sector continued to be the main driver of economic growth, although its pace moderated. It expanded by 7.19% in FY 2025 from 8.99% in FY 2024. The services sector retained its position as the largest

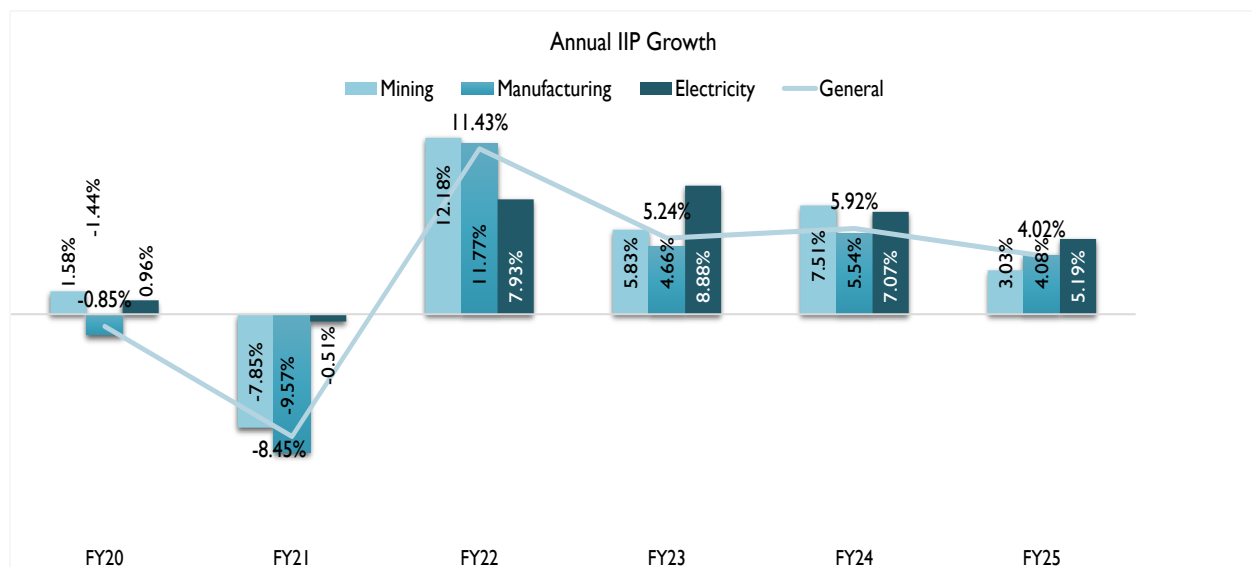
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contributor to GVA, rising from 54.32% in FY 2023 to 54.53% in FY 2024, with a further increase to 54.93% in FY 2025.

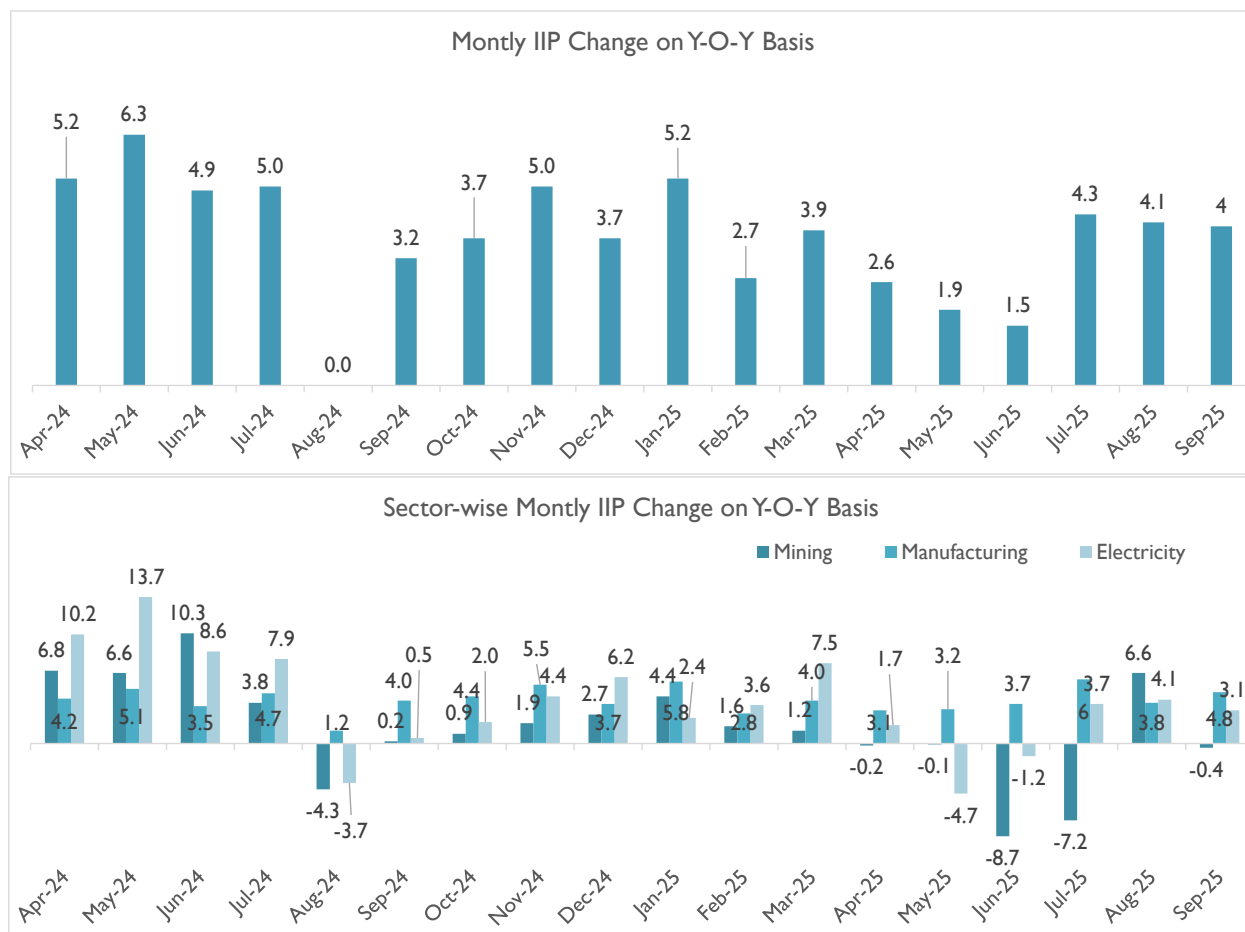
The agriculture sector saw an acceleration, with growth increasing from 2.66% in FY 2024 to 4.63% in FY 2025. However, its contribution to GVA declined marginally from 14.66% in FY 2024 to 14.41% in FY 2025. Overall, Gross Value Added (GVA) growth moderated to 6.41% in FY 2025 from 8.56% in FY 2024

Annual & Monthly IIP Growth

Industrial sector performance as measured by the IIP index exhibited moderation in FY 2025, recording a 4.02% y-o-y growth against a 5.92% increase in the previous year. The manufacturing index showed moderation and grew by 4.08% in FY 2025 against 5.54% in FY 2024. The mining sector index too moderated and exhibited a growth of 3.03% in FY 2025 against 7.51% in the previous years, while the Electricity sector Index also witnessed a moderation of 5.19% in FY 2025 against 7.07% in the previous year.



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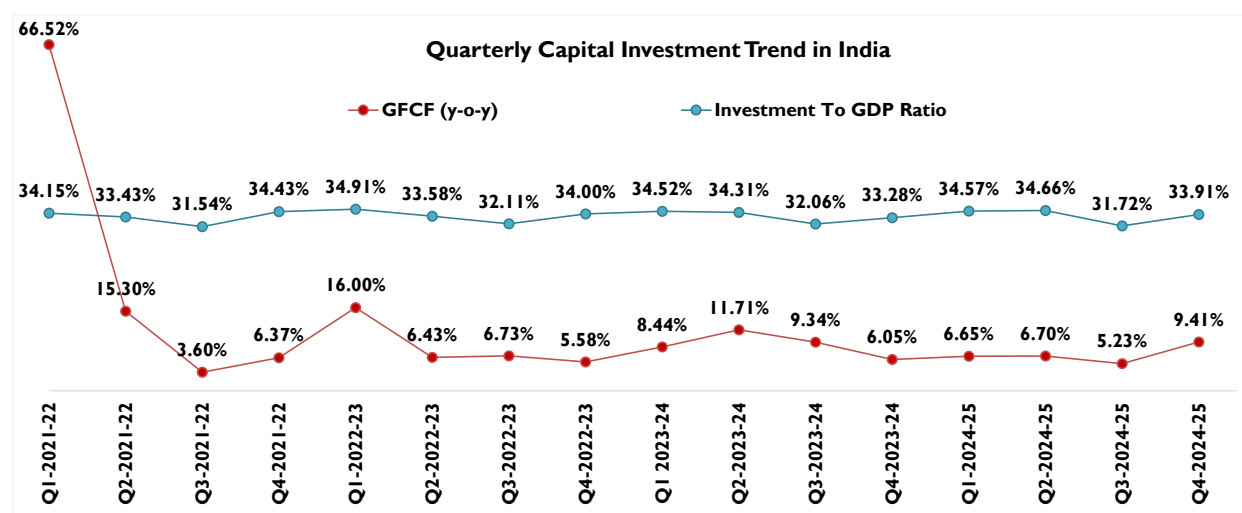
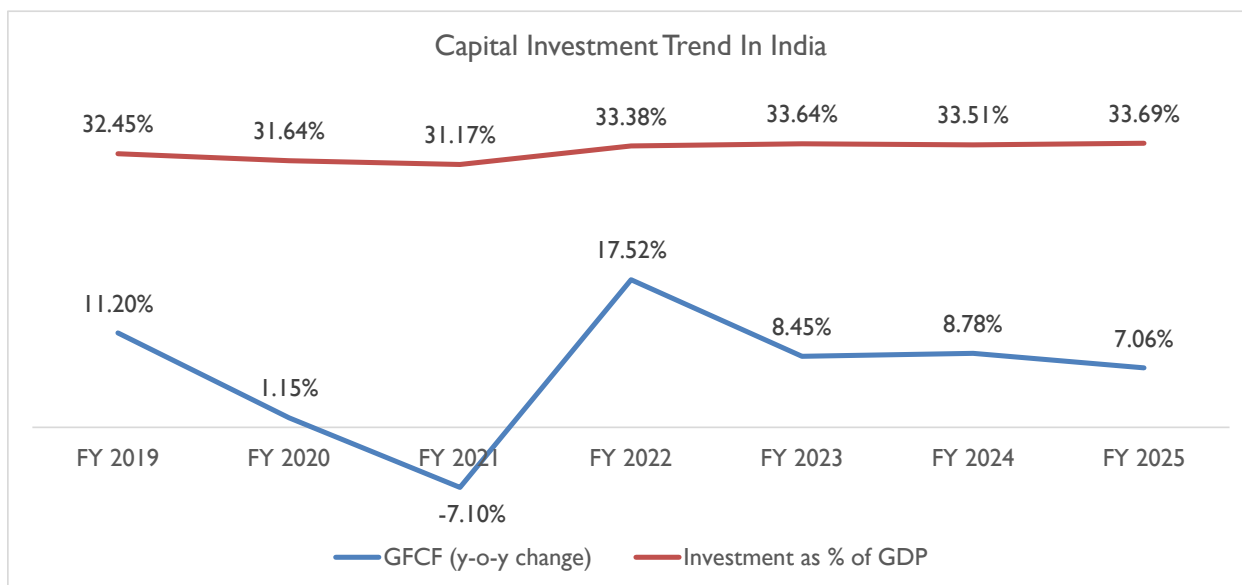


Source: Ministry of Statistics & Programme Implementation (MOSPI)

The IIP growth rate for the month of September 2025 is 4.0% which was 4.1% in the month of August 2025. The growth rates of the three sectors, Mining, Manufacturing and Electricity for May 2025 are (-)0.4%, 4.8% and 3.1% respectively.

Annual and Quarterly: Investment & Consumption Scenario

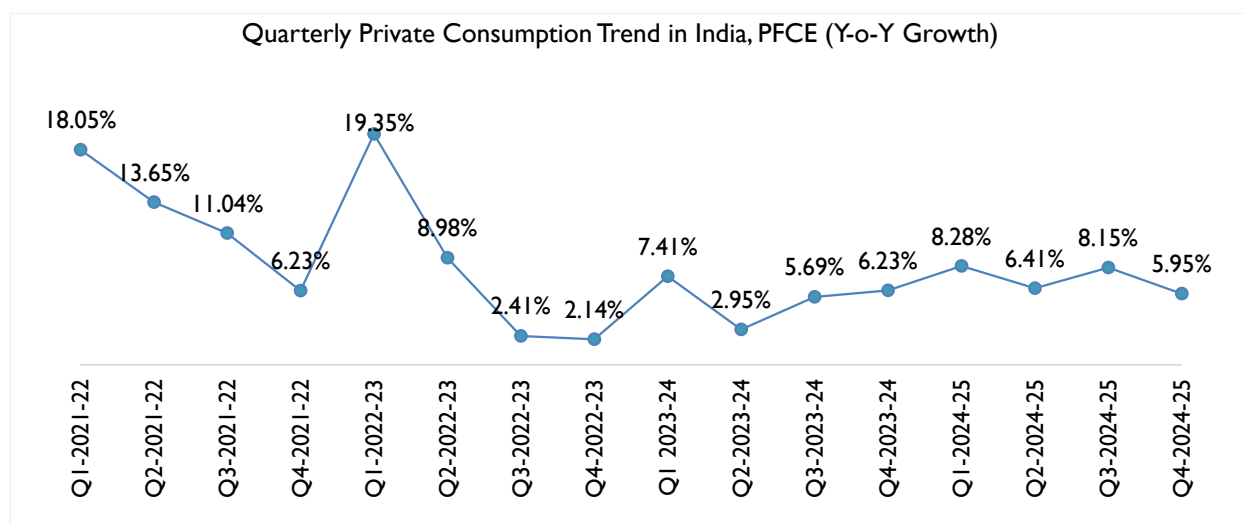
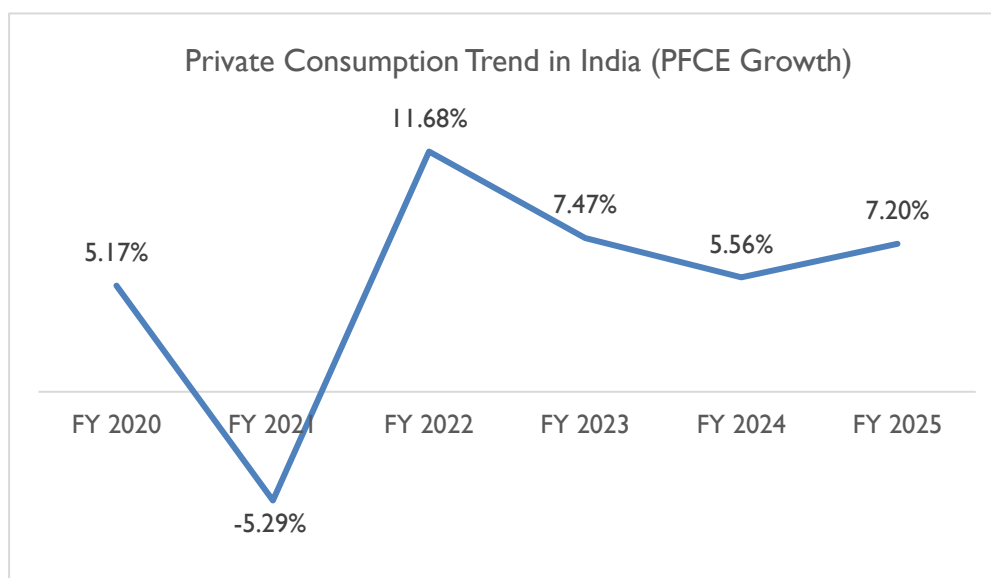
Other major indicators, such as Gross fixed capital formation (GFCF), a measure of investments, has shown fluctuation during FY 2025 as it registered 7.06% year-on-year growth against 8.78% yearly growth in FY 2024, taking the GFCF to GDP ratio to 33.69%.



Source: Ministry of Statistics & Programme Implementation (MOSPI)

Quarterly, GFCF exhibited a fluctuating trend in quarterly growth over the previous year same quarter. In FY 2024, the growth rate moderated to 6.05% in the March quarter against the previous two quarters as the government slowed on capital spending amidst the 2024 general election, while it observed an improvement in Q1 FY 2025 by growing at 6.65% against 6.05% in the previous quarter and moderated in the subsequent two quarters. On a yearly basis, the growth rate remained lower compared to the same quarter in the previous year during FY 2025. The GFCF to GDP ratio measured 33.91% in Q4 FY 2025.

Private Consumption Scenario



Sources: MOSPI

Private Final Expenditure (PFCE) a realistic proxy to gauge household spending, observed growth in FY 2025 as compared to FY 2024. However, quarterly data indicated some improvement in the current fiscal year as the growth rate improved over the corresponding period in the last fiscal.

Inflation Scenario

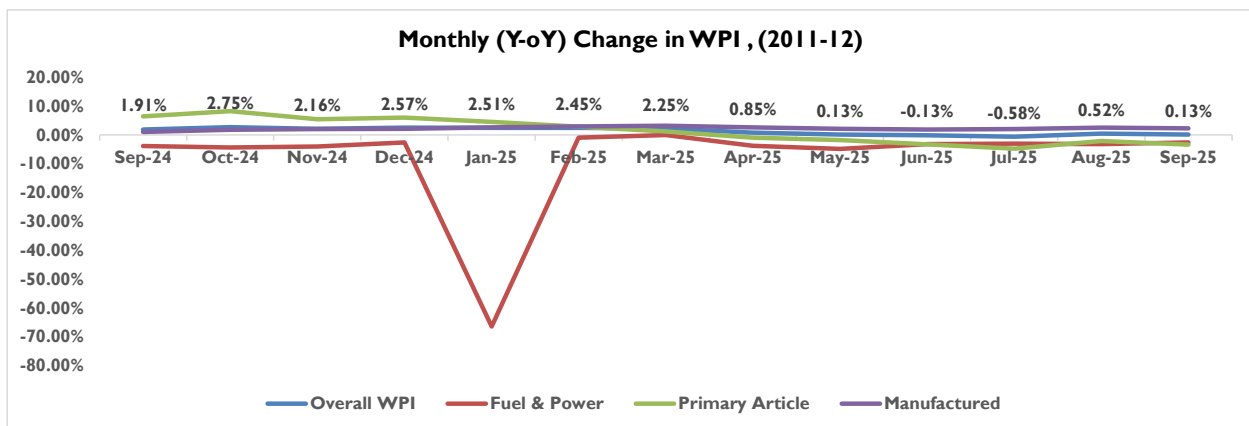
The inflation rate based on India's Wholesale Price Index (WPI) exhibited significant fluctuations across different sectors from September 2024 to September 2025. The annual rate of inflation based on the All-India Wholesale Price Index (WPI) number is 0.13% (provisional) for September 2025 (over September 2024). Positive rate of inflation in September 2025 is primarily due to an increase in prices of the manufacture of food products, other manufacturing, non-food articles, other transport equipment and textiles, etc.

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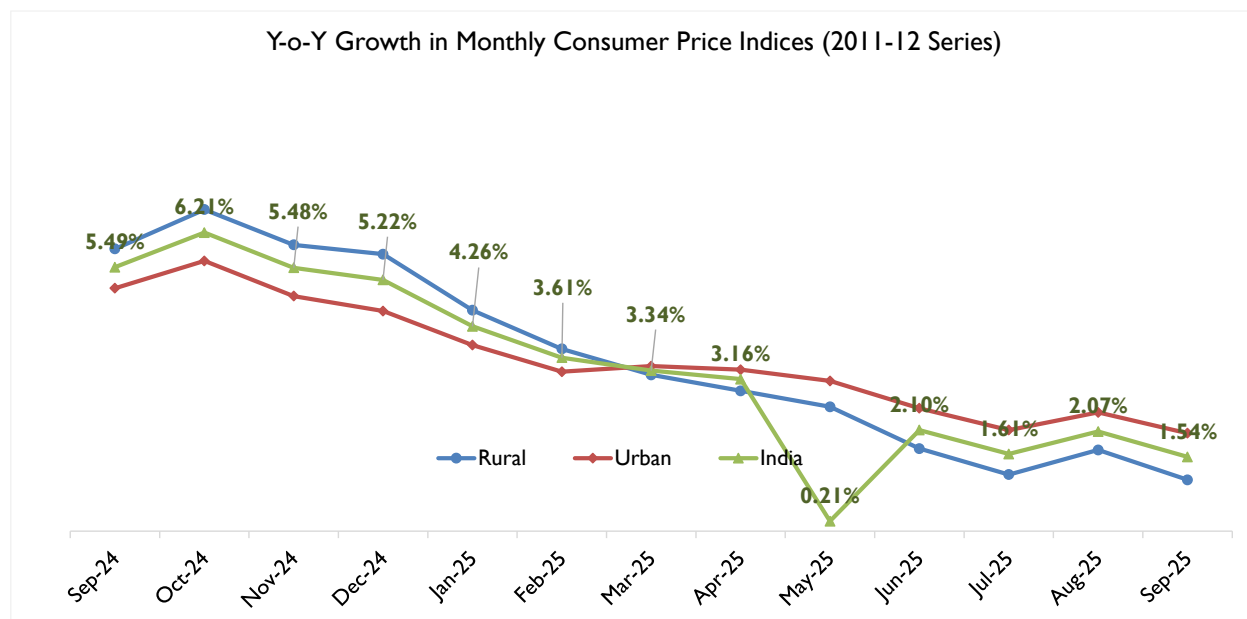
By September 2025, Primary Articles (Weight 22.62%): - The index for this major group decreased by 1.05 % from 191.0 (provisional) for the month of August 2025 to 189.0 (provisional) in September 2025. The price of food articles (-1.38%) and non-food articles (-1.06%) decreased in September 2025 as compared to August 2025. The price of minerals (1.36%) and Crude Petroleum & Natural Gas (0.64%) increased in September 2025 as compared to August 2025.

Moreover, Fuel & Power (Weight 13.15%): - The index for this major group decreased by 0.14% from 143.6 (provisional) for the month of August 2025 to 143.4 (provisional) in September 2025. The price of mineral oils (-0.54%) and coal (-0.15%) decreased in September 2025 as compared to August 2025. The price of electricity (1.20%) increased in September 2025 as compared to August 2025.

Furthermore, Manufactured Products (Weight 64.23%): - The index for this major group increased by 0.21% from 144.9 (provisional) for August 2025 to 145.2 (provisional) in September 2025. Out of the 22 NIC two-digit groups for manufactured products, 10 groups witnessed an increase in prices, 6 groups witnessed a decrease in prices, and 6 groups witnessed no change in prices. Some of the important groups that showed a month-over-month increase in prices were other manufacturing, food products, electrical equipment, textiles and other non-metallic mineral products, etc. Some of the groups that witnessed a decrease in prices were the manufacture of rubber and plastics products; motor vehicles, trailers and semi-trailers; pharmaceuticals, medicinal chemical and botanical products; leather and related products; and printing and reproduction of recorded media, etc., in September 2025 as compared to August 2025.



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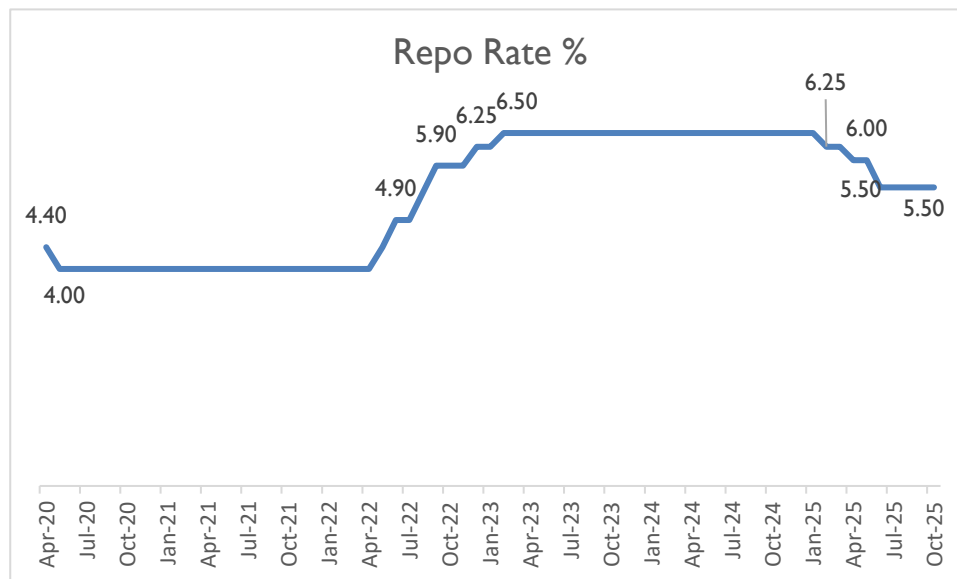
Source: MOSPI, Office of Economic Advisor

Retail inflation rate (as measured by the Consumer Price Index) in India showed notable fluctuations between September 2024 and September 2025. Year-on-year inflation rate based on All India Consumer Price Index (CPI) for the month of September 2025 over September 2024 is 1.54% (Provisional). There is a decrease of 53 basis points in the headline inflation of September 2025 in comparison to August 2025. It is the lowest year-on-year inflation since June 2017. Rural Inflation: A decrease in headline and food inflation in the rural sector was observed in September 2025. The headline inflation is 1.07% (Provisional) in September 2025, while it was 1.69% in August 2025. While in Urban inflation, A decrease from 2.47% in August 2025 to 2.04% (Provisional) in September 2025 is observed in the headline inflation of the urban sector. The decline in headline inflation and food inflation during the month of September 2025 is mainly attributed to a favourable base effect and to a decline in inflation of Vegetables, Oil and fats, Fruits, Pulses and products, Cereal and products, Egg, Fuel and light, etc.

CPI has been measured above the 6.00% tolerance limit of the central bank since July 2023. As part of its anti-inflationary stance, the Reserve Bank of India (RBI) hiked the repo rate by 250 basis points between May 2022 and 8 February 2023, holding it steady at 6.50% until January 2025. On 6 June 2025, the RBI reduced the repo rate by 50 basis points, bringing it to 5.50%, where it currently stands as per the October 2025 monetary policy review.



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Sources: CMIE Economic Outlook

Growth Outlook

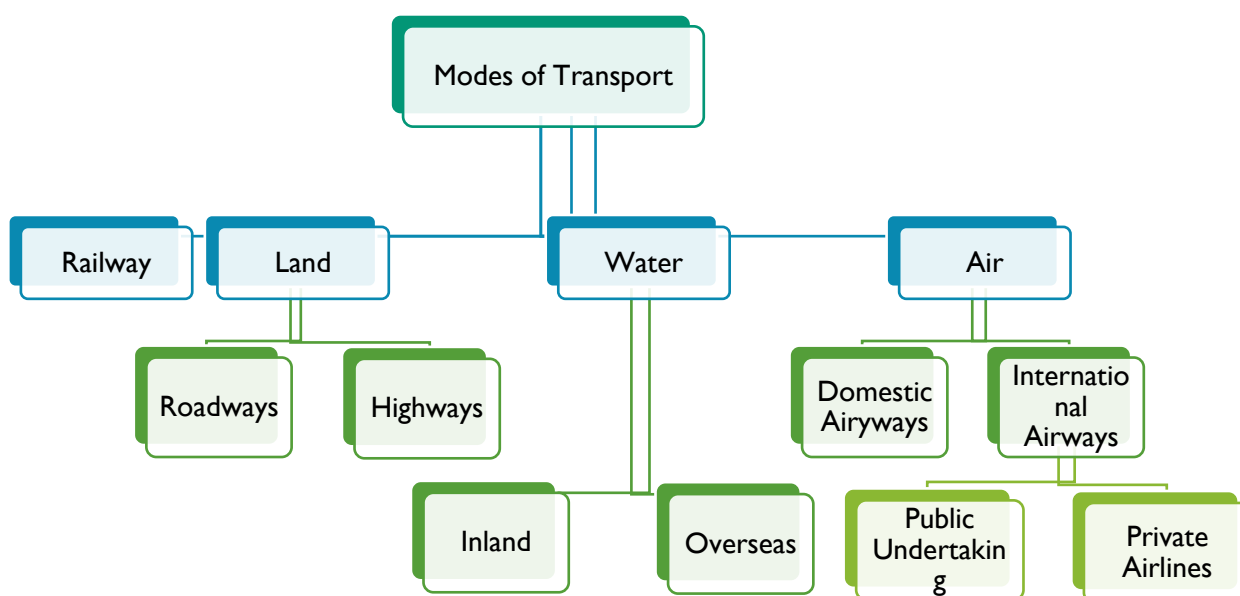
The Union Budget 2025-26 has laid the foundation for sustained growth by balancing demand stimulation, investment promotion and inclusive development. Inflation level is reaching within the central bank's target; the RBI may pursue further monetary easing that will support growth. The medium-term outlook is bright, fuelled by the emphasis on physical and digital infrastructure spending. With a focus on stimulating demand, driving investment and ensuring inclusive development, the budget introduces measures such as tax relief, increased infrastructure spending and incentives for manufacturing and clean energy. These initiatives aim to accelerate growth while maintaining fiscal discipline, reinforcing India's long-term economic resilience. The expansion of tax relief, i.e zero tax liability for individuals earning up to INR 12 lacs annually under the new tax regime, is expected to strengthen household finances and, consequently, boost consumption.

The external sector remains resilient, and key external vulnerability indicators continue to improve. However, tariff-related uncertainty is likely to weigh on exports and investment, prompting us to cut our CY26 GDP growth forecast to 6.2%.

Logistics Industry in India

An Industry Overview and Genesis of Logistics Market: Road Transport, Air Transport, Sea Transport and Rail Transport

The **logistics market in India** has evolved into one of the most critical enablers of the country's economic growth, acting as the backbone that connects producers, suppliers, and consumers across its vast geography. With the increasing pace of industrialization, urbanization, and globalization, the demand for efficient movement of goods has surged, giving rise to a complex and dynamic logistics ecosystem. According to a recent **NITI Aayog report**, India's logistics cost stands at around **14% of GDP**, a figure the government is actively working to reduce through infrastructure and policy reforms. The freight modal split reflects this shift: **road transport accounts for 60%, rail 30%, inland waterways and shipping 3%, and air 1%**, highlighting the multimodal nature of the sector.



Road transport dominates India's logistics landscape, accounting for the majority of domestic freight movement due to its flexibility, accessibility, and ability to provide door-to-door connectivity. The road network, consisting of over 6.3 million kilometres of highways and arterial roads, has expanded to approximately 63.45 lakh km by FY24, enabling the distribution of goods across urban and rural regions and supporting small-scale industries, e-commerce deliveries, and last-mile connectivity. In FY24 alone, around **214 million tonnes of coal** were transported via roads, illustrating its crucial role for energy and industrial supply chains. Government initiatives like the Bharatmala Project aim to improve road quality, reduce congestion, and enhance freight efficiency, further strengthening road transport's role in India's logistics ecosystem.



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Rail transport complements roadways by providing cost-effective and energy-efficient movement of bulk commodities over long distances. Dedicated Freight Corridors (DFCs) and modernized rail infrastructure have enabled faster transit of goods such as coal, cement, steel, fertilizers, and agricultural produce. In FY25, Indian Railways recorded **freight loading of over 1,617 million tonnes**, a ~2% increase over the prior year, which translated into **freight revenue of INR 1.75 trillion**. By February 2025, rail freight revenue for the year was estimated at **INR 2,78,600 crore**, indicating both scale and future potential. Rail logistics is particularly vital for inter-state and inter-regional trade, reducing dependency on long-haul trucking and mitigating highway congestion.

Air and maritime transport form the backbone of India's international logistics. Air transport is crucial for high-value, time-sensitive, and perishable cargo, including electronics, pharmaceuticals, and express parcels, ensuring rapid connectivity to global markets. Maritime transport, on the other hand, underpins the country's trade and energy security. In FY25, India's major ports handled **855 million tonnes of cargo**, marking a **4.3% year-on-year increase**. Container traffic, as well as petroleum, oil, and lubricants (POL) volumes, grew significantly, underscoring the expanding role of ports in supporting India's international trade. Initiatives like the **Sagarmala Project** focus on port modernization, coastal shipping, and multimodal integration, reducing logistics costs and improving competitiveness in global trade.

The logistics sector in India continues to evolve rapidly, driven by policy reforms, infrastructural investments, and technological innovations such as AI-enabled tracking, digital freight platforms, and integrated supply chain solutions. With industrialization, urbanization, and international trade expanding, new opportunities are emerging across road, rail, air, and maritime transport. The adoption of multimodal logistics, development of Dedicated Freight Corridors, port modernization under the Sagarmala Project, and growing investment in e-commerce and cold-chain infrastructure are reshaping the industry landscape. As the sector scales up, it is expected to adapt continuously to emerging technologies, changing trade patterns, and sustainability imperatives, reflecting India's dynamic economic growth trajectory.

Genesis of Indian Logistics

The evolution of India's logistics industry can be traced back to the economic liberalization of the 1990s. This period marked a shift toward global supply chain integration, attracting foreign investments and modernizing logistics operations. Before liberalization, the sector was highly fragmented, labour-intensive, and reliant on outdated infrastructure and technology.

The rapid expansion of e-commerce, manufacturing, and retail sectors has further fuelled the need for streamlined supply chains. Foreign investments have introduced advanced technologies and best practices, significantly enhancing the capabilities of Indian logistics providers.

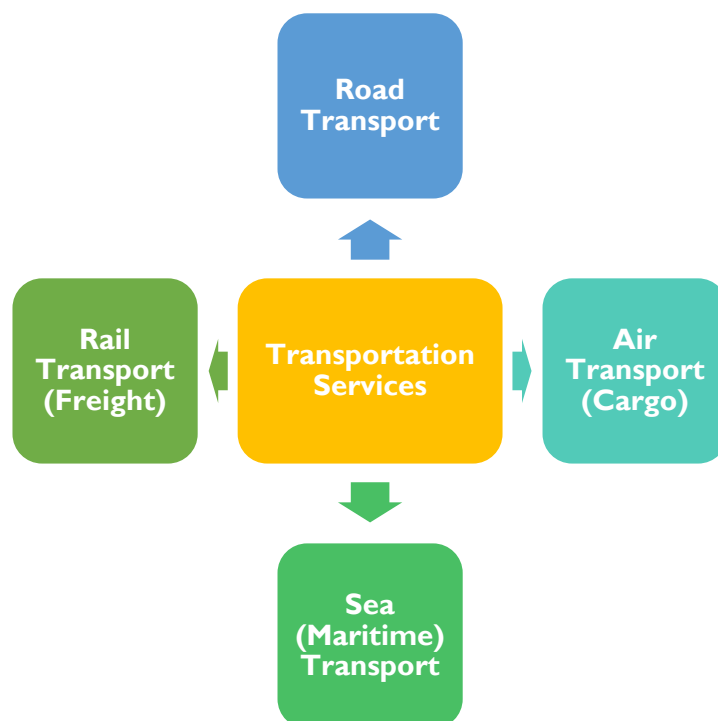


Recognizing the importance of infrastructure, the Indian government has heavily invested in roads, railways, ports, and airports. Key initiatives like the Goods and Services Tax (GST) and the Dedicated Freight Corridor (DFC) project have improved goods movement efficiency, reduced transit times, and strengthened India's logistics ecosystem.

- **Transportation Services:**

Transportation is the backbone of logistics, ensuring the movement of goods across the country. India's transportation network is shaped by historical developments, geography, and cultural influences. While transportation contributes significantly to economic growth, it also impacts the environment due to emissions and climate change. Advancements in technology have made transportation more cost-effective and efficient.

India's transportation system is classified into four primary modes:



I. Road Transport

Road transport forms the backbone of India's logistics sector, offering unmatched flexibility, last-mile reach, and accessibility to even the most remote regions. With an extensive road network of approximately **6.3 million kilometres**, one of the largest in the world, it plays a critical role in enabling domestic trade, efficient distribution, and smooth mobility across rural and urban markets.

Key Features of Road Transport:

- **Connectivity:** Links urban and rural areas, ensuring seamless movement of goods.
- **Flexibility:** Provides route flexibility and vehicle options to suit various logistics needs.



Types of Roads in India:

- **Golden Quadrilateral & Superhighways:** Connects major cities like Delhi, Kolkata, Chennai, and Mumbai, along with North-South and East-West corridors.
- **National Highways:** Maintained by the Central Public Works Department, connecting major cities and economic centres.
- **State Highways:** Links state capitals with district headquarters, managed by state authorities.
- **District Roads:** Connects district headquarters to smaller towns and villages.
- **Rural Roads:** Developed under schemes like the Pradhan Mantri Grameen Sadak Yojana.
- **Border Roads:** Constructed by the Border Roads Organisation to improve connectivity in strategic areas.

Key Government Initiatives

- **Bharatmala Pariyojana:** Focuses on developing economic corridors, border roads, expressways, and national highways.
- **PMGSY:** Strengthens rural connectivity by providing all-weather roads.
- **FASTag:** Introduces digital tolling for faster movement and reduced congestion.
- **Logistics Efficiency Enhancement Program (LEEP):** Aims to integrate transport modes and improve overall logistics performance.

2. Air Transport (Cargo)

Air transport is a rapidly expanding pillar of India's logistics ecosystem, enabling the fast movement of high-value, time-critical, and perishable goods. India has emerged as the world's third-largest domestic aviation market, supported by rising incomes, urbanisation, and demand for quicker delivery networks. During FY26 (April–July 2025), passenger traffic reached 96.54 million and air freight volumes touched 1.2 MMT, reflecting strong sectoral momentum. The country's airport network has grown from 74 airports in 2014 to 162 by September 2025, strengthening both domestic and global connectivity. Major hubs such as Delhi, Mumbai, Bengaluru, Chennai, and Hyderabad now anchor India's expanding air-cargo infrastructure, serving e-commerce, pharmaceuticals, electronics, perishables, and express logistics.

Development of Air Transport:



- Nationalized in 1953, leading to state-owned carriers like Indian Airlines and Air India.
- Private airlines and non-scheduled operators now enhance domestic air cargo connectivity.
- Helicopter services, such as Pawan Hans, facilitate transport to remote areas like the Northeast, Jammu & Kashmir, and Himachal Pradesh.

Government Initiatives Boosting Air Logistics

- **UDAN Scheme** to enhance regional connectivity and include cargo routes
- **Krishi UDAN** to support the transport of agricultural and perishable commodities
- **Air Cargo Logistics Promotion Board (ACLPB)** for process optimisation
- **Expansion of cargo terminals** with modern warehousing and handling facilities

Although air transport is more expensive than other modes, it offers unmatched speed and accessibility, making it crucial for industries requiring fast delivery.

3. Sea (Maritime) Transport

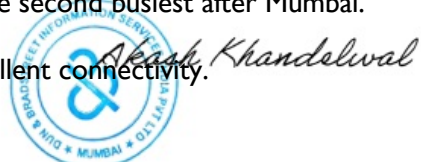
Maritime transport is one of the oldest and most cost-effective modes of transportation, ideal for moving bulk goods over long distances. India has an extensive coastline of **7,516.6 km**, with **12 major ports** and around **200 minor ports** that handle approximately **95% of India's international trade**.

Key Maritime Transport Features:

- Inland Waterways: India has 14,500 km of inland waterways, but only 5,685 km are navigable.
- Major National Waterways:
 - Ganga Waterway (Allahabad – Haldia, 1,620 km)
 - Brahmaputra Waterway (Sadiya – Dhubri, 891 km)
 - West Coast Canal (Kerala, 205 km)
 - Godavari-Krishna Waterway (1,078 km)
 - Brahmani-Mahanadi Waterway (588 km)

Major Sea Ports in India:

- Kandla (Deendayal Port): First port built after independence, located in Gujarat.
- Mumbai Port: Largest and most important natural harbour.
- Marmagao (Goa): Handles around 50% of India's iron ore exports.
- Chennai Port: One of the oldest artificial ports, the second busiest after Mumbai.
- Visakhapatnam Port: A deep inland port with excellent connectivity.



- Kolkata Port: A tidal river port requiring regular dredging.

Government Initiatives Boosting Maritime Transport

- **Sagarmala Programme:** Port modernization, coastal shipping, and port-led industrialization
- **National Waterways Development:** Promoting inland shipping and improving the navigability of rivers
- **Port Connectivity Improvement Project (PCIP):** Strengthening Road and rail links to major ports
- **Maritime Clusters & Industrial Corridors:** Establishing logistics parks and container terminals near ports
- **Digitization & Ease of Doing Business:** Single-window clearances and faster port operations.

4. Rail Transport (Freight)

Railways are the backbone of India's logistics sector, providing a **cost-effective, energy-efficient, and high-volume mode** for transporting goods over long distances. The network, divided into **16 zones**, plays a vital role in supporting industrial growth, agriculture, and trade. Rail freight is particularly suitable for bulk commodities such as coal, cement, steel, fertilizers, and agricultural produce, offering reliability even under adverse weather conditions.

Key Features of Rail Transport

- **High Capacity & Efficiency:** Ideal for long-distance, high-volume cargo.
- **Weather-Resistant:** Less affected by rain or extreme temperatures compared to road transport.
- **Cost-Effective:** Provides lower freight costs for bulk and heavy cargo.
- **Dedicated Freight Corridors (DFC):** Modernization initiatives to reduce congestion, increase speed, and improve logistics efficiency.

Factors Affecting Railway Development:

- Flat Terrain (Northern Plains): Ideal for railway expansion due to high population density and agricultural activity.
- Difficult Terrain (Hills & Deserts): Rail construction is challenging in hilly regions, deserts, and dense forests.



- **New Railway Projects:** The Konkan Railway has improved connectivity in western India but faces challenges like landslides.

Government Initiatives Boosting Rail Logistics

- **Dedicated Freight Corridors (DFC):** Modernizing rail freight with high-capacity, high-speed cargo routes
- **National Rail Plan (NRP) 2030:** Optimizing rail infrastructure to handle 50% of freight movement by rail
- **Private Freight Terminals (PFTs):** Encouraging private participation for faster loading/unloading and warehousing
- **Railway Electrification & Green Initiatives:** Reducing carbon footprint and improving energy efficiency
- **Digital Freight Operations & Tracking:** Implementation of real-time tracking, EDI, and automated scheduling for efficiency

Rail transport is a weather-resistant mode of transportation compared to road and air, making it a reliable choice for freight movement. Indian Railways continues to modernize with Dedicated Freight Corridors (DFC) and high-speed rail projects to enhance efficiency.

Freight Forwarding Services

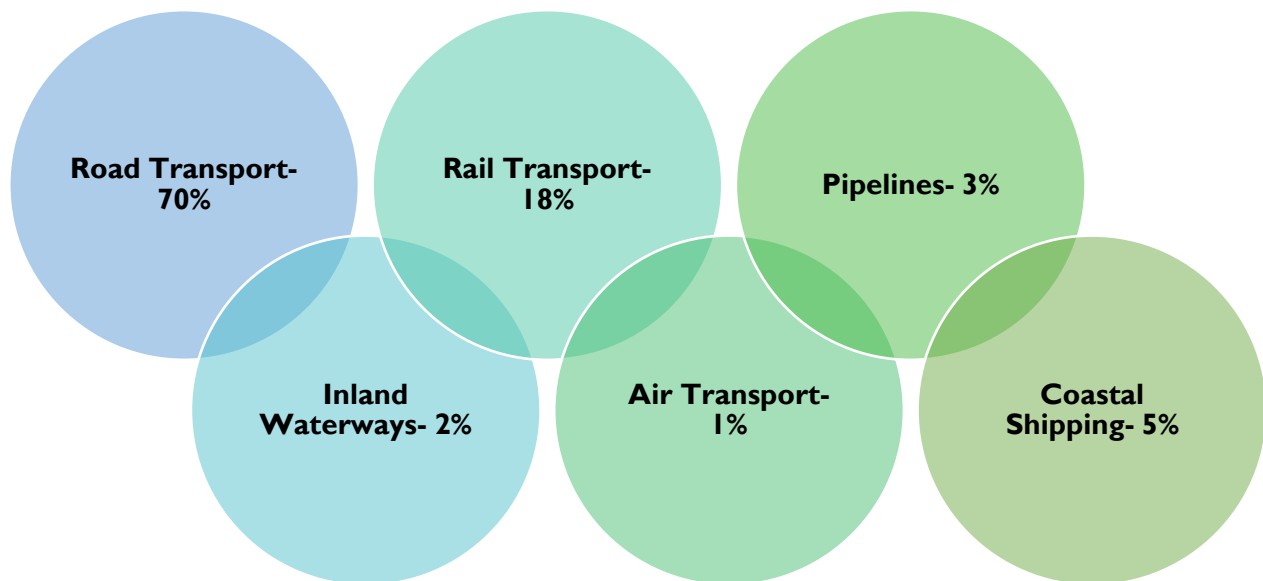
Freight forwarding involves the coordination and management of shipments on behalf of shippers, serving as a critical link in India's logistics ecosystem. Freight forwarders help businesses navigate the complexities of transportation by selecting optimal routes, coordinating multiple modes of transport, road, rail, air, and sea, and ensuring timely delivery. They handle key responsibilities such as documentation, customs brokerage, warehouse planning, cargo consolidation, and insurance facilitation, enabling the smooth movement of goods both domestically and internationally.

For businesses looking to expand, international trade offers the opportunity to reach a broader customer base, but global logistics can be challenging due to varying regulations and complex supply chains. Freight forwarders simplify this process by ensuring shipments comply with all regulatory requirements, are transported efficiently, and reach their destination on schedule. Their expertise not only reduces operational risks but also supports industries such as e-commerce, manufacturing, pharmaceuticals, and agriculture in maintaining reliable, cost-effective, and seamless supply chains.

India's freight transportation industry plays a crucial role in the nation's economy, facilitating the movement of goods across vast distances. The industry is dominated by road transport, followed by railways, waterways,



and other transport modes. Below is a detailed breakdown of the freight market, highlighting the share of each mode and its significance.



1. Road Transport

- **Market Share:** Approximately **70%** of India's total freight movement relies on road transport.
- **Key Factors for Dominance:**
 - **Versatility:** Road transport supports diverse cargo types, including perishable goods, consumer products, and industrial materials.
 - **Last-Mile Connectivity:** It provides seamless door-to-door delivery, making it the preferred choice for businesses.
 - **Expanding Infrastructure:** Government initiatives like the **Bharatmala Project** are enhancing highway networks to improve efficiency.

2. Rail Transport

- **Market Share:** Around **18%** of India's total freight traffic moves via railways.
- **Advantages:**
 - **Cost Efficiency:** Rail transport is **cheaper** for long-distance bulk shipments like coal, cement, and food grains.
 - **Sustainability:** Lower carbon emissions compared to road transport make it an **eco-friendly alternative**.



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3. Inland Water Transport (IWT)

- **Market Share:** Currently accounts for about **2%** of total freight movement.
- **Advantages:**
 - **Lower Costs:** Waterways offer a **cost-effective** alternative for bulk transportation over long distances.
 - **Energy Efficient:** Fuel consumption is significantly **lower** than road and rail transport.

4. Other Transport Modes (Coastal Shipping, Pipelines, Air Transport)

The remaining **9%** is distributed among various alternative freight modes.

➤ Coastal Shipping (5% of Total Freight Movement)

- **Usage:** Primarily handles **bulk cargo** like coal, iron ore, crude oil, and fertilizers.
- **Advantages:**
 - **Cost-effective** for long-distance freight movement along India's 7,500 km coastline.
 - **Energy-efficient** with lower carbon emissions compared to road and rail.

➤ Pipelines (3% of Total Freight Movement)

- **Usage:** Best suited for **liquids and gases**, including petroleum products, crude oil, and natural gas.
- **Advantages:**
 - **Safe and cost-efficient** for transporting large volumes over long distances.
 - **Minimal human intervention** reduces delays and operational risks.

➤ Air Transport (1% of Total Freight Movement)

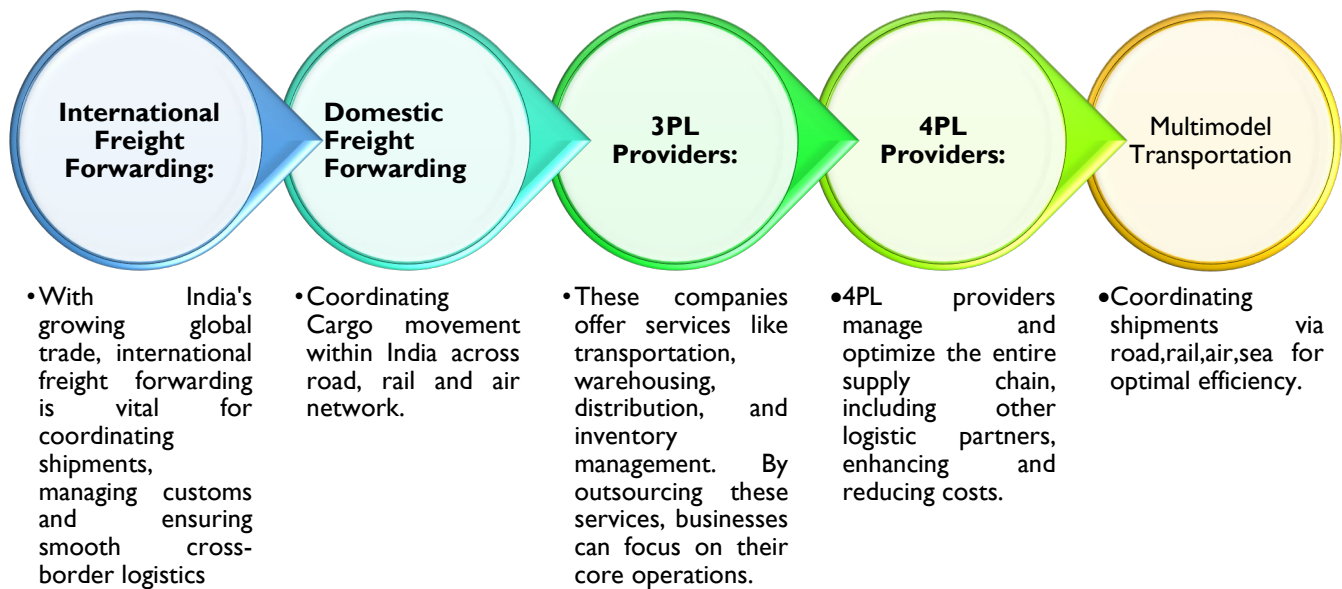
- **Usage:** Primarily for **high-value and time-sensitive goods** such as pharmaceuticals, electronics, and perishable items.
- **Advantages:**
 - **Fastest mode of transport**, reducing lead times for urgent shipments.
 - **High security and minimal damage risk** for delicate cargo.

While **road and rail** dominate India's freight market, **coastal shipping, pipelines, and air transport** serve specialized roles. **Coastal shipping** and **pipelines** hold significant potential for reducing logistics costs and congestion, while **air transport** will remain essential for express cargo and high-value goods.



Government initiatives and private sector investments will be key in improving infrastructure and increasing the efficiency of these transport modes.

Freight Forwarding and Logistics: Key Service Providers:



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Belly cargo space

Belly cargo space, the underfloor cargo hold of an aircraft, plays a crucial role in air freight operations. While freighters are dedicated to carrying cargo, passenger aircraft also utilize their belly capacity to transport goods, providing an additional revenue stream for airlines. The efficiency of belly cargo utilization depends on factors such as aircraft type, baggage allowance, and overall cargo demand.

Utilization of Belly Cargo Space in Passenger Aircraft

- Passenger airlines maximize their revenue by optimizing available belly space for cargo transport. However, there are limitations to how much cargo can be carried, primarily due to the allocation of space for passenger baggage.
- Typically, airlines offer a baggage allowance of **15 kg per passenger** in the domestic sector. For an aircraft carrying **200 passengers**, this translates to **3,000 kg of baggage**, reducing the available belly capacity for additional cargo. Despite this limitation, airlines have been making concerted efforts to enhance belly cargo efficiency, ensuring optimal utilization of available space.

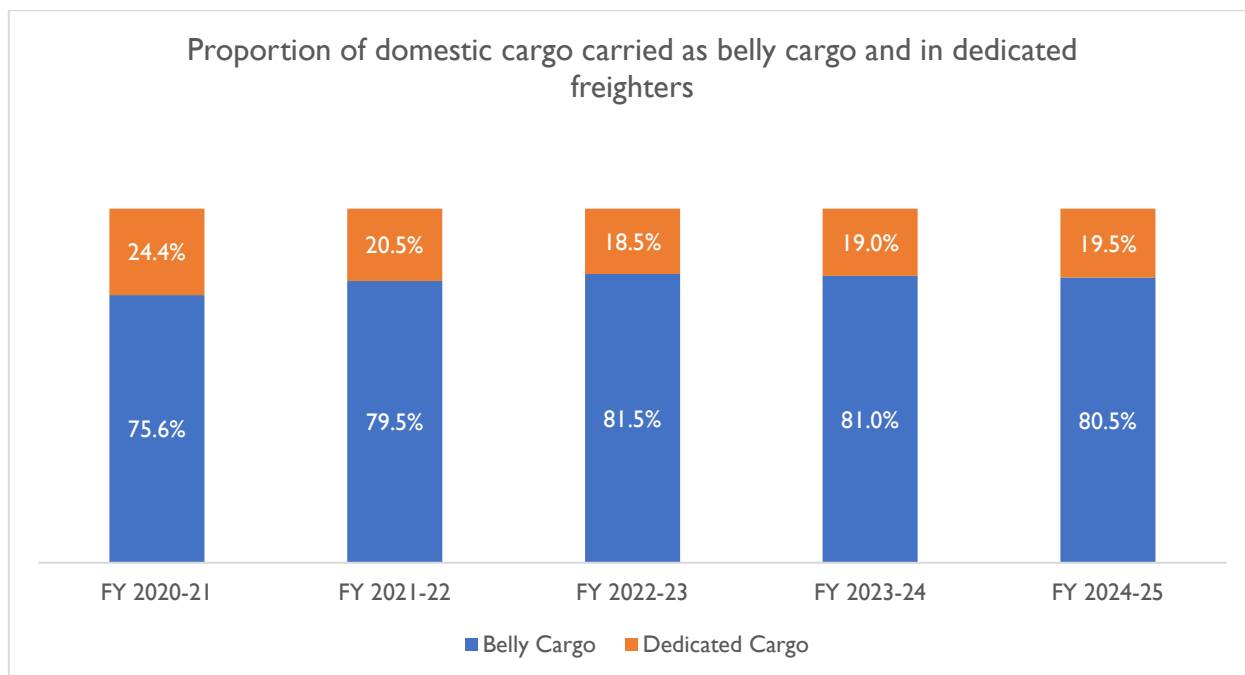
Belly Cargo Capacity in Different Aircraft Types

The belly cargo capacity varies based on the type of aircraft:

- **Narrow-body aircraft:** These aircraft, commonly used for domestic and short-haul international routes, can accommodate up to 6 tonnes of cargo in their belly compartments, depending on configuration and baggage load.
- **Wide-body aircraft:** Used for long-haul and international flights, wide-body aircraft have significantly larger belly capacities, often exceeding 20 tonnes.
- **Freight-dedicated aircraft** primarily focus on maximizing main deck capacity, with belly space serving as an additional cargo area.

Growth of Belly Cargo Utilization in India

In recent years, Indian airlines have significantly improved belly cargo utilization. There has been a steady increase in belly cargo utilization rates:



Note: as per the latest data of December 2025

Over the past five years, India's air cargo market has shown a clear dominance of **belly cargo**, which consistently accounts for the majority of shipments compared to **dedicated cargo**. In FY2020-21, belly cargo constituted **75.6%** of the market, rising to **81.5% by FY2022-23**, before slightly moderating to **80.5% in FY2024-25**. Conversely, dedicated cargo has maintained a smaller share, declining from **24.4% in FY2020-21** to a low of **18.5% in FY2022-23**, then edging up slightly to **19.5% in FY2024-25**. This trend reflects the continued reliance of air cargo operators on passenger flights for freight transport, while dedicated freighter services, though growing steadily, remain a smaller component of the overall air cargo ecosystem.

This growth highlights the importance of belly cargo in the Indian air freight sector. With increasing demand for e-commerce and time-sensitive shipments, airlines continue to optimize cargo operations to enhance revenue potential.

Challenges and Future Prospects

Despite its advantages, belly cargo space has some challenges:

- **Limited Capacity:** Passenger baggage reduces the space available for cargo, especially on high-density routes.
- **Load Balancing:** Cargo distribution must align with aircraft weight and balance requirements to maintain safety.

¹ Civil Aviation Statistics

- **Regulatory Compliance:** Cargo transport regulations, including weight restrictions and security screening, impact operational efficiency.

Going forward, advancements in cargo handling technology, automated loading systems, and real-time tracking solutions will further enhance belly cargo efficiency. Additionally, airlines may introduce flexible baggage policies to optimize space utilization while maintaining passenger convenience. Belly cargo space remains a vital component of air cargo operations, providing a cost-effective way to transport goods alongside passengers. As airlines continue to refine cargo strategies, the utilization of belly capacity is expected to grow, contributing to overall industry efficiency and revenue generation.



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India's Logistics Performance Index (LPI)

The country successfully ascended from the 44th to the 38th position in the World Bank's Logistics Performance Index (LPI) 2023, demonstrating notable progress. Logistic cost in India currently stands between 7.8-8.9% of GDP,² intriguingly closer to a level seen in developed nations.

Parameter	Rank 2023 (Out Of 139 Countries)	Rank 2018 (Out Of 160 Countries)	Rank 2016 (Out Of 160 Countries)
Overall LPI Rank	38	44	35
Custom	38	40	38
Infrastructure	47	52	36
International Shipments	22	44	39
Logistics Quality & Competence	38	42	32
Tracking & tracing	41	38	33
Timeliness	35	52	42

Source: World Bank

This advancement is largely due to strategic government initiatives such as the PM Gati Shakti National Master Plan and the National Logistics Policy, which have enhanced logistics efficiency and infrastructure. Investments in trade-related infrastructure and the adoption of digital technologies such as the Unified Logistics Interface Platform (ULIP) and the Logistics Data Bank have also played a crucial role. Additionally, the establishment of an Inter-Ministerial team and the National Committee for Trade Facilitation (NCTF) has contributed to this progress in addressing key logistics parameters. Notable improvements include India's climb in international shipment rankings from **44th to 22nd** and an enhancement in infrastructure scores from **52nd to 47th**.

² National Council of Applied Economic Research (NCAER) estimates in 2021-22



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Government Initiatives and Infrastructure Development

- **National Logistics Policy (2022)**
 - Launched in September 2022, the National Logistics Policy aims to optimize India's logistics landscape by improving multimodal transportation.
 - A key focus is to increase the railways' share in freight movement, currently at 18%, through investments in Dedicated Freight Corridors (DFCs) and expanded inland waterways.
 - As of April 2024, 96% of DFCs are complete, expected to enhance rail freight capacity and efficiency, thereby improving the modal mix.
 - The government's push for port privatization has led to improved infrastructure and efficiency at Indian ports, benefiting major logistics operators.
- **PM Gati Shakti – National Master Plan (2021)**
 - Launched in October 2021, PM Gati Shakti integrates **16** ministries to create a unified platform for planning and monitoring multimodal infrastructure.
 - The initiative focuses on reducing logistics costs (currently 14–16% of GDP) through better connectivity, seamless coordination, and faster project execution.
 - Over **1,600+** layers of data, covering roads, railways, ports, power, and land availability, are mapped to support efficient decision-making.
 - Connectivity projects such as industrial corridors, economic zones, and port linkages are prioritized to enhance freight efficiency and reduce transit delays.
- **Bharatmala Pariyojana**
 - A flagship highway development program aimed at building **~34,800 km** of highways to strengthen freight corridors and improve last-mile logistics connectivity.
 - Focuses heavily on the development of **economic corridors, feeder routes, border roads**, and expressways.
 - Has significantly reduced travel time on major freight routes and improved connectivity between manufacturing hubs and ports.
- **Sagarmala Program**
 - Targets modernization of port infrastructure and development of port-led industrial clusters to enhance coastal and EXIM logistics.
 - Covers more than 800+ projects across port connectivity, port modernization, and coastal community development.
 - Focuses on improving coastal shipping and reducing overall logistics costs by leveraging India's **7,500 km** coastline.
- **Logistics Costs and Modal Mix**



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- Logistic cost in India currently stands between **7.8-8.9% of GDP**³ intriguingly closer to a level seen in developed nations.
- The inefficiency in the sector is largely due to the skewed modal mix, where road transport accounts for **71%** of freight movement, leaving railways and waterways with a smaller share.
- To address these issues, the government has implemented Goods and Services Tax (GST) and invested heavily in road infrastructure, inland waterways, and DFCs.
- **Dedicated Freight Corridors (DFC)**
 - India is developing two major freight corridors, Eastern DFC (EDFC) and Western DFC (WDFC), to significantly increase freight speed and capacity.
 - As of 2024, **96% of the DFC network is completed**, with full operationalization expected to reduce transit time by up to 40%.
 - DFCs are expected to shift a substantial share of freight from road to rail, lowering logistics costs and improving reliability.
- **Airport Cargo & Connectivity Expansion**
 - The government is expanding air cargo capacity through upgrades under the UDAN and Bharatmala connectivity frameworks.
 - Dedicated cargo terminals and perishable cargo centers are being developed at major airports like Delhi, Mumbai, Hyderabad, and Bengaluru.
 - Airfreight capacity is rising to support e-commerce, high-value goods, and pharmaceutical exports.

Key Market Segments and Growth Trends

The Indian logistics sector is highly diverse, covering road transport, rail transport, air cargo, multimodal logistics, and industrial warehousing.

1. Domestic Express Logistics

- The domestic express logistics segment is projected to grow at a CAGR of **14%** from **FY23 to FY28**, driven by the expansion of e-commerce.
- Organized players currently control **80%** of the market and are expected to strengthen their dominance by leveraging government policies such as GST and the e-way bill.

2. Less-Than-Truckload (LTL) Segment

- The LTL segment in road transportation is expected to grow at a **CAGR of 10%** due to increasing demand for smaller, more frequent shipments that bypass warehouse storage to reach retailers directly.

³ National Council of Applied Economic Research (NCAER) estimates in 2021-22

3. Third-Party Logistics (3PL)

- The 3PL segment continues to expand as companies outsource transportation, warehousing, order fulfilment, and supply chain operations for cost efficiency.
- Demand is rising, particularly from sectors like automotive, e-commerce, FMCG, electronics, and pharmaceuticals.
- Technology adoption, including WMS, TMS, RFID, and real-time visibility solutions, is a major differentiator for leading 3PL providers.

4. Warehousing & Storage

- Warehousing is expanding rapidly as companies shift from fragmented storage facilities to larger, Grade-A warehouses.
- Automation, racking systems, and multi-level storage are becoming standard features, especially near industrial corridors and consumption hubs.
- E-commerce and retail consolidation are driving demand for fulfilment centres and last-mile storage nodes.

Growth Drivers of the Indian Logistics Market

• Expansion of E-Commerce and Digital Retail

- The boom in online shopping, hyperlocal delivery, and omni-channel retail has significantly increased demand for fast, reliable logistics services.
- E-commerce players are expanding fulfilment centers and last-mile networks even in Tier-2 and Tier-3 cities, driving logistics penetration deeper into the country.

• Infrastructure Modernization

- Massive investments in highways, expressways, multimodal logistics parks, and airports are enabling faster cargo movement across regions.
- Development of Dedicated Freight Corridors (DFCs), inland waterways, and port upgrades is strengthening India's multimodal transport network.

• Growth in Manufacturing and Industrial Activity

- Government programmes such as Make in India, PLI schemes, and industrial corridors are increasing production volumes, creating stronger demand for transportation, warehousing, and 3PL services.
- Sectors like automotive, electronics, pharmaceuticals, textiles, and FMCG are expanding their supply chains, directly boosting logistics requirements.

• Integration of Multimodal Logistics

- Increasing use of rail, road, air, and waterways in combination is reducing costs, improving transit times, and enhancing flexibility.



- Companies are moving toward dynamic route planning that can shift between modes based on cost, speed, and congestion

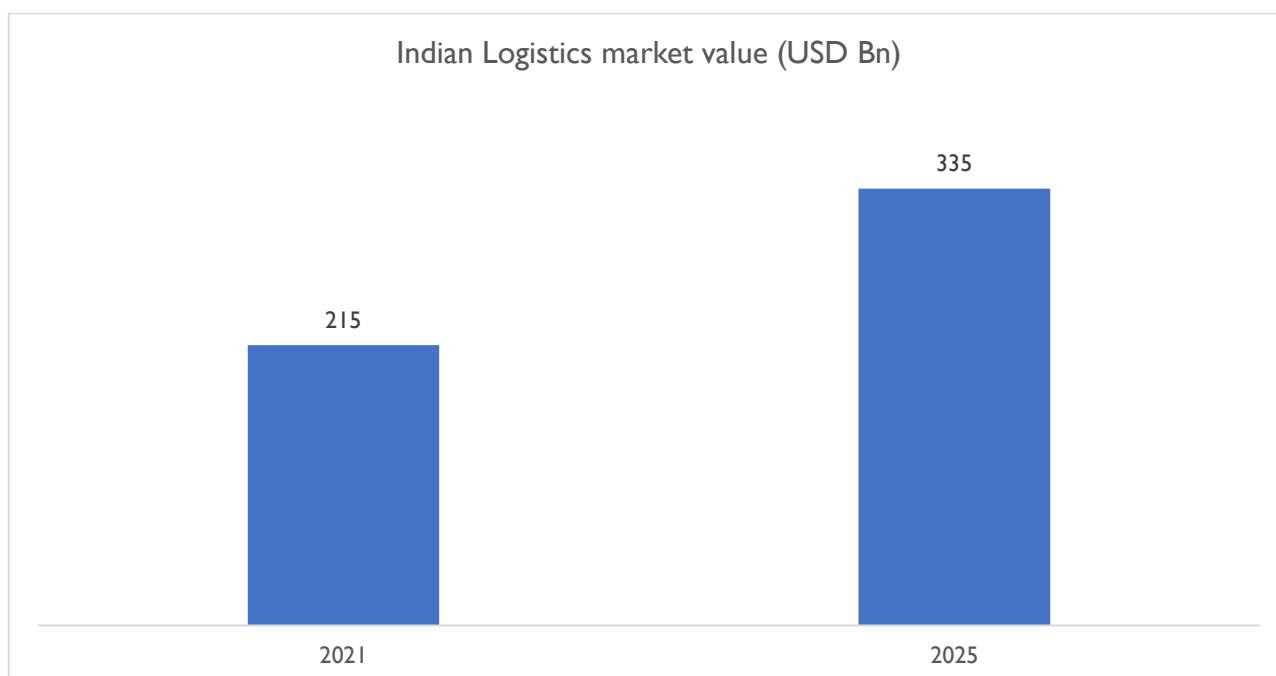


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Market Scenario: Estimated Market Size of Logistics Industry

The Indian logistics sector is among the largest in the world, offering significant opportunities for growth. This sector is an integral part of the national GDP value chain, as it connects various components of the economy and encompasses transportation, warehousing, and other supply chain solutions for both suppliers and end customers. An efficient logistics industry is a critical component as it secures better market access for goods and services and boosts consumption growth. Acknowledging the need to strengthen logistics infrastructure, the sector has attracted the attention of the government and even private participants.

India's logistics industry is thus poised to achieve significant milestones in 2024 due to a surge in e-commerce, government initiatives, and technological advancements. As the Indian economy expands, the logistics sector is expected to see unprecedented growth, driven by a range of factors including enhanced infrastructure and evolving market demands.



Source: Press Information Bureau, CII, D&B Research

The Indian logistics market has experienced robust growth over the past few years, with its market value reaching **USD 215 billion in 2021**. Driven by rapid industrialization, urbanization, and the expansion of e-commerce, the sector is projected to grow at a steady pace, reaching approximately **USD 335 billion by 2025** at a **compound annual growth rate (CAGR) of 11.72%**. This growth reflects increasing demand for efficient supply chain solutions, multimodal transportation, and advanced logistics services that connect producers, consumers, and international markets across India. The rising market size underscores the sector's critical role as a backbone of the country's economic development and trade competitiveness.

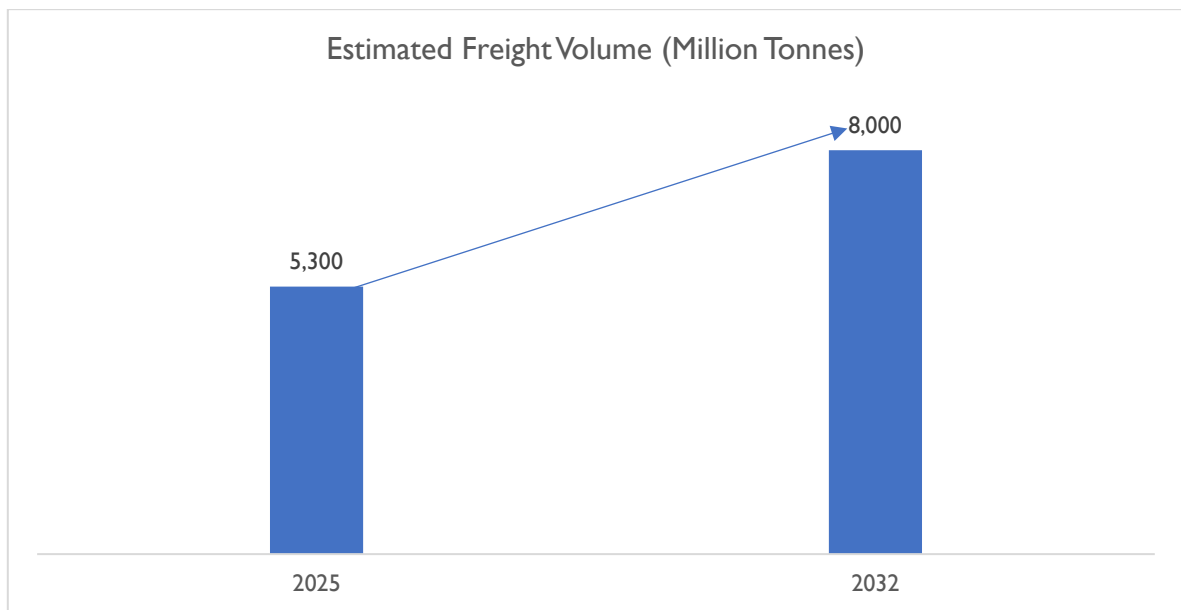
The boom in e-commerce has led to increased demand for efficient logistics solutions, particularly in last-mile delivery and rural expansion. This trend opens substantial investment opportunities in warehousing,



transportation, and technology-driven solutions. Government initiatives and major infrastructure projects such as Bharatmala and Sabarimala aim to streamline operations and improve connectivity. Technological advancements, including IoT, AI, blockchain, and automation technologies such as drones and driverless vehicles, are revolutionizing logistics operations by enhancing efficiency and reducing costs.

Estimated Volume of Freight Carried in India

The freight transportation industry in India has grown significantly due to increasing industrial activity, urbanization, and e-commerce expansion.



Current Scenario (2025):

- The total freight volume in India is estimated to be **5,300 million tonnes** as of 2025.
- This includes transportation through multiple modes such as road, rail, waterways, and air.
- The growth is driven by increasing demand in manufacturing, construction, and the retail sector.

Future Projection (2032):

- The total freight volume is expected to reach **8,000 million tonnes** by 2032.
- This indicates a substantial increase of **60% over eight years**.
- Government initiatives like the **PM Gati Shakti National Master Plan**, investment in dedicated freight corridors, and improved infrastructure are expected to support this growth.



Key Factors Driving Freight Growth

- **Economic Expansion:** India's GDP growth directly contributes to higher freight demand.
- **Urbanization & Industrialization:** Increased production and consumption lead to greater freight movement.
- **E-commerce Boom:** The rise of online shopping has increased demand for fast and efficient logistics.
- **Government Policies:** Infrastructure investments, multimodal logistics parks, and policy reforms support logistics efficiency.

India's freight transportation sector is poised for significant growth, with an expected increase from **5,300 MT in 2025 to 8,000 MT by 2032**. This reflects a growing economy, improved logistics infrastructure, and a shift toward more efficient supply chain mechanisms. However, challenges such as congestion, environmental concerns, and the need for sustainable logistics solutions must be addressed. Investments in green logistics, technology-driven solutions, and infrastructure upgrades will be key in ensuring efficient and sustainable freight movement in the coming years.

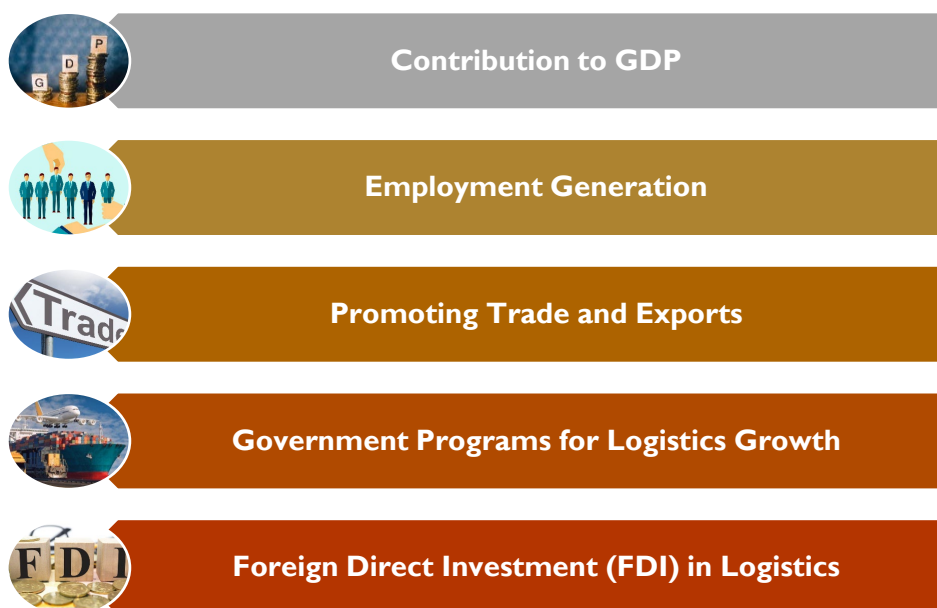
Factors Driving Freight Growth:

- **Rapid Industrialization and Manufacturing Expansion**
 - Expansion of manufacturing under initiatives like **Make in India**, PLI schemes, and industrial corridors is increasing the volume of goods transported.
 - Sectors like automotive, electronics, FMCG, chemicals, and pharmaceuticals are driving higher freight demand for both raw materials and finished products.
- **Growth of E-Commerce and Retail**
 - Rising online shopping, omni-channel retail, and hyperlocal deliveries are significantly increasing freight movement, especially in urban and semi-urban areas.
 - Demand for faster, frequent shipments is boosting LTL (Less-Than-Truckload) and last-mile freight services.
- **Infrastructure Development**
 - Investments in **highways, expressways, Dedicated Freight Corridors (DFCs), ports, airports, and multimodal logistics parks** are reducing transit times and improving reliability.
 - Better connectivity allows freight to move more efficiently across long distances, making rail, road, and waterways more viable for cargo.
- **Rising Organized Logistics and 3PL Adoption**
 - Businesses are increasingly outsourcing freight and supply chain operations to **organized 3PL and 4PL providers**, which improves efficiency and service levels.



- Professional logistics players offer route optimization, multimodal transport, and cargo tracking, further stimulating freight volumes.
- **Global Trade Expansion**
 - Rising imports and exports are increasing demand for freight services linking ports, airports, and inland hubs.
 - Integration into global supply chains is boosting containerized freight and multimodal transport demand.

Logistics Industry Contribution to Economic Growth



• **Contribution to GDP**

The logistics industry is a core support area of the Indian economy, generating ~**14-15%** of the GDP in 2025 released by the Ministry of Finance. Until a few years back, owing to the inefficiencies in warehousing, transportation, and supply chain management, the logistics costs remained disproportionately elevated at **13-14%** of GDP, against the international benchmark of **8-10%**. However, proving to be aware of this challenge, the Government of India brought out the National Logistics Policy (NLP) in 2022, which is steadily helping to bring down the cost of logistics in India, which currently measures 7.8-8.9% of GDP. This steady effort to reduce the logistic cost is anticipated to increase the competitiveness of Indian industries and add a further 1.5-2% to GDP growth every year.

Supporting this endeavour is the PM Gati Shakti National Master Plan, a revolutionary effort that unites infrastructure development among **16 critical ministries**, such as roads, railways, ports, and aviation. By aligning these projects, the plan seeks to do away with delays and minimize expenses, thus enhancing the overall efficacy of the logistics network.

For example, the Bharatmala Pariyojana, a flagship ministry of Road Transport and Highways highway development project, has already slashed freight travel time by **20%**, and the Dedicated Freight Corridors (DFCs) being constructed by Indian Railways are expected to save INR **1.3 lakh crore** in logistics costs annually once they become operational in full.

- **Employment Generation**

The logistics industry is one of India's biggest job creators, and it employs over **22 million** people, as documented by the Ministry of Commerce and Industry in 2025. This staff is spread throughout several sub-industries, where road transportation covers **70%** of employees, warehousing and cold storage take **15%**, and last-mile delivery solutions cover 10%. The growth of online platforms such as Flipkart and Amazon at a faster pace has increased demand for well-skilled logistics professionals even more in urban and semi-urban cities.

In order to meet the skills gap, the National Skill Development Corporation (NSDC) undertakes the Skill India Mission that has trained 5 lakh workers for logistics-specific functions up to 2023. The emerging fields like drone-based delivery, which is being encouraged under the Drone Policy 2022, and AI-powered warehouse management systems, which form a part of the overall Digital India program, are the focus areas for training programs. Through the provision of contemporary skills to the workforce, the government hopes to boost productivity and make sure that the logistics industry remains a key driver of inclusive economic growth.

- **Promoting Trade and Exports**

Adequate logistics are vital to boost India's trade competitiveness, as highlighted by recent progress toward reducing national logistics costs. Through strategic policy interventions such as the National Logistics Policy (2022), the PM Gati Shakti National Master Plan, and dedicated multimodal infrastructure schemes under the National Infrastructure Pipeline, the Government of India has projected that India's logistics cost is expected to drop to around 9% by December 2025, down from the earlier 14–16%. This improvement, driven by integrated planning, digital enablement, and enhanced multimodal connectivity, is set to make Indian manufacturing stronger, improve export competitiveness, and attract greater global investment, marking a significant step in India's economic transformation and easing a long-standing burden on the trade ecosystem. The Dedicated Freight Corridors (DFCs), the bedrock of India's logistics modernization, have decreased freight transit times considerably. For instance, the Delhi-Mumbai freight corridor has reduced the travel time from 14 to 8 hours and resulted in saving **INR 400 billion** annually, according to the Dedicated Freight Corridor Corporation of India (DFCCIL) in 2023. Further, the Sagarmala Programme, with its port-led development, has brought the cost of coastal shipping down by **40%**, improving the competitiveness of Indian exports in overseas markets. These gains are essential in helping the government realize its vision of USD 1 trillion in merchandise exports by 2030.



Akash Khandelwal

- **Government Programs for Logistics Growth**

The National Logistics Policy (NLP) 2022 is a comprehensive effort to make India's logistics sector more efficient. One of the main elements of this policy is the Unified Logistics Interface Platform (ULIP), which brings together 30+ digital platforms of railways, ports, and customs to track shipments in real-time. Another vital initiative is the creation of Multi-Modal Logistics Parks (MMLPs), which will cut freight costs by **10-12%**, as projected by NITI Aayog in its 2023 report.

The Sagarmala Programme, initiated by the Ministry of Ports, Shipping, and Waterways, is another revolutionary initiative to maximize port efficiency and lower logistics costs. With **600+** projects totalling INR 8.5 trillion, the program aims to increase port connectivity, upgrade infrastructure, and encourage coastal shipping. In the same vein, the Bharatmala Pariyojana, under the Ministry of Road Transport and Highways, is building 12,000 km of new highways to enhance freight mobility. These initiatives together seek to make India a global logistics hub.

- **Foreign Direct Investment (FDI) in Logistics**

Foreign Direct Investment (FDI) in the logistics sector has become a cornerstone for enhancing global trade efficiency, infrastructure development, and technological advancement. Between 2000 and 2024, total FDI inflows reached **USD 991 billion**, with a significant **67% (USD 667 billion)** of this investment occurring in the last decade (2014–2024), highlighting a sharp upward trend. Particularly notable is the manufacturing sector, which is closely linked to logistics and saw FDI equity inflows grow by 69% from **USD 98 billion** in 2004–2014 to **USD 165 billion** in 2014–2024. This surge reflects the increasing focus of global investors on integrated supply chains, driving demand for modern warehousing, cold storage, freight networks, and last-mile delivery services.

Emerging markets in Asia-Pacific, the Middle East, and Africa have attracted strong FDI due to rising consumption, infrastructure gaps, and government-led policy reforms. Countries like India, supported by initiatives such as “Make in India” and Gati Shakti, have become key destinations for foreign investment in logistics. At the same time, developed regions in North America and Europe are channelling FDI toward digital logistics infrastructure, sustainability, and automation. Despite challenges such as regulatory barriers and capital-intensive entry, the logistics sector remains a magnet for foreign capital, driven by its critical role in global economic growth and the need for resilient, tech-enabled supply chain systems.

Positioning of the Indian Logistics Industry in the Global Economy

India's logistics sector plays a critical role in the country's economic growth, global trade facilitation, and industrial competitiveness. The World Bank's Logistics Performance Index (LPI) 2023 report highlights significant advancements in India's logistics efficiency, with the country ranking 22nd globally in the International Shipments category and 38th in overall LPI score. This notable improvement is a testament to the government's continued emphasis on enhancing logistics infrastructure, digitization, and policy reforms.

The Logistics Performance Index (LPI) serves as a benchmark for evaluating the logistics capabilities of countries worldwide based on factors such as customs efficiency, infrastructure, international shipments, logistics quality, tracking and tracing, and timeliness. India's recent jump in rankings underscores its growing significance in global trade and supply chain networks.

One of the most noteworthy achievements is the reduction in port turnaround time, a critical indicator of logistics efficiency. As per the World Bank's LPI 2023 report, Indian ports recorded an average turnaround time of 0.9 days, outperforming several developed economies, including:

Sr no.	Country	Turnaround Time (Days)
1	India	0.9
2	Singapore	1
3	Malaysia	1
4	UAE	1.1
5	Indonesia	1.1
6	Ireland	1.2
7	Belgium	1.3
8	Germany	1.3
9	USA	1.5
10	Australia	1.7
11	Russia	1.8
12	Canada	2
13	South Africa	2.8

- **Maritime Amrit Kaal Vision 2047 – A Blueprint for Future Growth**

To further strengthen its logistics and maritime capabilities, India has developed the Maritime Amrit Kaal Vision 2047, aligning with the principles of the blue economy. This strategic roadmap outlines the long-term transformation of India's maritime sector, focusing on:



- Expansion of port capacity: Development of greenfield and brownfield ports to cater to rising trade volumes.
- Operational efficiency: Integration of automation, digitization, and advanced logistics technologies to improve supply chain management.
- Sustainability initiatives: Promotion of green logistics, establishment of hydrogen hubs, and enhancement of eco-friendly shipping practices.
- Boosting coastal tourism: Infrastructure development for island-based tourism and cruise shipping, strengthening India's position as a global maritime hub.
- Maritime capacity building: Investment in workforce training and skill development to create a future-ready logistics workforce.
- Shipbuilding and repair: Strengthening the domestic shipbuilding industry to reduce reliance on foreign manufacturing and boost India's shipping tonnage.
- Global engagement: Increasing participation in international maritime forums, reinforcing India's global maritime presence.

By executing this vision, India aims to position itself as a global logistics leader, ensuring seamless movement of goods, reduced transit costs, and enhanced trade efficiency.

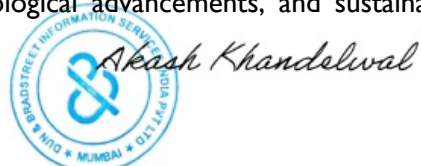
• Investment Commitments and Policy Reforms

The Global Maritime India Summit (GMIS) 2023 served as a major catalyst in attracting substantial investments in the logistics and maritime sectors. The summit resulted in **INR 10 trillion** worth of investment commitments, including:

- 360 Memorandums of Understanding (MoUs) signed, accounting for an investment of **INR 8.35 trillion**, including international collaborations.
- Additional investible projects worth **INR 1.68 trillion** were announced, further driving growth in logistics infrastructure.

These investments reflect India's commitment to modernizing its logistics ecosystem, ensuring seamless trade facilitation, and enhancing global supply chain integration.

India's ascent in the Logistics Performance Index (LPI) 2023 demonstrates its growing competitiveness in global logistics. The country's strong performance in port efficiency, logistics infrastructure development, and digital integration has positioned it as a critical player in international trade. With initiatives such as Maritime Amrit Kaal Vision 2047 and strategic investment inflows, India is well on track to become a global logistics powerhouse. Moving forward, continued policy reforms, technological advancements, and sustainability-



driven strategies will be essential in sustaining this momentum and elevating India's position in the global logistics landscape.

Air Transportation in Logistics

Overview of Air Transportation: Significance of air transportation in Logistics, Airport infrastructure Scenario in India

The Indian air transportation sector has emerged as one of the fastest-growing segments of the country's mobility and logistics ecosystem. Passenger traffic has consistently risen over the past few years, driven by rising disposable incomes, increased domestic and international travel, and the expansion of low-cost carriers (LCCs), which have enhanced affordability and accessibility. According to the **DGCA Annual Report 2024-25**, India's domestic airlines carried **431.98 lakh passengers** in FY 2024-25, marking a **10.35% year-on-year increase** over FY 2023-24. The Indian domestic market continues to expand, with regional connectivity improving under schemes like UDAN (Ude Desh ka Aam Naagrik), aimed at promoting air travel to tier-2 and tier-3 cities.

The air cargo segment has also witnessed growth due to the surge in e-commerce, express delivery requirements, and the movement of high-value and perishable goods. Dedicated cargo airports and logistics hubs in key cities such as Mumbai, Delhi, and Bengaluru have strengthened the sector's ability to handle large volumes. Monthly DGCA-reported data indicate that **February 2025 saw 140.44 lakh domestic passengers**, while **March 2025 recorded 145.42 lakh passengers**, with **IndiGo carrying approximately 9.31 million passengers** (64% market share) and the **Air India group transporting around 3.88 million (~26.7%)**. These figures reflect strong operational efficiency and high demand for both passenger and cargo services. Additionally, India's **passenger load factor in 2024 was 86.4%**, one of the highest globally, highlighting the sector's effective capacity utilization.

While the sector offers significant growth opportunities through airport modernization, greenfield projects, and digitalization of air traffic management and passenger services, it also faces challenges such as high fuel costs, regulatory compliance, competition from alternative transport modes, and environmental concerns. Despite these challenges, the Indian air transportation sector remains a crucial driver of economic growth, employment generation, and regional connectivity, while supporting trade, tourism, and emergency logistics. With sustained investment and policy support, the sector is projected to continue its robust growth trajectory in the coming years.

Air Cargo Segmentation by - Domestic / International

Domestic

Domestic air cargo refers to the movement of goods within a single country's borders, serving industries such as e-commerce, retail, healthcare, and perishables. Historically, this segment has shown steady growth,



propelled by rising consumer demand for faster deliveries, regional trade expansion, and infrastructure improvements. The growth of domestic air cargo has been closely linked to the rise of logistics providers, such as FedEx, UPS, and DHL, which have optimized air freight operations through hub-and-spoke models. Additionally, the increasing adoption of express parcel services and the integration of digital tracking technologies have enhanced efficiency, making air transport a preferred mode for time-sensitive shipments. The demand for domestic air freight surged with the boom in online retail and urbanization, compelling cargo airlines and logistics firms to expand their networks and invest in automation.

Domestic air cargo in India has shown **steady and resilient growth**, with a **CAGR of approximately 10%**, driven primarily by the rise of **e-commerce**, **increasing urbanization**, and **improvements in logistics efficiency**. The sector faces challenges such as **high operational costs** and **regulatory constraints**, but recent developments like **drone deliveries** and enhancements in **last-mile logistics** are helping to strengthen the domestic air freight ecosystem.

International

International air cargo involves the transportation of goods across countries and continents, serving global industries such as electronics, pharmaceuticals, automotive, and cross-border e-commerce. Historically, this segment has exhibited fluctuating growth, influenced by global trade agreements, economic downturns, and geopolitical developments. The emergence of free trade zones, advancements in freight aircraft capacity, and the liberalization of air cargo markets have significantly bolstered international air freight movement. Major global hubs such as Hong Kong, Frankfurt, and Dubai have played a pivotal role in facilitating international cargo transit. However, international air cargo is highly sensitive to external factors, including fuel price volatility, trade disputes, and regulatory restrictions. Despite these challenges, the segment has remained resilient, adapting to new consumer trends such as the rise of cross-border online shopping and the demand for just-in-time inventory replenishment.

International air cargo growth is more **fluctuating**, with a **CAGR of around 6%**, influenced by global **economic cycles and trade dynamics**. Key drivers include **global trade agreements**, **evolving supply chains**, and **international economic policies**. Challenges such as **geopolitical instability** and **fuel price volatility** can affect performance, while recent initiatives focusing on **e-commerce-driven international shipping** and **sustainability measures** are shaping the future of India's global air freight operations.

Significance of Air Transportation in Logistics

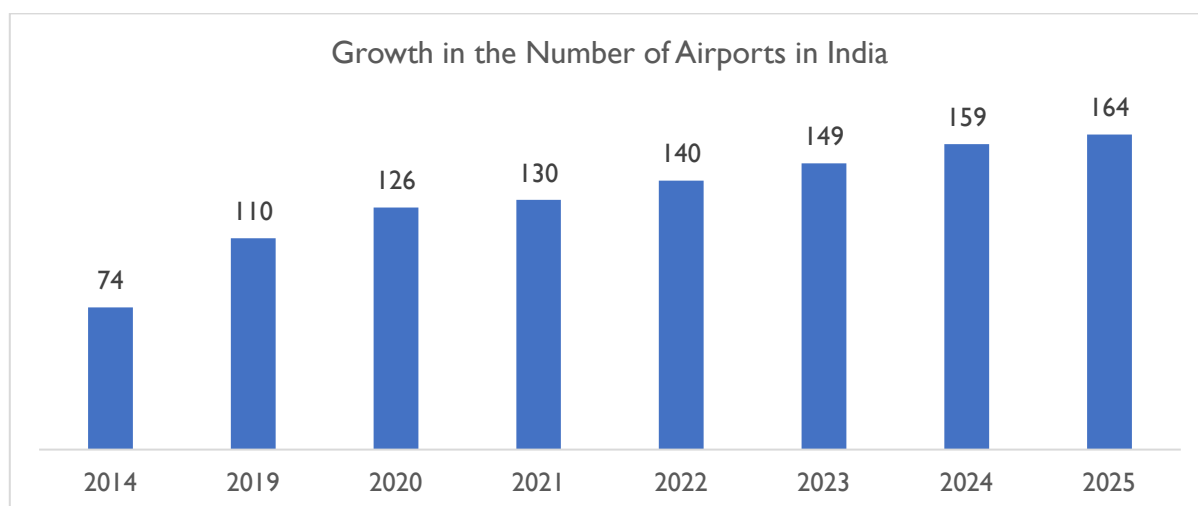
- **Speed and Efficiency:** Air transport is the fastest mode of freight movement, often delivering international shipments within 24-48 hours. This rapid transit is crucial for time-sensitive goods such as perishables, pharmaceuticals, and high-value items, ensuring they reach markets promptly and in optimal condition.



- **Global Reach:** Air freight enables businesses to access international markets swiftly, facilitating global trade and commerce. The ability to transport goods quickly across continents allows companies to respond effectively to market demands and maintain competitive advantages.
- **Supply Chain Optimization:** Utilizing air transport allows businesses to adopt just-in-time delivery models, reducing the need for extensive warehousing and minimizing inventory costs. This efficiency enhances overall supply chain management, allowing for better responsiveness to consumer needs and market fluctuations.
- **Strategic and Emergency Use:** Air transport is crucial for **emergency relief, disaster management, and defence logistics**. It allows rapid deployment of personnel, equipment, and aid during natural disasters or crises.
- **Integration with Multimodal Networks:** Modern airports are increasingly linked with logistics parks, rail, and road networks, enabling smooth multimodal cargo movement. This reduces transit times and increases overall supply chain efficiency.
- **Boosts International Trade:** Air cargo supports India's trade by connecting the country to **global markets**. Sectors like pharmaceuticals, electronics, and perishables rely heavily on-air transport for timely exports and imports.

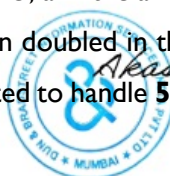
Airports Infrastructure Scenario in India: Current Landscape

As of 2025, India boasts 164 operational airports, a significant increase from 74 in 2014. This expansion reflects the country's commitment to enhancing its aviation infrastructure to support growing passenger and cargo demands.



Source: Airports Authority of India (AAI), as on 30.11, 2025

India's aviation industry has experienced significant growth in the past 10 years. The number of operational airports in the country has doubled from **74 in 2014 to 164 in 2025**, and the aim is to increase this number to **350-400 by 2047**. The domestic air passengers have more than doubled in the past decade, with Indian airlines significantly expanding their fleets. Indian air cargo is projected to handle **5.8 million tonnes by 2029**.



The Indian international air cargo market outperformed both the global and Asia Pacific average and is on track for around **19%** growth in 2024.

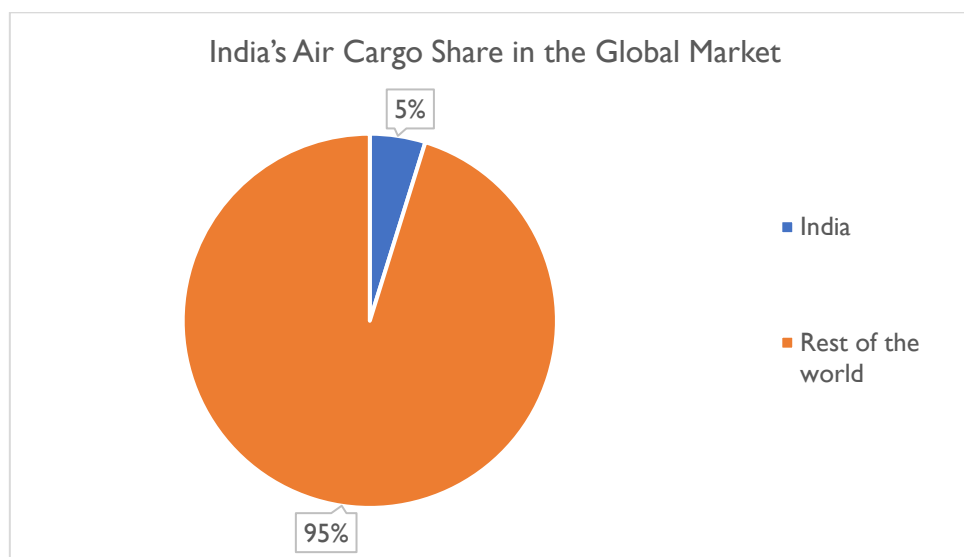
Over the next five years, Indian air cargo traffic is expected to grow from **3.7 million** tonnes today to between **5 and 5.8 million** tonnes by 2029 or about 6-9% per year. The forecast range for Indian air cargo volumes in the next five years is between **6.2% and 9.1%**. This is higher than recent performance, but in line with longer-term performance through 2019, according to the India Air Cargo Outlook 2025-2029 by Trade Data Service by the Trade and Transport group published in January 2025.

Total Cargo carried by Air in terms of volume

Air cargo serves as a critical pillar of global trade, ensuring the efficient and timely movement of goods across international markets. Over the years, the total volume of cargo transported by air has demonstrated a steady upward trajectory, driven by various factors such as economic expansion, technological advancements, and shifting trade patterns.

Global air cargo volumes are projected to reach **72.5 million tonnes** in 2025, reflecting a **5.8% year-on-year growth**. This rise is largely driven by the expansion of e-commerce and strategic adjustments in supply chains, particularly in response to disruptions in traditional maritime routes. Businesses are increasingly turning to air freight to ensure **faster and more reliable delivery**, offsetting delays in ocean shipping.

India's aviation sector is emerging as a significant player in global air cargo logistics. In FY 2024-25, Indian airports handled approximately **3.7 million tonnes of air cargo**, with **2.32 million tonnes** transported internationally and **1.39 million tonnes** domestically. This positions India's share at roughly **5% of the global air cargo market**, highlighting the country's growing importance in import and export operations.



The sector's recovery post-pandemic is evident in both passenger and freight traffic. Improved airport infrastructure, the rise of regional connectivity, and the expansion of air cargo facilities have strengthened India's capacity to handle increasing volumes. As a result, air transport is becoming a key component of the



country's logistics and trade ecosystem, providing faster, more efficient solutions for both domestic and international shipments.

Key Drivers of Air Cargo Growth

- **E-commerce Expansion:** With the rapid rise of online shopping, the demand for express delivery services has significantly increased, contributing to higher air cargo volumes.
- **Supply Chain Disruptions:** Challenges such as the Red Sea crisis have led businesses to pivot towards air freight for urgent and time-sensitive shipments.
- **Post-Pandemic Recovery:** The strong rebound in passenger air traffic has also positively impacted cargo movement, as more flights translate into enhanced freight capacity.
- **Infrastructure Development:** Improvements in cargo-handling facilities and airport expansions have further facilitated efficient air freight movement.
- **Rising Global Trade:** India's growing integration with **international markets** increases import and export volumes, particularly for pharmaceuticals, electronics, and perishables, which often rely on air cargo for speed and safety.
- **Cold Chain and Specialized Cargo:** Rising demand for **temperature-sensitive goods** such as pharmaceuticals, frozen food, and perishable products is driving the development of specialized air cargo solutions with cold-chain facilities.

As global trade continues to evolve, the air cargo industry is expected to witness sustained growth. The projected 5.8% annual increase in cargo volumes indicates a promising future, with air freight becoming an increasingly vital component of global logistics. In India, the government's ongoing investments in aviation infrastructure and logistics hubs will further strengthen the country's role in the international air cargo ecosystem. Given the current trajectory, air transport will remain a crucial enabler of economic growth, ensuring seamless global trade and connectivity.

Services Offered by Air Cargo in India

India's air cargo industry offers a range of services tailored to meet varying delivery timeframes and budget considerations.

- **Priority Services:** Priority cargo services provide **fast-track processing**, from acceptance to loading, ensuring minimal dwell time. This is crucial for temperature-sensitive goods, machinery parts, and high-value shipments that require immediate uplift.
- **Same-Day and Next-Day Delivery:** Major airports and cargo operators offer **same-day and next-day delivery** for intercity shipments, enabling faster turnaround for businesses. This service is widely used for medical supplies, perishables, and urgent business shipments that require guaranteed delivery windows. For instance, Delhivery launched a 'guaranteed' same-day delivery service across 15 key cities in India, enabling Direct-to-Consumer (D2C) brands to fulfil orders *on the day they are received.*



- **Express Services:** Air cargo in India provides **express delivery solutions** designed for time-critical shipments, ensuring rapid transport of parcels, documents, and small consignments. These services cater heavily to e-commerce platforms, pharmaceuticals, and high-value goods that demand speed and reliability.
- **Temperature-Controlled and Perishable Delivery:** Specialized cold-chain air cargo services support the delivery of **perishables, pharmaceuticals, vaccines, and food products**. With temperature-controlled storage and transport, the system ensures product integrity from origin to destination.
- **Freighter-Based Bulk Delivery:** Dedicated freighter aircraft support **bulk cargo delivery**, transporting heavy machinery, automotive components, electronics, and industrial consignments across domestic and international routes. These services enable faster replenishment cycles for industries.

Facilities/Advantages of Air Transport

Air transport plays a crucial role in modern logistics and global trade. Its unique characteristics make it an indispensable mode of transportation, offering unparalleled efficiency and reliability. Below are the key advantages and facilities that define the significance of air transport.

Advantages of Air Transport

Fastest Mode of Transportation

- One of the most significant benefits of air transport is its exceptional speed, making it the fastest mode of transportation.
- Delivers goods within hours, ideal for urgent and time-sensitive cargo.
- Frequent flight schedules further enhance delivery efficiency, ensuring the timely shipment of goods.
- Greatly reduces transit time for critical shipments like medical supplies and electronics.

Reliable and High Service Quality

- Strict schedules with predictable arrival and departure times.
- Advanced tracking systems ensure minimal delays and high reliability.
- Helps reduce inventory holding and buffer stock needs.

Global Reach and Extensive Connectivity

- Connects India quickly to major global trade hubs.
- Enables seamless international export–import operations.
- Supports rapid movement across continents.

Suitable for High-Value & Sensitive Cargo



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- Enhanced security protocols ensure the safety of fragile and expensive goods.
- Ideal for electronics, diamonds, pharmaceuticals, and luxury items.
- Controlled handling reduces damage risks.

Minimal Physical Constraints

- Unlike road or rail transport, air transport is not restricted by geographical barriers or infrastructure limitations.
- This enables uninterrupted services, making it a highly reliable option for businesses that require seamless logistics.

High Reliability

- Air transport is known for its punctuality and adherence to strict schedules, reducing the risk of delays.
- The chances of cargo misplacement or loss are minimal due to stringent security and tracking measures.

Long-Distance Connectivity

- No other mode of transport covers vast distances as efficiently as air transport.
- It is the most viable solution for international trade, bridging continents and enabling the swift movement of goods and passengers.

Access to Remote & Difficult Terrains

- Provides essential connectivity to hilly, island, and landlocked regions.
- Crucial for supply chains in Northeast India, Ladakh, disaster-hit areas, and remote towns.
- Ensures movement where road/rail access is limited or slow.

Facilities of Air Transport

Cargo Handling & Storage Facilities

- **Dedicated Air Cargo Terminals** – Major international airports are equipped with specialized cargo terminals for streamlined processing and efficient handling.
- **Temperature-Controlled Warehousing** – Storage facilities for perishable goods such as fresh produce, pharmaceuticals, and vaccines ensure product integrity.
- **Advanced Security Measures** – X-ray scanners, screening procedures, and tracking systems enhance cargo safety and prevent unauthorized access.

Passenger & Aircraft Maintenance Facilities



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- **Modern Airport Infrastructure** – Airports feature advanced runways, air traffic control (ATC) systems, and well-equipped maintenance hangars.
- **Passenger Amenities** – Comfortable waiting lounges, baggage handling services, and duty-free shopping enhance the travel experience.
- **Aircraft Servicing & Refuelling** – Airports provide on-site maintenance, fuelling stations, and technical support to ensure uninterrupted operations.

Multimodal Transport Connectivity

- **Integrated Transport Networks** – Efficient coordination between air, rail, and road networks allows seamless cargo and passenger movement.
- **Logistics & Express Cargo Hubs** – Leading logistics providers such as SpiceXpress, Blue Dart, and FedEx operate specialized hubs to enhance air cargo efficiency.

Digital & Technological Facilities

- **Real-Time Cargo Tracking Systems** – Advanced digital platforms enable end-to-end visibility of shipments, improving transparency and coordination.
- **E-Air Waybill (e-AWB) Adoption** – Digital documentation reduces processing time, enhances accuracy, and supports faster cargo clearance.
- **Automated Handling Equipment** – Use of conveyor belts, cargo loaders, and robotic systems ensures quick and safe cargo movement.

Operational & Support Facilities

- **24/7 Operational Capability** – Airports function round-the-clock, enabling continuous cargo and passenger movement.
- **Efficient Customs & Clearance Support** – Dedicated customs desks and fast-track clearance reduce processing delays for imports and exports.
- **Freighter Parking Bays** – Designated areas for cargo aircraft to allow quick turnaround and loading operations.

Business & Commercial Facilities

- **Cargo Village & Warehousing Zones** – Integrated logistics clusters offering warehousing, distribution, and value-added services.
- **Free Trade Warehousing Zones (FTWZs)** – Allow duty-free storage, repackaging, and consolidation for exporters and importers.
- **Airport Hotels & Business Lounges** – Support business travellers and freight professionals with meeting areas and stay options.



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Air transport remains a cornerstone of global logistics, offering unmatched speed, reliability, and connectivity. Its well-established infrastructure and integrated services continue to drive economic growth and facilitate international trade.

Regulatory Landscape specific to Aviation

India's aviation industry is governed by a comprehensive regulatory system aimed at providing safety, security, and efficiency in diverse areas such as commercial air transport, cargo aviation, and general aviation.

Bodies that regulate this sector

- **Ministry of Civil Aviation (MoCA)**

The Ministry of Civil Aviation (MoCA) is the main government agency that makes policies and regulates civil aviation in India. It manages the growth of aviation infrastructure, bilateral air service agreements, and airline operations. One of its major initiatives is the National Civil Aviation Policy (NCAP) 2016, which looks to promote regional connectivity under the UDAN (Ude Desh Ka Aam Nagrik) scheme by offering low fares for flying in smaller cities.

The policy also allows **100%** Foreign Direct Investment (FDI) in scheduled airlines with up to **49%** possible under the automatic route and over that through government approval. Apart from this, MoCA has also launched the DigiYatra program, which is a digital travel system based on biometrics aimed at streamlining the passenger processing in airports by eliminating the need for manual checks and bolstering security. The ministry also controls fares on some routes under the Route Dispersal Guidelines (RDG) for connecting remote and unserved areas.

- **Directorate General of Civil Aviation (DGCA)**

The Directorate General of Civil Aviation (DGCA) is India's central aviation safety regulatory, licensing, and airworthiness authority. It functions under the Aircraft Act, 1934, and the Aircraft Rules, 1937, enforcing Civil Aviation Requirements (CARs) that cover every aspect of civil aviation. The DGCA issues pilot licenses, such as Commercial Pilot Licenses (CPL) and Airline Transport Pilot Licenses (ATPL), and oversees medical standards compliance. It also issues Air Operator Certificates (AOC) to airlines, insisting on strict compliance with safety regulations, maintenance practices, and operating procedures.

The DGCA manages Flight Duty Time Limitations (FDTL) to avoid pilot fatigue and improve flight safety. It also monitors consumer protection practices, insisting that airlines pay compensation to passengers in case of flight cancellations, delays, or denied boarding according to specified guidelines. The DGCA also certifies aircraft types and ensures that the aircraft flying within India are internationally compliant with standards of safety approved by bodies such as the FAA and EASA.

- **Bureau of Civil Aviation Security (BCAS)**



The central organization for ensuring aviation security in India is the Bureau of Civil Aviation Security (BCAS). It functions under the Aviation Security Programme (ASP) and implements regulations according to ICAO Annex 17 (Security Standards). The BCAS develops the National Civil Aviation Security Programme (NCASP), which provides security procedures for airports, airlines, and passengers. Major duties are the conduct of Pre-Employment Background Checks (PEBC) for airport employees to secure sensitive areas against unauthorized entry. The BCAS requires rigorous passenger and baggage checks at airports to identify banned items and possible dangers. It also carries out anti-hijacking procedures under the Anti-Hijacking Act, 2016, such as sending armed sky marshals on specific flights. The BCAS also carries out routine security audits and surprise checks in airports to monitor compliance with security standards and thwart possible violations.

- **Airports Authority of India (AAI)**

The Airports Authority of India (AAI) is charged with the management and development of civil aviation infrastructure in the country, such as air traffic control (ATC), navigation facilities, and airport management. Set up under the Airports Authority of India Act, 1994, the AAI is in charge of over 125 airports, ranging from international, domestic, to regional airports. Its key activities involve the provision of air traffic management (ATM) services, the safe movement of aircraft during take-off, landing, and in-route flight, respectively. The AAI also executes airport modernization schemes, terminal expansion, runway extension, and cargo terminals to meet increased passenger and freight traffic. In accordance with the Regional Connectivity Scheme (RCS-UDAN), the AAI develops and upgrades smaller airports to improve connectivity in remote locations. In addition, the AAI controls landing and parking fees at non-major airports, whereas the Airports Economic Regulatory Authority (AERA) controls tariffs at major airports such as Delhi and Mumbai.

- **Airports Economic Regulatory Authority (AERA)**

Airports Economic Regulatory Authority (AERA) is a statutory body set up under the AERA Act, 2008, to regulate major airport tariffs and charges in India. Its main function is to maintain fair prices for aeronautical services such as landing fees, parking charges, and passenger service charges to keep airport operators from monopolizing activities. AERA fixes tariffs for airports with more than 1.5 million annual passenger traffic, including Delhi, Mumbai, Bangalore, and Hyderabad. The authority reviews airport revenue and expenditure from time to time to fix reasonable charges that reconcile the interests of airlines, passengers, and airport operators. Regulating economic aspects of airport operations, AERA encourages competition, efficiency, and investment in India's aviation infrastructure.



Government policies for this sector

- **Cargo Aviation Regulations**

Open Sky Policy for Air Cargo: This policy, initiated in 1990, allows Indian as well as foreign airlines satisfying prescribed operation and safety norms to operate scheduled and non-scheduled cargo services to/from any airport having customs/immigration facilities. The policy was revised in September 2020 to limit foreign ad hoc and non-scheduled freighter charter services to six main airports: Bengaluru, Chennai, Delhi, Kolkata, Hyderabad, and Mumbai. The policy modification is to increase the demand for domestic freight charter services.

- **National Civil Aviation Policy (NCAP 2016)**

The NCAP 2016 provides a foundational framework for the growth of air cargo in India, both on domestic and international routes. It emphasizes the promotion of cargo facilities by co-locating them at airports, giving them infrastructure status, and streamlining customs procedures. The policy mandates digital processing of cargo through systems like the Advance Cargo Information (ACI) and envisions a **single-window clearance system** at cargo terminals that integrates multiple regulatory bodies (Customs, FSSAI, Drug Controller, etc.). Additionally, NCAP seeks to develop cargo villages near airports and supports Free-Trade Warehousing Zones (FTWZ) to facilitate air transshipment.

- **Air Cargo Logistics Promotion Board (ACLPB)**

Under NCAP, the Air Cargo Logistics Promotion Board (ACLPB) is tasked with developing a detailed action plan to enhance cargo throughput and reduce dwell time. Swaniti Initiative+ I The board funds initiatives to promote transshipment at Indian airports and supports the adoption of industry best practices like FTWZs. BYJU'S It also works to streamline governance, recommending ideas like freighter-priority space norms and favourable user charges at airports.

- **Drone Regulation for Cargo Delivery (Drone Rules 2021)**

The **Drone Rules, 2021**, notified by the Ministry of Civil Aviation, create a regulatory framework that encourages the use of drones for cargo delivery. The rules expand the permissible weight for drones to **500 kg**, enabling the use of cargo drones and even “drone taxis.” Approvals have been simplified: several permissions have been removed, and the **Digital Sky Platform** offers a self-generated permission system with minimal human intervention. The policy also proposes the development of **dedicated “drone corridors” for cargo deliveries** and sets up a Drone Promotion Council to foster a business-friendly drone ecosystem.

- **New Aircraft Act — Bharatiya Vayuyan Adhiniyam, 2024**



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Coming into force on **1 January 2025**, the Bharatiya Vayuyan Adhiniyam, 2024, replaces the decades-old Aircraft Act of 1934. This new law modernizes India's aviation regulatory framework by explicitly regulating aircraft design, manufacturing, and maintenance, areas not clearly covered earlier. It also retains key regulatory bodies (DGCA, BCAS, and AAIB) while strengthening their powers, streamlining licensing, and creating a more robust appeals mechanism.

- **Customs Reforms for Air Cargo and Transshipment**

In April 2025, the Central Board of Indirect Taxes & Customs (CBIC) simplified import procedures for **Unit Load Devices (ULDs)**, allowing temporary import via a continuity bond. In the same move, India **waived transshipment permit fees**, reducing compliance costs for airlines and cargo agents. Unique ID tracking for ULDs and tighter BCAS (security) compliance were also introduced to improve cargo visibility and security.

- **National Civil Aviation Policy (NCAP) 2016**

The **NCAP 2016** was introduced to promote the aviation sector, including air cargo. It includes several measures to improve cargo operations:

- **Air Cargo Logistics Promotion Board (ACLPB):** Established to facilitate faster movement of cargo by reducing clearance times and enhancing coordination between various regulatory bodies.
- **Air Cargo Community System (ACCS):** A digital platform to streamline documentation and reduce paperwork delays, ensuring quicker cargo processing.
- **Service Delivery Modules (SDMs):** Implemented to monitor cargo handling performance and minimize inefficiencies.

These reforms have contributed to **reducing air cargo dwell times and increasing efficiency at airports**, making air freight a more attractive option for businesses.

- **Make in India Initiative**

The Make in India program was launched to boost domestic manufacturing, which has directly impacted air cargo demand.

- **Encouragement of exports:** By expanding industrial production, the program has led to increased shipments of electronics, pharmaceuticals, and textiles, sectors that heavily rely on air freight.
- **Development of the Maintenance, Repair, and Overhaul (MRO) sector:** This has reduced India's reliance on foreign MRO facilities for aircraft maintenance, increasing domestic cargo movement.



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- Promotion of indigenous manufacturing in high-tech industries, which has driven air freight shipments of precision components and machinery to global markets.
- **Infrastructure Development for Air Cargo**

The Indian government is investing **approximately USD 1.83 billion in airport modernization and cargo infrastructure expansion** by 2026. Some key developments include:

- Upgrading **150** airports to improve cargo handling efficiency, including the establishment of cold storage facilities for perishable goods.
- Expansion of dedicated air freight corridors, ensuring faster movement of high-value goods.
- Development of satellite freight cities near major airports to improve connectivity and reduce congestion.

- **National Logistics Policy (NLP)**

The **National Logistics Policy**, launched to enhance supply chain efficiency, has played a crucial role in strengthening India's air cargo sector:

- Reducing logistics costs by integrating different modes of transport (air, rail, road, and waterways).
- Development of multimodal logistics parks to facilitate seamless cargo movement.
- Simplification of regulatory procedures and digitalization of logistics operations to boost efficiency.

- **Open Sky Policy**

The **Open Sky Policy** has **liberalized cargo movement**, allowing **foreign airlines to operate unrestricted freighter services**.

- This has increased competition, leading to better pricing and improved services in the air cargo sector.
- The policy has also helped Indian exporters access more international markets through expanded air connectivity.

- **Regional Connectivity Scheme (RCS) – UDAN Cargo**

The **UDAN (Ude Desh Ka Aam Nagrik) Cargo Scheme** aims to **enhance air connectivity for smaller cities and improve last-mile logistics**.



- Establishment of air cargo terminals at smaller airports, reducing dependence on metropolitan hubs.
- Encouraging e-commerce logistics through air transport, benefiting businesses in remote locations.
- Boosting agricultural exports by providing better access to international markets for perishable goods.

- **Stakeholder Collaboration and Industry Consultations**

The government is actively **engaging with logistics companies, airlines, and exporters** to simplify air cargo operations.

- The goal is to handle **10 million** tonnes of air cargo annually by 2030.
 - Policies are being refined based on feedback from industry leaders, ensuring practical implementation of reforms.
- **Moratorium on Repayments** – Regulated entities (REs) may offer a moratorium on term-loan instalments and defer interest on working-capital (WC) facilities falling due between 1 September and 31 December 2025. Interest will accrue on a simple-interest basis, without compounding, and may be converted into a Funded Interest Term Loan (FITL) repayable between 31 March and 30 September 2026.
 - **Extension of Export Credit Tenure** – The permissible credit period for pre-shipment and post-shipment export credit has been extended to 450 days for credit disbursed up to March 31, 2026. For packing credit facilities already availed by exporters before August 31, 2025, where dispatch could not take place, REs may allow liquidation from any legitimate alternate source, including domestic sale proceeds or substitution of export contracts.
 - **Flexibility in Working-Capital Management** – To maintain liquidity during the effective period, REs may recalculate drawing power by reducing margins or reassessing working-capital limits.
 - **Regulatory Forbearance on Asset Classification** – The moratorium/deferment period will be excluded from days-past-due (DPD) calculations under IRACP norms; granting relief will not be treated as restructuring; credit bureaus (CICs) are directed to ensure borrowers' credit histories are unaffected.
 - **FEMA Relaxations for Export Realisation** – Under the Foreign Exchange Management (Export of Goods and Services) (Second Amendment) Regulations, 2025, the period for realisation and repatriation of export proceeds has been extended from nine months to 15 months, and the shipment period against advance payments has been increased from one year to three years.
 - **Niryat Protsahan (Financial Enablers):** As part of the Government of India's ongoing efforts to boost export competitiveness and facilitate trade, the Niryat Protsahan (Financial Enablers) scheme has

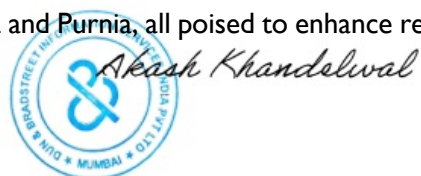


been implemented to provide targeted financial support to exporters. The scheme offers interest subvention on pre- and post-shipment export credit, along with enhanced access to working capital, particularly benefiting MSMEs and small-scale exporters. Through this initiative, exporters can avail of concessional credit facilities, reduced cost of funds, and easier compliance mechanisms, enabling them to improve liquidity, expand market reach, and compete more effectively in the global market. The scheme aligns with India's broader export promotion strategy and supports the achievement of long-term trade growth objectives.

- **Niryat Disha (Non-Financial Enablers):** As part of the Government of India's broader strategy to enhance export readiness and deepen global market integration, the Niryat Disha (Non-Financial Enablers) scheme has been instituted to provide holistic non-financial assistance to exporters. The initiative focuses on strengthening quality compliance, market access, logistics, and capacity-building support—especially for MSMEs and exporters in remote regions. Under this scheme, exporters receive facilitation for testing and certification, international branding and packaging guidance, subsidized participation in trade fairs and buyer-seller meets, reimbursements for inland transport from distant districts, and enhanced warehousing and logistics solutions. Additionally, the program drives skill development through training at the cluster and district levels, and strengthens export facilitation cells to build a resilient and competitive export ecosystem. By addressing critical infrastructural and procedural barriers, Niryat Disha aligns with India's Foreign Trade Policy 2023 and supports sustainable, inclusive export growth and greater participation in global value chains.

Government Initiatives and Investments

- The Indian government plans to invest approximately **INR 920 billion** (USD 11 billion) to increase the number of operational airports to 200 by 2025. This investment aims to bolster connectivity, stimulate economic growth, and accommodate the rising air traffic in the country.
- In 2025, India's aviation sector will reach new milestones with the development of **secondary airports** like **Navi Mumbai International Airport** and **Noida International Airport**. These airports aim to **reduce congestion at primary hubs**, expand capacity, improve connectivity, and cater to rising passenger demand.
- In 2025, Union Civil Aviation Minister **Kinjarapu Ram Mohan Naidu** highlighted plans to develop the **Northeast region** as a strategic logistics and aviation hub. With **16 operational airports** and growing international traffic from Guwahati, this initiative aims to **connect India more effectively with Southeast Asia**.
- In September 2025, **Air India SATS Airport Services (AISATS)** became the first Indian ground handling company to secure the **DGCA Safety Clearance under CAR**. This aligns operations with **ICAO standards**, making India the **second country globally** to formally regulate ground handling services.
- **Adani Airports** plans to invest **USD 7 billion** to expand its operations in India, focusing on enhancing its **existing airport portfolio** and **landside infrastructure**. This investment comes as India faces a significant **aviation infrastructure gap**, with only **149 commercial airports** serving a population of **1.4 billion**.
- In July 2025, the Digi Yatra app hit a milestone of **15 million** downloads, facilitating over 61 million contactless passenger journeys across **24 airports** since its December 2022 launch, **with plans to expand to four more airports, support international travel pilots** and multiple Indian languages, and reach 80 percent of domestic flyers by 2028.
- On February 4, 2025, the Central Government granted in principle approval for aviation expansion in Bihar, sanctioning greenfield airports at Rajgir, Bhagalpur and Sonapur, redevelopment of the brownfield airport at Raxaul, and new civil enclaves at Bihta and Purnia, all poised to enhance regional air connectivity under the UDAN framework.
- UDAN scheme fosters regional connectivity with **519** operational routes, encompassing **53** tourism and 48 helicopter routes, benefiting **133.86** lakh passengers. Over **2.56 lakh** flights have operated, involving **13** airlines, including startups, shaping a transformative impact on regional growth and air connectivity.
- In February 2025, the Central Government granted in principle approval for aviation expansion in Bihar, sanctioning greenfield airports at Rajgir, Bhagalpur and Sonapur, redevelopment of the brownfield airport at Raxaul, and new civil enclaves at Bihta and Purnia, all poised to enhance regional air connectivity under the UDAN framework.



Drivers of Air Cargo Traffic in India



- **Rapid Expansion of E-commerce and Online Retail in India:** The rapid rise of India's e-commerce sector already valued at nearly **USD 125 billion in 2024** and projected to reach **USD 345 billion by 2030** and further to around **USD 550 billion by 2035** is fundamentally reshaping the logistics landscape and emerging as one of the strongest demand drivers for air cargo. With online retail penetrating deeper into Tier-II and Tier-III cities, order volumes are not only increasing but also becoming more geographically dispersed. This creates a pressing need for faster, more reliable, and more time-definite transportation channels. Air freight therefore becomes essential for bridging long intercity distances, ensuring quick replenishment for sellers, and managing high consumer expectations around same-day and next-day delivery. Categories such as electronics, fashion, personal care, and premium consumer goods where speed and handling quality matter further amplify reliance on air cargo networks.

In addition, India's e-commerce ecosystem is witnessing unprecedented investment in fulfilment networks, automated warehouses, and integrated logistics systems by major players like Amazon, Flipkart, Meesho, Nykaa, and logistics specialists such as Delhivery and Blue Dart. These companies increasingly use air cargo to maintain lean inventories, shorten delivery cycles, and serve high-order-density regions efficiently. Cross-border e-commerce imported electronics, accessories, high-end fashion, and niche consumer items is also rising sharply, pushing international air freight volumes upward. Festivals, flash sales, and seasonal demand spikes further intensify air cargo utilisation across the year. Together, these developments reinforce air freight as a strategic pillar of India's digital commerce economy, making the rapid expansion of e-commerce one of the most transformative and sustained growth drivers of the country's air cargo traffic industry.



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- **Rising Pharmaceutical Export–Import Volumes Boosting Air Freight Demand:** India's pharmaceutical sector one of the strongest globally plays a pivotal role in driving air cargo traffic due to the sensitive, high-value, and time-critical nature of pharma shipments. India currently ranks among the top three pharmaceutical producers in the world by volume and is recognized as the largest global supplier of generic medicines. With pharmaceutical exports touching **USD 30.47 billion in FY 20243–25, growing 9.2% year-on-year**, the industry has become a cornerstone of India's external trade. A substantial share of these exports such as vaccines, biologics, and active pharmaceutical ingredients (APIs) must move through highly controlled environments, requiring strict temperature integrity and minimal transit time. Air cargo is therefore indispensable, as it provides the speed, precision, and regulatory compliance necessary for maintaining product efficacy and meeting international health and safety standards.

Simultaneously, India's imports of specialized APIs, high-value biologics, medical devices, and clinical trial materials are increasing rapidly, further boosting inbound air cargo volumes. The rise of contract manufacturing, global supply chain integration, and India's growing role as a hub for vaccine and biosimilar production has intensified demand for certified cool-chain air freight corridors. Airports such as Mumbai, Hyderabad, Bengaluru, and Delhi have invested heavily in GDP-compliant cargo terminals, thermal insulation, automation, and temperature-controlled ULDs to serve pharma exporters and multinational drug companies. This expanding infrastructure, combined with the scale and sophistication of India's pharmaceutical industry, makes pharma trade one of the most consistent and strategic demand drivers for India's air cargo industry ensuring high load factors, premium yields, and year-round cargo movement.

- **Expanding Cargo Volumes Driving Infrastructure and Network Growth:** India's air cargo industry is witnessing a clear and sustained rise in freight movement, signalling strong structural momentum rather than short-term volatility. Recent data shows that international cargo volumes continue to climb steadily, **with January 2025 handling around 184,273.8 tonnes, a 7.1% increase over the previous year**. Over the longer ten-month period from April 2024 to January 2025, international air freight reached **1.93 million tonnes**, marking a significant 17.2% year-on-year surge. Domestic air cargo volumes are rising in parallel, **with 113,662.9 tonnes handled in January 2025 alone, reflecting healthy 6.9% annual growth**. This broad-based increase across both domestic and international segments highlights a rising baseline demand fuelled by export–import trade, e-commerce, pharmaceuticals, perishables, and other time-sensitive cargo categories.

This consistent volume growth acts as a major driver for the sector by creating a powerful reinforcement cycle of investment, efficiency, and expanding capacity. Higher cargo movement encourages airports, ground handlers, and logistics operators to upgrade their physical and technological infrastructure building new cargo terminals, expanding cold-chain capabilities, deploying



automated handling systems, and improving customs processes. These enhancements make air freight more efficient and reliable, attracting even more shippers dealing in high-value or time-critical goods. The steady rise in volumes also enables airlines and freight forwarders to stabilise yields, optimise networks, and increase cargo capacity on key routes. Collectively, these developments strengthen India's position as a growing hub for domestic and international air logistics, making rising cargo volumes one of the most important drivers of long-term growth in the country's air cargo traffic industry.

- **Increasing Demand for Perishable and Temperature-Sensitive Goods:** The demand for perishable and temperature-sensitive goods is emerging as a major driver for India's air cargo industry. With the growing export of fruits, vegetables, seafood, dairy products, and frozen foods, there is an increasing need for transport solutions that ensure freshness, quality, and safety. Air cargo offers the speed and reliability required to move such products across long domestic and international distances, minimizing spoilage and maintaining strict temperature requirements. The expansion of India's cold-chain infrastructure, including temperature-controlled storage facilities and refrigerated handling at airports, has further strengthened the industry's capacity to handle perishable shipments efficiently.

Domestic trends are also fuelling this growth, with the rise of fresh e-grocery deliveries and online food marketplaces requiring fast and dependable logistics. Consumers increasingly demand high-quality fresh produce delivered directly to their homes, often within 24 hours, which creates additional reliance on air freight. At the same time, international demand for Indian agricultural exports such as mangoes, pomegranates, seafood, and dairy products has been steadily rising, requiring rapid and secure transport solutions to reach global markets in optimal condition. Together, these factors make perishable and temperature-sensitive goods one of the fastest-growing segments for air cargo, driving continuous investment in specialized aircraft, cold-chain technology, and logistics processes.

- **Trade Route Disruptions and Digital Innovation:** The Indian air cargo market is undergoing a period of significant transformation, driven by both global disruptions and evolving trade dynamics. The ongoing Red Sea crisis has disrupted traditional maritime shipping routes, prompting international traders and freight forwarders to divert substantial volumes of cargo from sea to air to ensure continuity in critical supply chains. This shift has created new business opportunities for the Indian air cargo sector, particularly in express and time-sensitive shipments that cannot tolerate delays. At the same time, regional trade agreements such as the Regional Comprehensive Economic Partnership (RCEP), India-ASEAN Free Trade Agreement, and bilateral pacts with countries like Australia and the UAE are reshaping cargo flows, establishing preferential trade corridors that favor air transport of high-value, low-weight goods. These developments are positioning India as a strategic



transshipment hub for intra-Asian and global trade networks, further strengthening the role of air cargo in the country's logistics ecosystem.

Technological advancements and sustainability initiatives are also driving growth and efficiency in India's air cargo industry. IoT-enabled cold chain tracking systems are being widely implemented, allowing real-time monitoring of temperature, humidity, location, and handling conditions across the logistics chain. Such solutions are critical for sensitive shipments, including pharmaceuticals, biotechnology products, and perishable foods, ensuring precise environmental control and regulatory compliance. Meanwhile, airlines and cargo operators are increasingly adopting sustainable aviation fuel (SAF) to reduce carbon footprints, meet global environmental standards, and maintain operational efficiency. Investments in SAF infrastructure, along with partnerships with biofuel producers, are helping the industry align with sustainability mandates while supporting cost-competitive, reliable cargo operations. Together, these trends in trade dynamics, technology adoption, and sustainability are reshaping the outlook for India's air cargo market, making it more resilient, efficient, and strategically positioned for future growth.

- **Infrastructure Development and Policy Support Boosting Market Growth:** Infrastructure development and government policy support are playing a crucial role in driving growth in India's air cargo market. The National Air Cargo Policy seeks to enhance cargo handling capacity and establish India as a global transshipment hub. Significant investments in dedicated cargo terminals, free trade zones, and multimodal logistics parks are strengthening the overall supply chain network, positively impacting air cargo operations. Industry projections indicate that the sector is expected to attract **USD 4.99 billion in investments** over the next four years, while government plans **earmark USD 1.83 billion by 2026** for airport infrastructure upgrades and enhancement of aviation navigation services.

Digitalization and modernization initiatives are further boosting operational efficiency and capacity in the sector. Programs such as DigiYatra and advanced digital cargo tracking systems are reducing turnaround times and improving cargo visibility. Airport privatization and modernization efforts, particularly at major hubs like Delhi, Mumbai, and Bengaluru, are increasing cargo handling capabilities and overall market share. Additionally, government measures to promote indigenous aircraft manufacturing and leasing incentives under the Atmanirbhar Bharat initiative are contributing to long-term market stability. Regional connectivity schemes like UDAN are expanding air freight services to Tier-2 and Tier-3 cities, enhancing logistics efficiency and supporting rising demand for both domestic and international air cargo transport.

- **Rise of the Defence and Aerospace Sector:** India's increasing focus on strengthening domestic defence and aerospace manufacturing is significantly boosting air cargo movement. With the government promoting indigenous production through initiatives like Make in India and the Defence



Production & Export Promotion Policy (DPEPP), the flow of military components, aircraft parts, and unmanned systems (drones) has grown considerably. As defence exports rise and global collaborations expand, the need for fast, secure, and specialized air freight logistics has intensified. Air cargo enables the timely transport of high-value, sensitive defence equipment, supporting both domestic deployment and global distribution.

- **Growth in Space Technology and Satellite Exports:** India's rapidly expanding space ecosystem powered by ISRO, NewSpace India Limited (NSIL), and private startups is contributing to increased air freight of satellite parts, propulsion systems, and scientific payloads. With India emerging as a global hub for cost-effective launch services, the export of satellite hardware has increased markedly. Many of these components require precision handling, environmental control, and rapid transportation, making air cargo the preferred mode. The commercial expansion of satellite launches, space-tech research collaborations, and international payloads further fuel the demand for specialized aerospace logistics solutions.
- **Surge in Luxury Goods and Consumer Electronics:** The rise of premium consumer markets in India has driven a surge in imports of luxury goods, high-end fashion, smartphones, wearables, and premium automobiles. Global brands rely heavily on-air freight to maintain lean inventory cycles and ensure rapid restocking across major Indian cities. Air cargo supports the fast-moving nature of electronics and luxury sectors, where product life cycles are short and market competition is intense. Quick replenishment through air freight helps companies address demand spikes, seasonal launches, and festival-driven sales cycles.
- **Regional Airports Enhancing Air Cargo Connectivity:** Government initiatives such as UDAN and the expansion of regional airport infrastructure are transforming tier-2 and tier-3 cities into emerging cargo hubs. These airports are now equipped to handle increasing freight volumes, improving access for industries located outside major metros. By enabling direct cargo flights and better multimodal connectivity, regional airports reduce transportation time, lower overall logistics cost, and decongest major airports like Delhi, Mumbai, and Bengaluru. This decentralization strengthens India's nationwide air cargo network and supports balanced economic development.
- **Expansion of Dedicated Freighter Services:** Airlines are increasingly investing in dedicated freighters, cargo aircraft conversions, and specialized cargo terminals to meet rising market demand. India has also seen the development of exclusive cargo airports and express logistics hubs, which specifically support high-volume and time-critical shipments. Dedicated freighter operations improve reliability, offer flexible scheduling, and enhance capacity for e-commerce, pharmaceuticals, perishables, and industrial goods. This infrastructure modernization is enabling smoother cargo flow and improving India's competitiveness in global air logistics.
- **Consumer Shift Towards Faster Deliveries:** The rise of e-commerce, quick-commerce, and omni-channel retailing has led to an unprecedented consumer demand for same-day and next-day



deliveries. Platforms like Amazon, Flipkart, and Meesho are increasingly relying on air freight to ensure faster movement of parcels between metropolitan and regional hubs. Air cargo reduces delivery timelines from days to hours, making it essential for companies that compete on speed and reliability. This shift in consumer expectations continues to push logistics providers to expand their air-based express delivery networks.

- **AI and Automation in Air Cargo Operations:** The adoption of artificial intelligence, predictive analytics, robotics, and automated warehousing is transforming India's air cargo operations. AI-driven systems optimize cargo loading, track shipments in real time, and improve demand forecasting for air freight companies. Automated sorting systems and digital documentation reduce delays, enhance cargo visibility, and minimize human error. These technologies also help streamline customs processes and enable efficient end-to-end logistics management, contributing to faster turnaround times and improved cost efficiency for the sector.
- **Growth of Semiconductor and EV Battery Manufacturing:** With the government's strong push to build domestic semiconductor fabs and EV battery manufacturing facilities under PLI schemes, the movement of high-value chipmaking equipment, wafers, and battery components has increased substantially. These items are extremely sensitive, often requiring controlled environments and rapid transportation, making air cargo the ideal mode. As India integrates deeper into global electronics and EV supply chains, the demand for specialized air freight services capable of handling precision, fragile, and high-technology goods will continue to rise.



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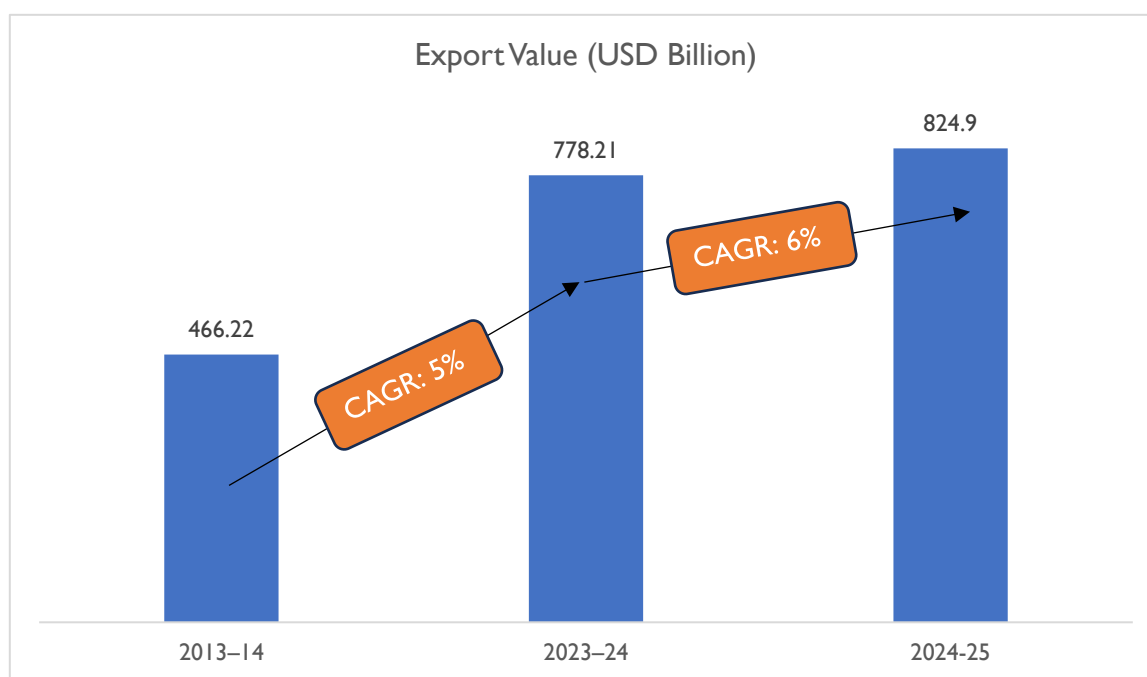
India's Foreign Trade Growth

Over the past decade, India's foreign trade has witnessed remarkable growth, driven by strong performance across both merchandise and services exports. In **FY 2013-14**, India's total exports were valued at approximately **USD 466.22 billion**, rising steadily to **USD 778.21 billion** in **FY 2023-24**, marking a strong **5%** increase over ten years. This growth trajectory continued into **2024-25**, with exports reaching **USD 824.9 billion** and achieving **6% year-on-year growth**, even as global trade faced pressure from geopolitical tensions, commodity price volatility, and slower demand across advanced economies. This growth reflects a diversified export basket, including electronics, pharmaceuticals, engineering goods, textiles, iron ore, and processed agricultural products, highlighting India's increasing integration into global trade networks and gradual movement up the value chain. Alongside exports, imports have also grown significantly, reflecting rising domestic demand for raw materials, capital goods, and technology. This balanced growth underscores India's expanding role in international commerce and its increasing competitiveness in multiple sectors.

The growth momentum continued into H1 **FY 2026**, with exports during April–September 2025 reaching **USD 413.30 billion**, up **4.45%** compared to the same period in 2024. Merchandise-only exports during the same period were **USD 220.12 billion**, a **3.02%** increase year-on-year, while non-petroleum, non-gems & jewellery exports recorded a robust **7.04%** growth, reflecting strong demand for value-added manufactured goods. Key sectors such as pharmaceuticals, engineering goods, and textiles contributed significantly to this rise, while services exports, including IT and software services, continued to strengthen India's global trade footprint. The consistent growth amid global uncertainties highlights the resilience and adaptability of India's export ecosystem.

The expansion of foreign trade has significant implications for India's logistics and air cargo sectors. Rising exports and imports are increasing demand for efficient, time-sensitive, and high-value transport solutions, including air freight for perishables, pharmaceuticals, and high-tech products. Airports and cargo operators are investing in advanced infrastructure, cold-chain capabilities, and digital tracking systems to handle growing volumes with speed and reliability. The continued growth in both domestic and international trade underscores the need for enhanced cargo handling capacity, modernized logistics networks, and multimodal connectivity, positioning India as a strategic hub in global supply chains. This trend is expected to sustain long-term investment in the air cargo sector and strengthen India's competitiveness in global trade.

India's Export Growth



Source: Ministry of Commerce & Industry – PIC Delhi

India's export ecosystem has shown consistent and broad-based expansion over the past decade, with total exports rising from **USD 466.22 billion in 2013-14 to USD 778.21 billion in 2023-24**, registering a solid **5% CAGR** supported by stronger manufacturing capacity, deeper integration with global value chains, and policy interventions aimed at boosting competitiveness and market diversification. This growth trajectory continued into **2024-25**, with exports reaching **USD 824.9 billion** and achieving **6% year-on-year growth**, even as global trade faced pressure from geopolitical tensions, commodity price volatility, and slower demand across advanced economies. The performance reflects the increasing resilience of India's export sectors, driven by sustained momentum in services, pharmaceuticals, engineering goods, technology-enabled industries, and other high-value segments. Overall, the data underscores India's gradual shift toward more value-added and innovation-led exports, along with its strengthening presence in global markets despite an uncertain external environment.

In 2024-25, merchandise exports stood at **USD 374.1 billion**, while services exports contributed **USD 387.5 billion**, demonstrating a well-balanced expansion. Key sectors like electronics, pharmaceuticals, engineering goods, iron ore, and textiles played a vital role in this surge. Strengthened by strategic policy measures, enhanced competitiveness, and broader market access, India's export ecosystem is now more resilient and deeply integrated into the global economy.

The momentum has continued into FY 2024-25, with cumulative exports during April-December 2024 estimated at **USD 602.64 billion**, a **6.03%** increase from **USD 568.36 billion** in the same period of 2023.

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Strengthened by strategic policy measures, enhanced competitiveness, and broader market access, India's export ecosystem is now more resilient and deeply integrated into the global economy.

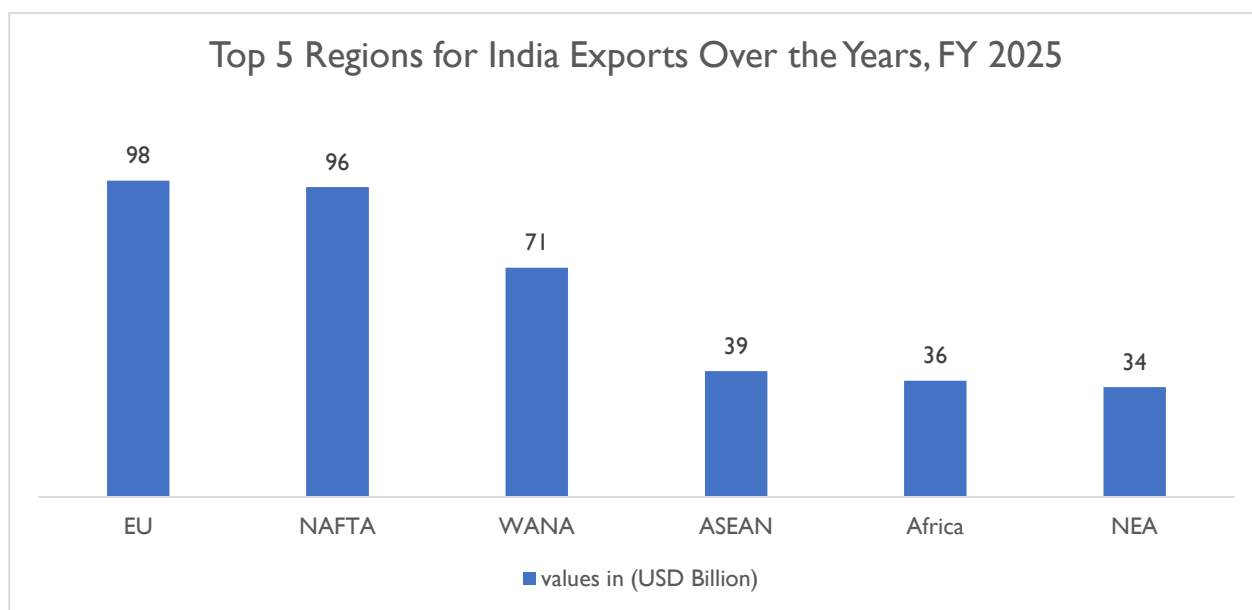
India's merchandise exports have continued to grow steadily in 2025, supported by strong performance across a diverse range of sectors. Key categories such as electronic goods, petroleum products, engineering goods, rice, marine products, and drugs & pharmaceuticals have shown significant year-on-year growth, reflecting both domestic manufacturing strength and robust global demand. The following table highlights the export values for September 2024 and September 2025 along with the corresponding growth rates:

Product / Category	Exports Sep 2024 (USD Billion)	Exports Sep 2025 (USD Billion)	YoY Growth (%)
Non-Petroleum	177.03	189.49	7.04
Electronic Goods	2.07	3.12	50.54
Petroleum Products	4.3	4.96	15.22
Engineering Goods	9.83	10.11	2.93
Rice	0.69	0.92	33.18
Marine Products	0.63	0.78	23.44
Drugs & Pharmaceuticals	2.56	2.62	2.56

Leading Export Regions Over the Years

In 2004-05, India's exports were predominantly directed to regions like North America, the European Union, North-East Asia, West Asia-Gulf Cooperation Council, and ASEAN. By 2013-14, there was a marked increase in export values across these regions, with North America, the EU, and West Asia seeing notable growth.

Fast forward to 2024-25, and the export landscape shows continued expansion, with North America leading as the largest destination. The EU, West Asia, and ASEAN also experienced robust growth, illustrating India's diversified and strengthened global trade relationships over the years.

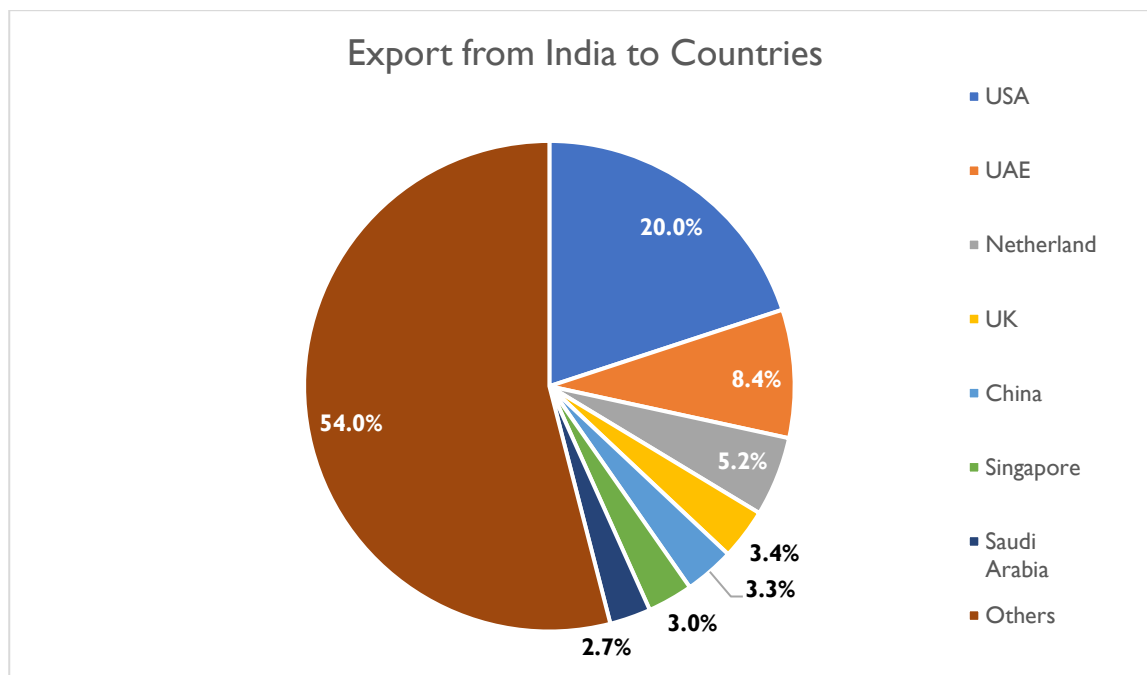


Source: Govt. of India “Niryat”

- **North America (NA)** emerged as the fastest-growing export destination for India, with exports nearly **USD 96 billion in 2024-25**. This dramatic rise reflects strengthening bilateral trade relations, rising demand for Indian manufactured goods, IT services, pharmaceuticals, and consumer products in the US and Canada. NA's growth also underscores India's growing competitiveness in high-value and technology-intensive exports.
- **European Union (EU)** also showed remarkable growth, with exports **USD 98 billion in 2024-25**. The sustained rise indicates strong trade partnerships, particularly in sectors such as pharmaceuticals, engineering goods, textiles, and automotive components. The EU's stable regulatory environment and consistent demand for quality products have contributed to this long-term growth trend.
- **West Asia-North Africa saw moderate growth**, with exports **USD 71 billion in 2024-25**. This region's growth is primarily driven by petroleum-related products, engineering goods, and agricultural exports. While the increase is less steep than NA or EU, it demonstrates India's continued economic engagement with West Asian markets, especially through energy, infrastructure, and construction-linked exports.
- **ASEAN (Southeast Asia)** experienced a smaller but steady growth, **USD 39 billion in 2024-25**. ASEAN countries remain important for India's exports of textiles, machinery, chemicals, and electronics. The moderate increase indicates growing regional integration and trade under agreements like the India-ASEAN Free Trade Agreement.
- **Africa** has emerged as a key growth market for India's exports, recording shipments worth **USD 36 billion in FY 2024–25**. This reflects strengthening trade linkages driven by rising demand across pharmaceuticals, engineering goods, consumer products, and automobiles. India's expanding diplomatic and commercial engagement with African economies has further supported this momentum, positioning the region as an increasingly important destination in India's global export strategy.
- **Northeast Asia (NE/Asia)** exports stand at **USD 34 billion in 2024-25**, which may be attributed to shifting global supply chains, changing trade policies, or stronger competition from other exporting countries. The decline highlights the need for India to strengthen its trade competitiveness in these markets, especially in electronics, IT services, and manufactured goods.



Key Export Destinations in fiscal year 2024-25



In FY **2025**, India's export profile is characterized by a strong presence in advanced and emerging economies alike, with the **United States** remaining the single largest destination, accounting for **20.0%** of total exports. This reflects sustained demand for Indian goods across sectors such as engineering, pharmaceuticals, electronics, textiles, and chemicals. The **UAE**, with a **8.4%** share, continues to serve as a critical trade gateway, driven by both direct consumption and its role as a regional re-export hub linking India to West Asia and Africa.

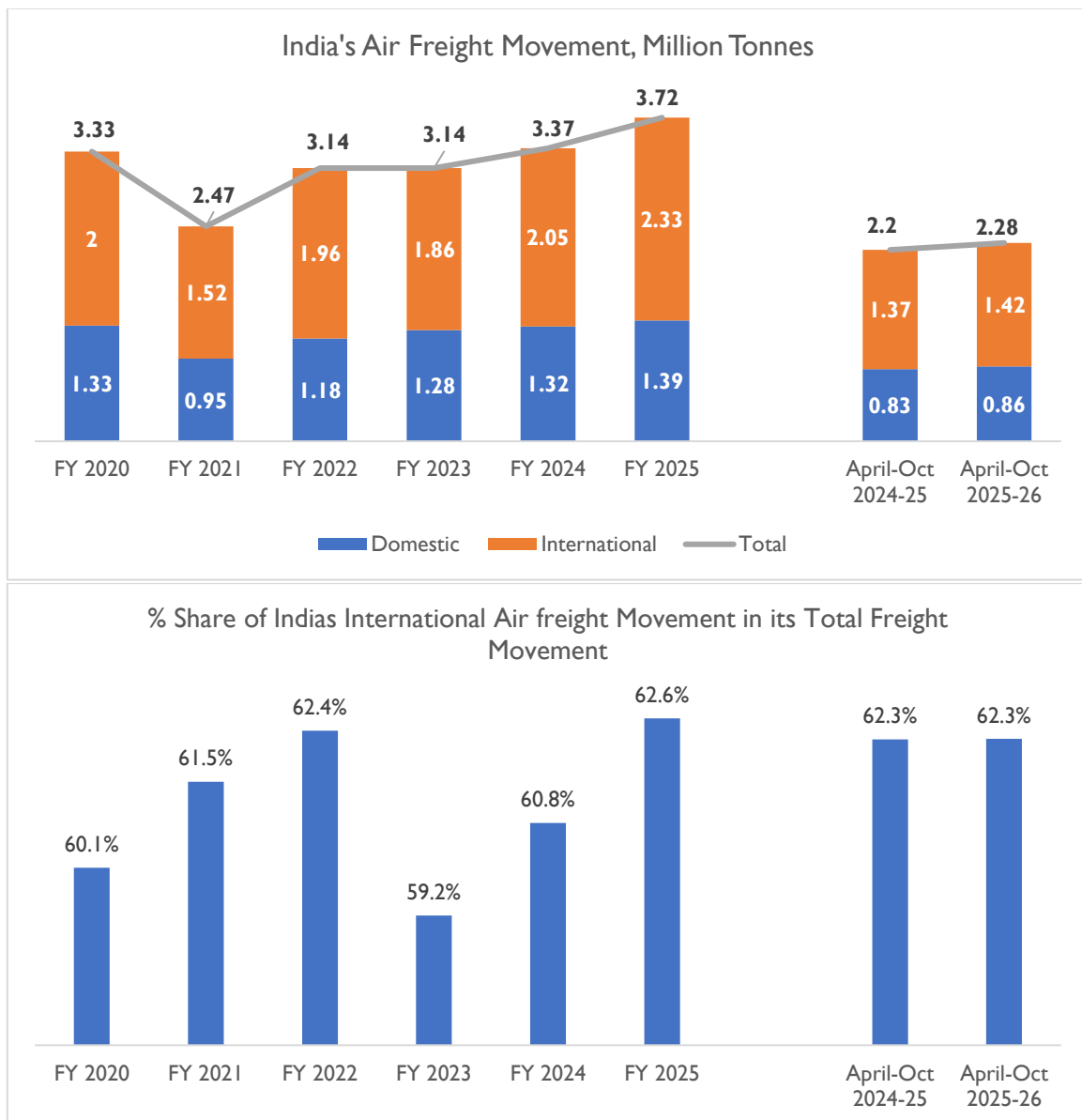
European markets also play a significant role in India's export ecosystem, led by the **Netherlands (5.2%)**, a major logistics and entry point into the EU, and the **UK (3.4%)**, supported by strong trade ties and diversified demand. Asian markets such as **China (3.3%)** and **Singapore (3.0%)** contribute steadily, reflecting supply chain interlinkages and regional trade flows, while **Saudi Arabia (2.7%)** highlights growing engagement with the Gulf region. Importantly, **other countries collectively account for 54.0% of exports**, underscoring India's broad geographical diversification and reduced dependence on a limited number of markets, which enhances the resilience of its export-led growth.



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Trends in Export Logistics: Air and Ocean Freight in India

○ Air Freight:



Sources: Airport Authority of India

India's air freight movement has maintained a steady upward trajectory between FY 2020 and FY 2025, growing at a **CAGR of 2.2%** as volumes increased from **3.33 million tonnes** to **3.72 million tonnes**. The momentum strengthened notably in FY 2025, with overall air freight rising by **11%** year-on-year, compared to **7%** growth in the previous fiscal year. This trend highlights the recovery of economic activity, expansion of e-commerce logistics, and increasing reliance on faster transportation modes amid global supply chain realignments. International freight continued to dominate India's air cargo landscape, consistently contributing between **59.3% and 62.5%** of total volumes during the FY 2020–25 period, underscoring India's deep integration with global trade routes and cross-border supply chains.

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India's international export share in total air freight movement has remained consistently high over recent years, reflecting the country's growing integration into global supply chains. From FY 2020 to FY 2025, the share fluctuated within a relatively narrow band, ranging from 59% to 63%, with a noticeable peak of 63% in FY 2025. This upward trend underscores the rising importance of international cargo in India's air logistics sector. More recently, during the April–October periods of both 2024–25 and 2025–26, the share held steady at 62%, indicating stable export performance amid evolving global trade dynamics. Overall, international freight continues to dominate India's total air cargo volumes, consistently accounting for nearly two-thirds of all air freight movement.

Economic Activity:

The steady rise in overall freight volumes reflects a broad-based expansion in India's economic activity. Growth in air cargo typically signals increased manufacturing output, stronger consumption patterns, and higher demand for fast and reliable logistics. When industries operate at higher capacity whether in pharmaceuticals, electronics, or perishables, the need for timely movement of goods intensifies, leading to increased freight traffic. This correlation between freight growth and economic momentum indicates that businesses are scaling production and distribution, contributing to healthier supply chain cycles across the country.

Global Trade Dynamics:

The strong performance of international air freight points toward favourable global trade conditions and India's strengthening position in cross-border commerce. Higher international cargo volumes may stem from rising exports, increased import of high-value goods, entry into new global markets, and the signing of new bilateral or multilateral trade agreements. Additionally, global companies are increasingly diversifying supply chains, and India is emerging as a key alternative hub amid shifting geopolitical conditions. These factors collectively enhance international trade flows, helping air freight gain momentum.

Domestic Market Growth:

The relatively moderate yet steady growth in domestic air freight highlights a stable internal market that is gradually expanding. Domestic cargo volumes are closely linked to factors such as consumer demand, expansion of e-commerce penetration in Tier-II and Tier-III cities, development of regional logistics infrastructure, and efficiency of multimodal transport systems. While domestic demand may not be growing as sharply as international segments, its consistency suggests resilience in internal trade and a slow but sustained improvement in manufacturing and distribution networks within the country.

Sectoral Performance:

A deeper examination of sector-specific contributions is crucial to fully understand the drivers behind freight growth. Different industries influence freight trends in varied ways. For instance, the rise of electronics manufacturing boosts demand for high-value, time-sensitive air cargo, while agricultural and perishable



segments rely on air freight for export quality maintenance. Similarly, the surge in pharmaceuticals and medical supplies post-pandemic has significantly increased cold-chain dependent air shipments. Meanwhile, e-commerce and express logistics continue to reshape domestic cargo patterns. Identifying which sectors are contributing most helps assess both current performance and future growth potential of the air freight market.

Impact on Air Cargo Traffic:

Increased Demand for Express Deliveries: The surge in e-commerce has escalated the need for expedited shipping solutions. Consumers now anticipate shorter delivery windows, prompting e-commerce platforms and logistics providers to utilize air cargo for its speed and efficiency. This shift is evident as express shipments are projected to constitute a significant portion of air cargo business in the coming decades.

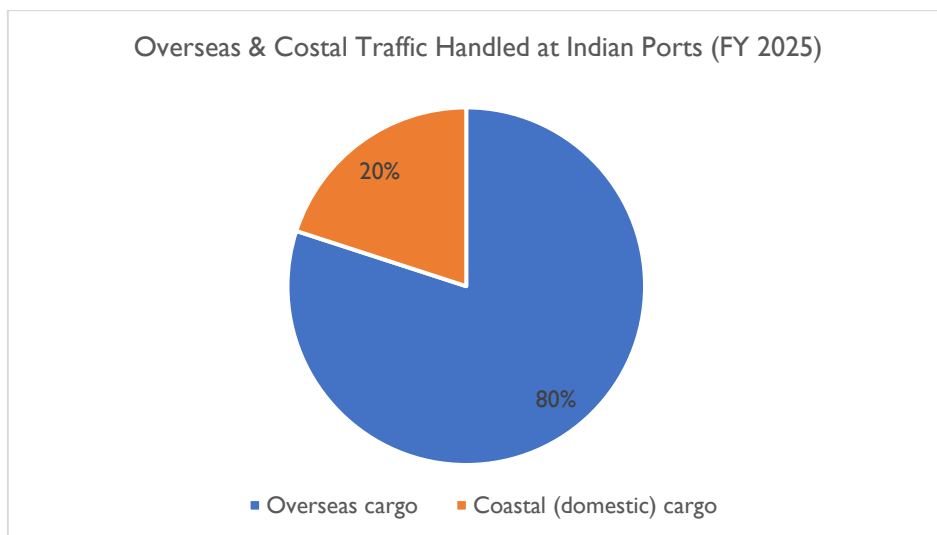
Cross-Border E-Commerce Growth: The expansion of cross-border online sales has further amplified the demand for air cargo services. Air freight is often the preferred mode for international shipments due to its ability to reduce transit times, thereby meeting customer expectations for prompt deliveries.

Infrastructure and Fleet Expansion: To accommodate the burgeoning e-commerce sector, logistics companies and airlines are investing in expanding their air cargo capacities. This includes acquiring additional aircraft, enhancing cargo handling facilities, and optimizing logistics networks to ensure efficient operations

Cargo Movement at Indian Ports

Maritime transport in India is fundamentally driven by the expansion of global trade and economic output. With an extensive coastline stretching approximately **7,517 km** (including island territories), ocean routes are instrumental to India's trade logistics. Notably, around **95% of the country's foreign trade by volume and 70% by value** is conducted via maritime channels. These routes not only facilitate international trade but also ensure connectivity between the Indian mainland and its island regions.

India's maritime ecosystem encompasses ports, shipping services, shipbuilding and repair, as well as inland waterway networks. This sector supports a wide array of operations across **overseas trade, coastal trade, and inland transport**. In terms of cargo volume, Indian ports handled a mix of **international (overseas) and domestic (coastal) cargo**, accounting for approximately **80% and 20%**, respectively, of total seaborne traffic:



Sources: Ministry of Shipping

Despite being one of the most cost-effective and environmentally sustainable modes of transportation, **coastal shipping in India remains underdeveloped**. To address this, several initiatives are underway to boost coastal cargo handling capacity, particularly at minor ports. For instance, **in June 2024, the Kerala Maritime Board launched infrastructure enhancements** including the installation of berths and cargo-handling cranes at ports such as **Kollam, Bepore, Azhikkal, and Vizhinjam**.

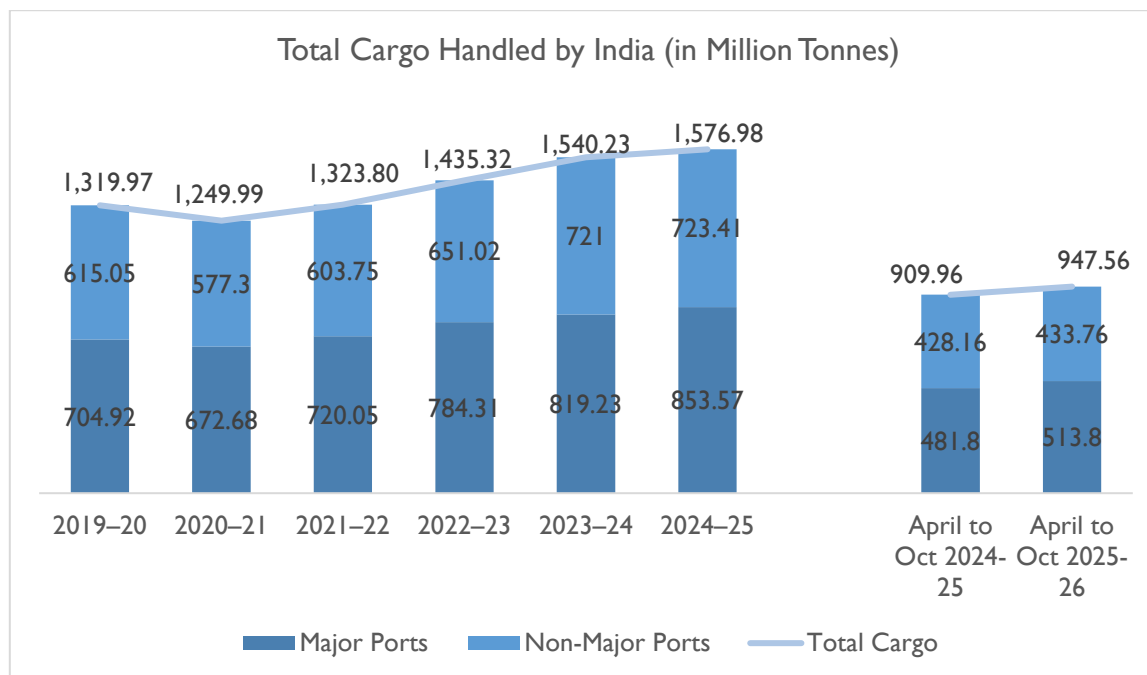
India's shipping services follow global maritime practices and are broadly classified into **tramp and liner services**. **Tramp services** operate on an ad-hoc basis without a fixed route or schedule, responding to market demand by transporting cargo wherever required. In contrast, **liner services** function on fixed schedules and predetermined routes, typically involving large ocean vessels offering regular service between ports.

In addition to cargo movement, a range of **marine support services** including **tugboat operations, mooring and stevedoring, lighterage, barge operations, and dredging** are essential for ensuring the smooth flow of vessel traffic. These auxiliary services are especially critical in India, where **port infrastructure remains underdeveloped**. Notably, in government-operated ports, these services are predominantly outsourced to **third-party providers**, with limited involvement from port authorities or shipping companies.



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Historical Cargo Handling



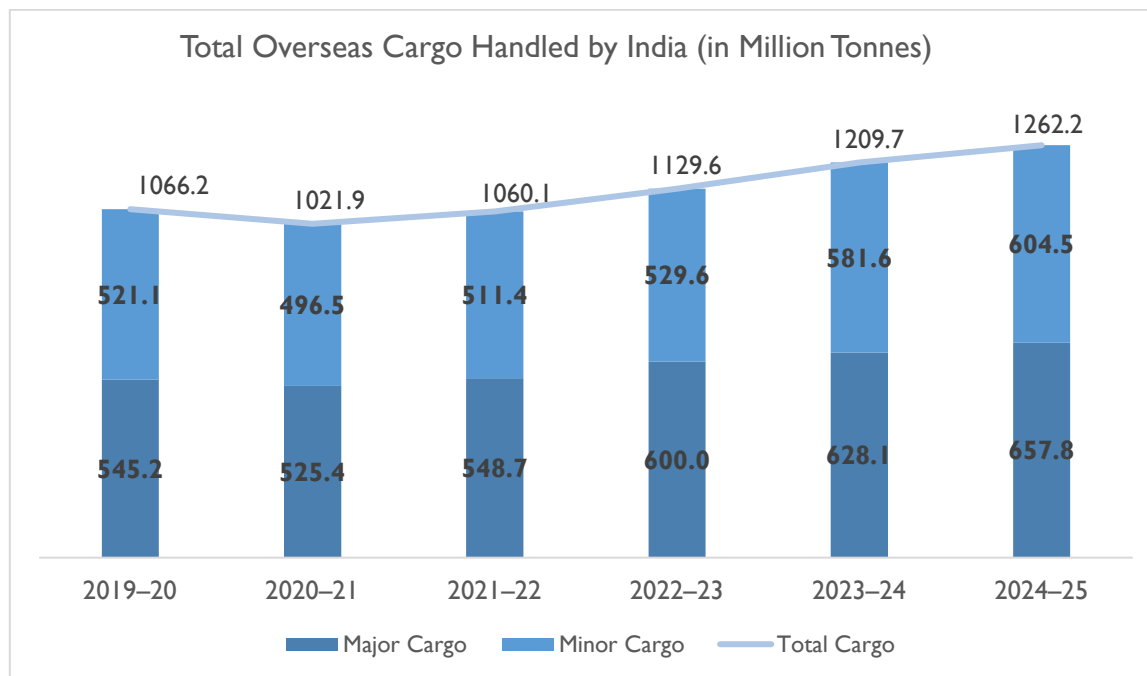
Source: Source: - Port Data Management Portal, Ministry of Ports, Shipping and Waterways

Between **FY 2020-25**, cargo volumes handled by Indian ports (Major and Minor) observed steady growth from FY 2022 onwards. In FY 2021, cargo volumes declined by **5.3%**, primarily due to the **COVID-19 pandemic**, which disrupted global supply chains, halted industrial activity, and impacted port operations. This marked the only year in the period with negative growth. A strong recovery was observed in **FY2022**, as cargo volumes rebounded to **1,576.98 Million tonnes**, registering a **5.98%** CAGR growth. This resurgence was driven by the revival in domestic industrial activity, improvement in global trade flows, and increased movement of key commodities such as coal, fertilizers, and crude oil, in tandem with infrastructure growth post-pandemic. However, the y-o-y growth momentum moderated in **FY2024-25 to 7.3% and 2.4% respectively from 8.4% in FY 2023**. The relatively slower pace suggests a **normalization of growth rates** post the high-recovery phase, and impact of **geopolitical tensions, volatile global trade routes, and tightening economic conditions in key markets**.

During the period from April to October 2024–25, India's total seaport cargo throughput stood at 909.96 million tonnes, comprising 481.8 million tonnes handled by major ports and 428.16 million tonnes by non-major ports. In the corresponding period of 2025–26, total cargo volume rose to 947.56 million tonnes, reflecting steady growth across both segments: major ports increased their handling to 513.8 million tonnes, while non-major ports contributed 433.76 million tonnes. This upward trend underscores the ongoing capacity expansion and efficiency improvements across India's port infrastructure, supporting rising trade volumes and reinforcing the country's maritime logistics capabilities in line with national infrastructure and connectivity goals.


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India's International Ocean (Overseas) Cargo Volume:



Source: - Port Data Management Portal, Ministry of Ports, Shipping and Waterways

Between FY 2020 and FY 2025, India's overseas cargo volumes handled across both major and minor ports demonstrated a generally upward trajectory, particularly strengthening from FY 2022 onward. During this period, the country's port ecosystem showcased resilience, efficiency gains, and steady integration with global maritime trade networks. Although the growth pattern was not uniform throughout the years, the long-term trend highlights India's rising role in international logistics and commodity movement.

The only major setback occurred in **FY 2021**, when overseas cargo volumes fell by **4.2%**, a direct consequence of the **COVID-19** pandemic. This decline was driven by widespread disruptions, including international vessel delays, container shortages, cargo handling constraints, and reduced industrial production. The pandemic also strained global supply chains, triggering port congestion, restricted manpower availability, and temporary shutdowns across key export-oriented sectors. This year became a clear outlier within the five-year window, marking the only period of negative growth.

A strong and broad-based recovery unfolded in FY 2022, when cargo volumes rebounded sharply, reaching **1,262.24 million tonnes** and registering a **5.4% CAGR** between **FY 2021** and **FY 2025**. This revival was supported by a resurgence in domestic industrial activity as factories resumed full operations, global demand strengthened, and maritime trade routes gradually normalized. The increase in cargo handling was particularly



driven by higher movement of critical bulk commodities such as coal, fertilizers, iron ore, crude oil, and containerized goods, reflecting increased consumption and manufacturing needs. Investments in port infrastructure under the Sagarmala Programme, digitalization of port operations, and improved hinterland connectivity further enabled smoother cargo flow.

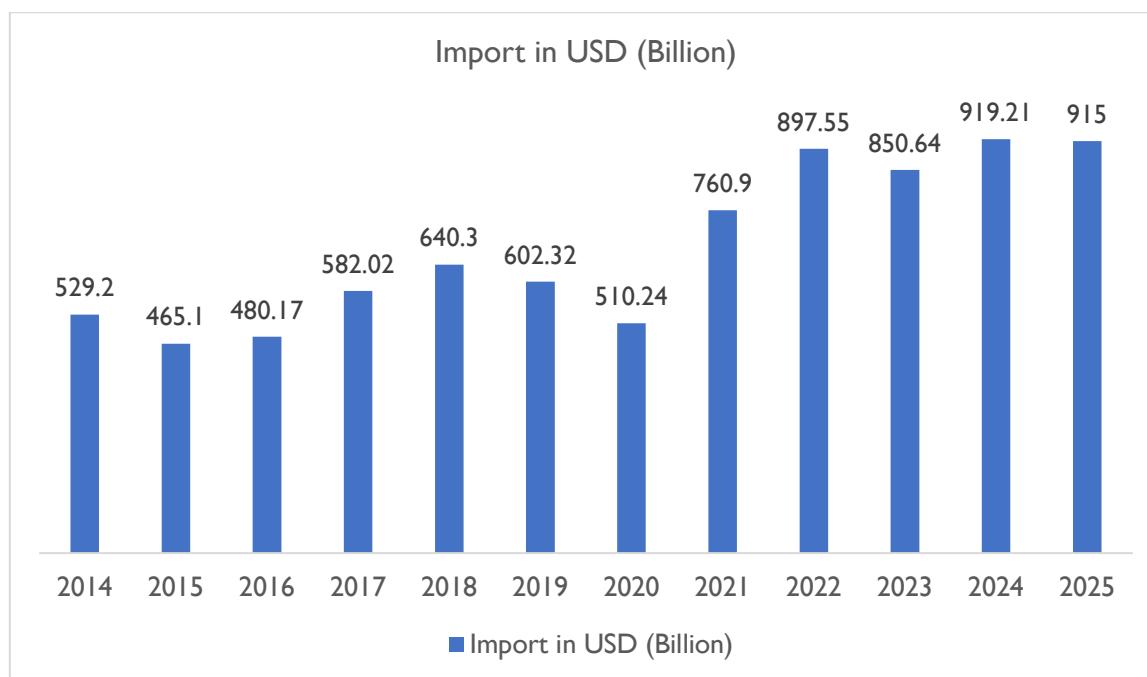
By FY 2024-25, India's ports had not only recovered from pandemic-induced disruptions but also surpassed pre-COVID levels, indicating structural improvements in port efficiency and growing integration with global supply chains. This steady rise signifies India's strengthening maritime capabilities and its growing significance as a strategic logistics hub in international trade.

India's Import Growth and its share in global exports.

India's import landscape has evolved significantly, reflecting both the country's expanding economic needs and its growing integration with global markets. While long-term annual imports have risen sharply, from about **USD 529.2 billion** in 2014 to nearly **USD 919.21 billion** in 2024, short-term monthly trends continue to show natural fluctuations driven by global demand, commodity cycles, and domestic consumption patterns. Recent figures highlight this volatility: imports declined to **USD 50.96 billion** in February 2025, down from **USD 59.42 billion** in January 2025. This movement indicates the influence of factors such as changing crude oil prices, variations in industrial demand for intermediate goods, currency shifts, and broader global economic conditions. Despite these temporary month-to-month changes, the long-term trend clearly shows a steady expansion of India's import requirements, driven by rising domestic consumption, industrialization, and the need for raw materials, technology, and capital goods to support the country's development trajectory.

India's main imports are: mineral fuels, oils and waxes and bituminous substances (27 per cent of total imports); pearls, precious and semi-precious stones and jewellery (14 percent); electrical machinery and equipment (10 percent); nuclear reactors, boilers, machinery and mechanical appliances (8 percent); and organic chemicals (4 percent). India's major import partners are: China (16 per cent of total imports), the United States (6 percent), the United Arab Emirates (6 percent), Saudi Arabia (5 percent) and Switzerland (5 percent).

Detail Import data from 2014 to 2024:



Source-World Bank as per latest data

The table and graph show India's annual import values from 2014–15 to 2023–24, highlighting how total imports have moved year by year during this period.

- **2014-2016:** India's import trajectory between **2014 and 2016** reflected the impact of global commodity cycles. Imports fell sharply from **USD 529.2 billion in 2014 to USD 465.1 billion in 2015**, primarily due to the steep crash in global crude oil prices and subdued industrial demand. A mild recovery followed in 2016, with imports rising to **USD 480.17 billion**, indicating gradual stabilization in energy prices and a modest revival in domestic consumption.
- **2017-2019:** Between 2017 and 2019, India experienced a period of strong economic expansion, which led to a steady increase in import demand. Imports rose significantly, touching **USD 640.3 billion in 2018**, supported by higher crude prices, increased manufacturing activity, and a growing consumer base. However, in 2019, imports dropped to **USD 602.32 billion**, reflecting the early signs of a global economic slowdown, US–China trade tensions, and tightening domestic liquidity conditions.
- **2020:** The year **2020** marked an unprecedented disruption in India's import landscape, with total imports sharply declining to **USD 510.24 billion**. The outbreak of COVID-19 and subsequent nationwide lockdowns severely restricted industrial production, consumer activity, and cross-border trade. Major import-dependent sectors such as electronics, machinery, automotive, and crude oil experienced steep drops in demand due to supply-chain breakdowns, mobility restrictions, and collapsing global commodity prices.



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- **2021-2022:** The downturn of 2020 was followed by a powerful rebound in **2021 and 2022**, as economic activity normalised and global trade flows recovered. Imports surged to **USD 760.9 billion in 2021**, driven by the reopening of industries, restoration of supply chains, and a sharp rise in domestic demand for raw materials, electronics, and machinery. This momentum further escalated in 2022, with imports climbing to a historic high of **USD 897.55 billion**. Elevated global commodity prices particularly crude oil, coal, metals, and agricultural commodities, significantly increased India's import bill.
- **2023:** In 2023, imports moderated to **USD 850.64 billion**, impacted by global recessionary pressures, tighter monetary policies worldwide, and a decline in demand for certain commodities.
- **2024:** India's imports rebounded to **USD 919.21 billion**, reflecting strong domestic consumption, rising non-oil purchases, and a revival in investment-driven demand. The uptick underscores India's continued deepening of its integration with global supply chains, even amid broader global economic uncertainties.
- **2025:** India's imports experienced a slight decline to **USD 915 billion**, edging down from the previous year's robust USD 919.21 billion. This marginal dip nonetheless reflects the economy's continued resilience, underpinned by strong domestic consumption, healthy demand for non-oil goods, and sustained investment activity. Despite global economic uncertainties and fluctuating trade dynamics, India maintained a deeply integrated position within global supply chains, demonstrating the underlying stability and adaptive capacity of its import-driven trade framework.

Overall, India's import trajectory over the past decade reflects the combined influence of global commodity cycles, domestic economic conditions, and geopolitical developments. Imports declined between 2014 and 2016 largely due to falling crude oil prices, while strong economic expansion pushed them upward during 2017–2019. The sharp contraction in 2020 was driven by the COVID-19 pandemic and widespread disruptions, but this trend quickly reversed as imports surged in 2021 and reached a historic peak in 2022 amid robust post-pandemic recovery and rising global prices. Although 2023 witnessed a moderate decline due to global uncertainty, inflationary pressures, and tempered demand, imports rose again in 2025 to **USD 915 billion**, indicating renewed economic momentum and higher demand for energy, intermediates, and capital goods. Looking ahead, India's import performance will continue to depend on global trade policies, commodity price movements, domestic manufacturing growth, and currency stability.



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Globalization of Supply Chains

The globalization of supply chains has dramatically elevated the role of air cargo as companies pursue faster, more resilient, and geographically diversified logistics networks. With production shifting across continents, from East Asian manufacturing clusters to consumer hubs in North America, Europe, and increasingly the Global South, air freight has become indispensable for moving time-critical, high-value, and perishable goods. Sectors such as electronics, pharmaceuticals, fast fashion, and e-commerce now depend on air cargo to maintain lean inventories, achieve rapid replenishment, and ensure service-level consistency across dispersed global markets. The rise of integrated express operators and digital freight platforms has further optimized cross-border flows, enabling end-to-end visibility and multimodal connectivity.

Amid this global transformation, **India is emerging as a pivotal node** in the international air logistics ecosystem. The country's growing footprint in global supply chains, driven by the "China+1" strategy, its manufacturing base in India, and rapidly expanding export sectors, has increased demand for high-capacity, tech-enabled air cargo infrastructure. Major airports such as Delhi, Mumbai, Bengaluru, and Hyderabad are being upgraded into multimodal cargo hubs, supported by initiatives like the **National Air Cargo Policy Outline (NACPO)**, the **Unified Logistics Interface Platform (ULIP)**, and **PM Gati Shakti**. India's pharmaceutical and biotech exports, fresh horticulture shipments, and electronics manufacturing (especially smartphones and semiconductors) increasingly rely on air freight to serve distant markets efficiently.

Simultaneously, India's booming e-commerce market and rising cross-border online retail have triggered a surge in express and parcel volumes, placing the country alongside global consumption markets. With dedicated freighter operations expanding, growing participation of global integrators, and policy emphasis on improving transshipment and cold-chain capacity, India is steadily positioning itself not just as a consumption market but as a **strategic air cargo gateway** for Asia and beyond. As global supply chains become more volatile, due to geopolitical tensions, natural disruptions, and shifting trade patterns, India's strengthening air cargo ecosystem is set to play a larger role in enhancing resilience and supporting global economic integration.

- **Integration into Global Manufacturing Networks:** India's deeper integration into global manufacturing circuits, across electronics, pharmaceuticals, automotive components, and textiles, has elevated the requirement for high-speed logistics. As production becomes globally distributed, the timely movement of raw materials and finished goods is critical to maintaining competitiveness. Air freight has become the preferred mode for high-value, time-sensitive shipments due to its reliability and speed. With industry projections indicating annual air cargo growth of **6%–9%** and total volumes expected to reach **5–5.8 million tonnes by 2029**, the sector's expansion mirrors India's rising dependence on rapid cross-border logistics to support export-oriented and technology-driven industries.
- **Support for Just-in-Time (JIT) Production Models:** The adoption of JIT manufacturing in sectors like electronics, automotive, and precision engineering has directly increased reliance on efficient air logistics. These industries require components to arrive exactly when needed to prevent assembly-line



delays or inventory bottlenecks. Air cargo enables ultra-short lead times, flexible scheduling, and reduced warehouse holdings, which are essential for maintaining lean supply chains. By minimizing production downtime and optimizing inventory turnover, air freight has become an integral pillar of India's modern manufacturing ecosystem as it integrates more tightly with global value chains.

- **Diversification of Global Sourcing Strategies:** As multinational corporations diversify supply bases to mitigate risks from geopolitical tensions, trade fragmentation, and pandemic-driven shocks, India is gaining traction under the global "China+1" strategy. This shift strengthens India's role as both a sourcing destination and a manufacturing hub for high-value goods. Air cargo is vital for supporting this transition, ensuring fast and secure movement of sensitive shipments such as semiconductor components, life-saving pharmaceuticals, aerospace parts, and specialized machinery. The dependability of air logistics helps global companies maintain supply continuity while deepening India's position in diversified global supply chains.
- **Government-Driven Logistics Modernization:** India's logistics transformation is being accelerated by policy initiatives designed to enhance multimodal connectivity and reduce systemic inefficiencies. The National Logistics Policy (NLP) and PM Gati Shakti framework are central to this effort, promoting integrated planning across air, road, rail, and port infrastructure. These initiatives focus on digitization of logistics processes, infrastructure harmonization, streamlined clearances, and improved cargo handling capacity at airports. Such reforms strengthen India's global logistics competitiveness by enabling faster cargo turnaround, higher visibility in supply chain workflows, and better alignment of air cargo operations with global standards.



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Growth in Pharmaceutical, Manufacturing and Agricultural

The growth of India's air cargo traffic is significantly driven by key industries such as **pharmaceuticals, manufacturing, and agriculture**. These sectors rely heavily on-air freight due to the need for speed, efficiency, and global market integration. Air cargo plays a crucial role in ensuring timely delivery, maintaining product quality, and expanding India's trade presence worldwide.

I. Pharmaceuticals Sector and Air Cargo Demand

- **Why Air Cargo is Important for Pharmaceuticals?**

- Pharmaceutical products such as vaccines, biologics, insulin, and speciality drugs are highly sensitive to temperature variations. Even a minor deviation from required conditions can render them ineffective or unsafe. Air cargo offers advanced cold-chain solutions, ranging from refrigerated containers to ultra-low-temperature storage, which are crucial for maintaining product integrity during long-distance transit. Medicines and medical supplies often need to reach hospitals, pharmacies, and distribution centers rapidly. Air freight significantly reduces delivery time compared to sea or road transport. This speed is essential not only for commercial fulfilment but also during emergencies, humanitarian missions, and global health crises.
- After COVID-19, the world's dependency on fast, secure delivery of vaccines and pharmaceutical raw materials surged. India became a global supplier of vaccines, boosting the need for efficient air cargo. The pandemic reshaped logistics networks, making air freight an integral part of pharma supply chains for both routine and emergency shipments.

- **Key Growth Indicators:**

- **Surging Exports:** India's pharmaceutical exports have emerged as a significant global force, reaching **USD 30.47 billion in FY2024–25**, with a **year-on-year CAGR of 9.21%**, reflecting the sector's robust growth trajectory and increasing prominence in international markets. This strong growth reinforces global confidence in India's capabilities in producing high-quality, affordable medicines. As export volumes increase, so does the demand for reliable and fast cargo services, most of which move by air due to the nature of the products.
- **Major Global Supplier:** India played a pivotal role in this global landscape. As the third-largest producer of drugs by volume, India contributed 20% to the global export of generic drugs. The Indian pharmaceutical industry is projected to grow at a CAGR of over 10%, with its market size expected to reach \$130 billion by 2034. India's strength lies in cost-efficient production, a skilled workforce, and a robust regulatory framework. This vast international outreach requires an extensive, well-



coordinated air freight system to connect Indian manufacturers with markets in the US, Europe, Africa, and Asia.

- **Increase in Specialized Air Cargo:** In 2023, pharmaceutical shipments via air cargo grew by **22%**, driven by rising exports of high-value, temperature-controlled products. Airlines and logistics companies are increasingly investing in dedicated pharma corridors, equipped with real-time temperature monitoring, thermal blankets, insulated containers, and secured storage, to cater to this growing demand.

2. Manufacturing Sector and Air Freight Growth

The manufacturing sector plays a crucial role in strengthening India's air freight ecosystem. As global supply chains evolve and production cycles become more time-sensitive, manufacturers increasingly rely on air cargo to maintain efficiency, reduce downtime, and meet international delivery commitments.

- **Why Air Cargo is Essential for Manufacturing?**

- Modern manufacturing, especially in sectors like automotive, electronics, aerospace, and precision engineering, operates on just-in-time and lean inventory models. In these systems, even small delays in component delivery can halt entire production lines. Air cargo provides the speed and dependability required to ensure uninterrupted operations by delivering critical components exactly when needed.
- Many manufactured goods fall into the high-value or sensitive category such as semiconductor chips, telecom equipment, medical devices, EV components, and industrial machinery parts. These products require rapid and secure movement to prevent damage, avoid production delays, and meet contract deadlines. Air freight offers lower risk, better tracking, and faster transit, making it the preferred mode for such shipments.
- Indian factories are increasingly integrated into global production networks. Whether supplying parts for electronics in East Asia, automotive systems in Europe, or textiles for global retail brands, the need for fast international connectivity is essential. Air freight ensures Indian manufacturers can meet global timelines and maintain competitiveness.

- **Key Growth Indicators:**

- **Growth in Export-Oriented Manufacturing:** India's export sector has reached new heights, with **'Make in India' merchandise exports (excluding petroleum products) hitting USD 374.1 billion in FY2024–25**, marking a **6% increase over the previous year**, while **services exports reached USD 387.5 billion**. Strong global demand, diversification in sourcing, and favourable trade agreements have driven this sustained growth. Key sectors such as electronics, automotive components, engineering goods, textiles, and chemicals are leading the export surge, reflecting India's



expanding industrial capabilities. To meet strict delivery timelines and rising global expectations, manufacturers are increasingly leveraging **air freight**, ensuring faster, more reliable transport of high-value and time-sensitive goods.

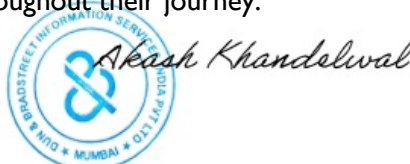
- **Air Freight Usage by Global Brands:** Global companies are increasingly turning to Indian suppliers and manufacturers, and many are boosting their air cargo shipments from India to avoid delays in sea transport. A notable example is Inditex, the parent company of Zara, which increased its air freight shipments from India by **37%**. This shift helps them match rapid fashion cycles, respond to dynamic consumer demand, and reduce supply chain bottlenecks caused by ocean freight disruptions.
- **Make in India & PLI Schemes Boosting Air Cargo:** Government initiatives such as Make in India, Atmanirbhar Bharat, and the Production Linked Incentive (PLI) schemes have significantly strengthened domestic manufacturing across sectors like electronics, pharmaceuticals, textiles, semiconductors, and renewable energy. As output increases, industries require faster international logistics support, both for exporting finished goods and importing specialized components. This has led to a substantial rise in air freight demand from industrial clusters, SEZs, and export hubs.

3. Agriculture Sector and Air Freight Expansion

The agricultural sector has emerged as one of the fastest-growing contributors to India's air cargo demand. As global consumers increasingly prefer fresh, high-quality, and exotic food products, the need for rapid and reliable transport has grown significantly. Air freight plays a crucial role in ensuring that India's perishable agricultural goods reach international markets on time, with minimal spoilage and maximum value realization.

• Why Air Cargo is Vital for Agriculture?

- Agricultural commodities such as fruits (grapes, mangoes, pomegranates), vegetables, seafood, dairy items, meat products, and floriculture goods have limited shelf life and require fast, temperature-controlled logistics. Air cargo ensures they reach global markets, such as the Middle East, Europe, and Southeast Asia, within hours rather than weeks, preserving freshness and nutritional value.
- Fast delivery through air freight allows Indian exporters to supply products at peak freshness, helping them tap into premium international markets. This not only increases competitiveness but also enables farmers and exporters to secure higher prices for their goods. Air cargo also facilitates the export of seasonal and specialty items that need to arrive within strict market windows.
- With India diversifying its agricultural basket from traditional staples to exotic produce, organic foods, and ready-to-eat items air freight is becoming essential. It reduces transit time drastically and helps exporters meet stringent quality standards set by importing countries. This is especially vital for products that must remain in controlled temperatures throughout their journey.



- **Key Growth Indicators:**

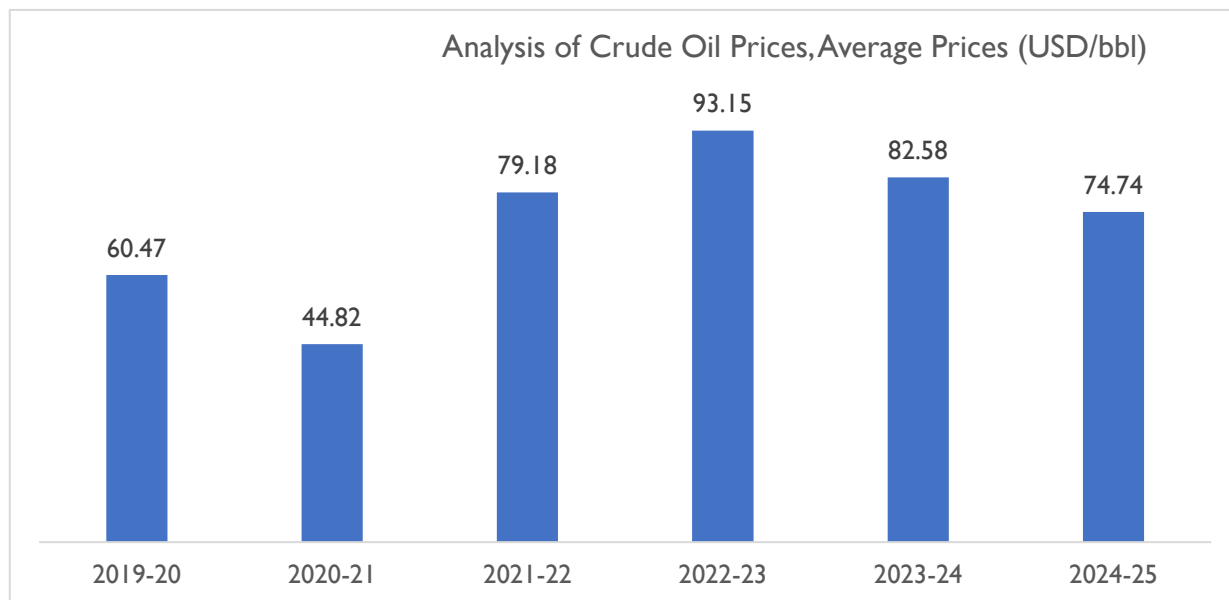
- **Rising Agricultural Exports:** India's agricultural export performance continues to show steady growth, with exports increasing by **6.4%**, rising from **USD 48.8 billion in 2023–24** to **USD 51.9 billion in 2024–25**. This underscores the sector's significant contribution to the country's trade and its expanding presence in global markets. The growth reflects India's deeper integration into international food supply chains and rising global demand for Indian produce. Enhanced logistics, particularly through air cargo, have enabled exporters to minimize losses, maintain higher quality standards, and expand to new markets more efficiently.
- **Government's Agriculture Export Policy:** The Government of India's Agriculture Export Policy is focused on doubling agricultural exports by 2030 through improved logistics, better quality control, and expanded market access. The policy emphasizes minimizing wastage, strengthening cold-chain infrastructure, and promoting export clusters all of which directly boost demand for air freight. By aligning farmers, FPOs, exporters, and logistics providers, the policy aims to position India as a leading global food supplier.
- **Development of Agri-Air Corridors:** To support perishable exports, the government is developing dedicated agri-air corridors connecting major production hubs to international gateways. These corridors help streamline customs processes, reduce transit delays, and enhance cold-chain efficiency. Examples include improved connectivity for products like mangoes from Maharashtra, grapes from Nashik, flowers from Bengaluru, and seafood from coastal clusters. Dedicated corridors significantly reduce spoilage, increase export frequency, and boost the confidence of global buyers in Indian produce.



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Impact of Crude Oil Volatility on Logistics Cost Structures

An analysis of Crude Oil Prices' impact on the Overall logistics industry and the post-COVID impact of Crude oil Prices on the logistics sector.

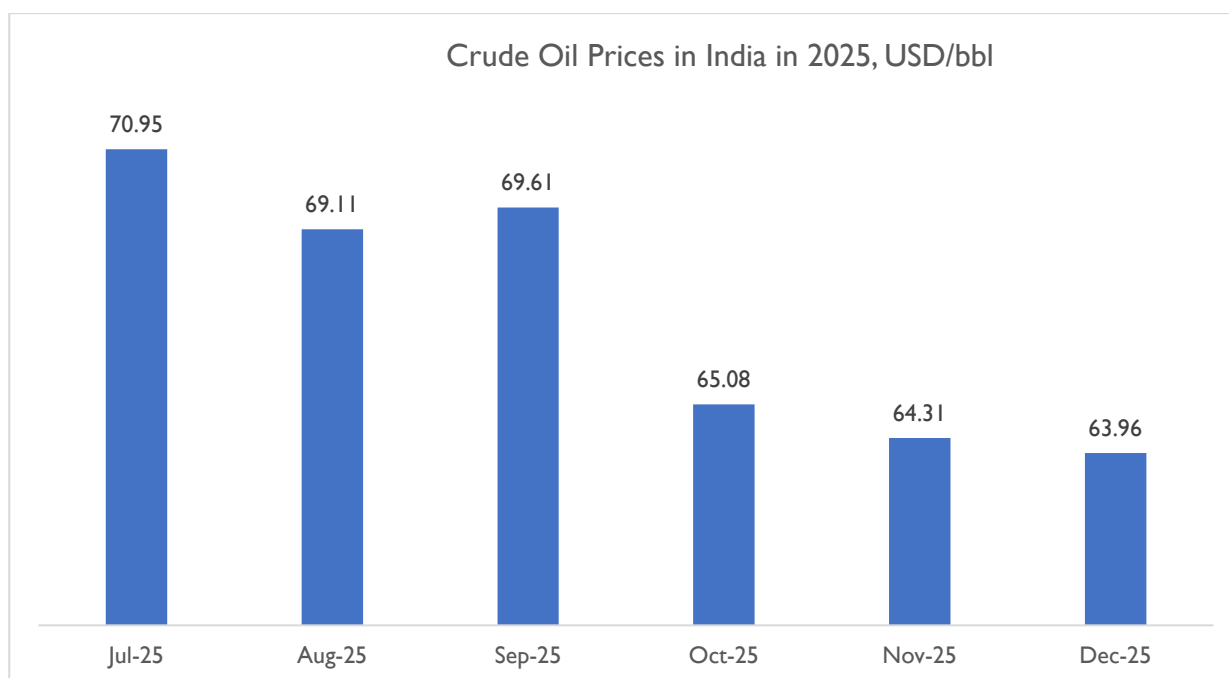


Crude oil prices play a pivotal role in shaping the logistics industry, as fuel costs directly impact transportation expenses for shipping, trucking, aviation, and rail. A surge in oil prices leads to increased freight rates, affecting supply chains, consumer prices, and overall business profitability. The data from 2019 to 2025 reveals significant fluctuations, with a sharp dip in 2020-21 due to COVID-19, followed by a steep rise in subsequent years. These price movements have forced logistics companies to adopt fuel-efficient technologies, optimize routes, and explore alternative energy sources to mitigate cost pressures.

Crude oil prices have exhibited significant volatility over the past few years, reflecting global supply-demand dynamics, geopolitical factors, and economic recovery trends. The average price per barrel fell to **USD 44.82 in 2020-21** amid the pandemic-induced demand shock, before rebounding sharply to **USD 79.18 in 2021-22** and further to **USD 93.15 in 2022-23** as global economic activity recovered. Prices moderated to **USD 82.58 in 2023-24** and **USD 74.74 in 2024-25**, reflecting a combination of stable supply, energy market adjustments, and easing inflationary pressures. Year-to-date data for **December 2025** indicates an average price of **USD 63.96 per barrel**, signaling a continuing downtrend as global oil markets stabilize and alternative energy adoption rises. This trend highlights the sensitivity of crude oil markets to both macroeconomic conditions and energy transition dynamics.



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Crude oil prices have shown a gradual decline in the second half of 2025, reflecting moderating global demand and improving supply conditions. Prices fell from **USD 70.95 per barrel in July** to **USD 63.96 per barrel in December**, with monthly averages of USD 69.11 in August, USD 69.61 in September, USD 65.08 in October, and USD 64.31 in November. This consistent downward trend over the last six months of the year highlights the easing of energy market pressures and a stabilizing crude oil environment as the global economy adjusts to post-pandemic consumption patterns and shifts toward alternative energy sources.

The COVID-19 pandemic had a profound impact on crude oil prices, with an unprecedented collapse in demand during 2020-21, driving prices to record lows. This decline provided temporary relief for logistics firms but was overshadowed by global supply chain disruptions, labour shortages, and reduced consumer demand. However, the post-pandemic recovery saw a dramatic rebound in oil prices due to supply chain constraints, geopolitical tensions, and increased energy demand. The logistics industry faced higher operating costs, forcing companies to adjust freight pricing, implement fuel surcharges, and enhance supply chain resilience through digital transformation.

Looking ahead, while crude oil prices have shown signs of stabilization in 2024-25, they remain a key factor in logistics profitability and inflationary pressures. The transition towards sustainable logistics, including electric vehicles (EVs) and alternative fuels, is gaining traction to counter oil price volatility. Additionally, governments and businesses are focusing on energy efficiency and policy interventions to ensure cost-effective logistics operations. The long-term impact of fluctuating crude oil prices will continue to shape industry strategies, with an emphasis on sustainability, resilience, and cost optimization.

Challenges for Air Cargo Traffic in India

India's air cargo sector has witnessed significant growth in recent years, driven by rising e-commerce, manufacturing exports, and the pharmaceutical and perishable goods markets. Despite this expansion, the sector faces multiple structural and operational challenges that constrain its efficiency and competitiveness. While demand for fast, reliable, and temperature-sensitive cargo transport continues to rise, gaps in infrastructure, high operational costs, regulatory hurdles, and workforce limitations pose serious obstacles. Understanding these challenges is critical for developing strategies to enhance the capacity, resilience, and overall performance of India's air cargo industry.

➤ **Inadequate Infrastructure & Capacity Gaps at Airports**

Many airports in India, especially those outside major metropolitan areas, lack dedicated cargo terminals, modern warehousing, cold-storage facilities for perishables and pharmaceuticals, and automated handling systems. This results in inefficient cargo movement, long dwell times, and limited throughput. Even major hubs face constraints such as insufficient apron space, warehouse capacity, night-handling facilities, and cargo-specific infrastructure, which struggle to keep up with rising demand.

➤ **High Operational Costs & Cost-Competitiveness Issues**

Air cargo in India is relatively expensive due to high ground handling charges, airport tariffs, fuel costs, and security or handling surcharges. These high operational costs reduce the competitiveness of air cargo, particularly for medium-to-low value goods. The reliance on "belly-hold" space in passenger aircraft rather than dedicated freighters adds volatility, as cargo capacity may shrink during travel downturns or off-peak seasons, complicating scheduling and further increasing costs.

➤ **Over-concentration on Few Major Airports ("Hub Bottlenecks")**

Approximately **90%** of India's air cargo is handled by a small number of major airports, creating congestion and limiting scalability. This over-concentration restricts the ability to expand cargo operations nationwide and makes the system vulnerable to disruptions at these hubs. Such bottlenecks impact the resilience of the supply chain and reduce efficiency across the sector.

➤ **Regulatory Complexity and Customs / Documentation Delays**

Air cargo operators frequently face overlapping regulatory requirements across customs, aviation, safety, and other authorities. These complexities lead to delays, unpredictable clearance times, and compliance challenges. Inconsistent procedures, manual paperwork, and limited automation exacerbate the problem, particularly affecting time-sensitive or perishable shipments.

➤ **Lack of Skilled Workforce & Low Technology/Automation Adoption**

The air cargo sector suffers from a shortage of trained personnel, especially for specialized consignments like cold-chain pharmaceuticals, perishables, and high-value electronics. Additionally, the adoption of advanced



technologies, such as real-time tracking, automated warehouse systems, cargo management software, and cold-chain logistics, remains limited. This reduces operational efficiency and restricts the ability to scale services effectively.

➤ **Fragmented Logistics Ecosystem & Weak Multimodal/Hinterland Connectivity**

India's logistics sector is highly fragmented, with many small and unorganized players, making coordination, standardization, and integration difficult. Poor connectivity between air cargo hubs and road or rail freight networks, particularly in remote or Tier-2/Tier-3 regions, limits effective first-mile and last-mile movement. This reduces the reach, reliability, and overall efficiency of air cargo services for exporters and importers across the country.

Disruptions in Ocean Shipping

The global shipping industry has faced multiple disruptions in recent years, leading to significant shifts in freight transport trends. These disruptions have had a direct impact on India's air cargo industry, as businesses seek faster and more reliable alternatives to ocean shipping. Factors such as **port** congestion, geopolitical conflicts, capacity shortages, and supply chain delays have driven an increasing number of exporters and importers to choose air freight over traditional sea routes.

Factors responsible for disruption in Ocean Shipping

- **Port Congestion and Capacity Constraints**
 - Major seaports worldwide, including those in Asia, have experienced severe congestion, resulting in long delays for cargo shipments.
 - Container shortages and lack of available shipping slots have further increased wait times, affecting supply chains that depend on predictable schedules.
 - To mitigate these delays, businesses have turned to air freight, which, despite being more expensive, offers a faster and more reliable delivery alternative.
 - Reports indicate that ocean freight disruptions have had a direct impact on air cargo demand, with companies increasing their reliance on air shipments to avoid supply chain bottlenecks.
- **Geopolitical Conflicts and Trade Route Disruptions**
 - Geopolitical tensions and conflicts, such as those affecting the Red Sea and Suez Canal, have caused shipping diversions, longer transit times, and increased costs.
 - The recent crisis in the Red Sea region has significantly impacted sea trade routes, forcing businesses to seek alternative transport methods.



- Global brands, including Inditex (Zara's parent company), have increased air freight shipments from India by 37% to avoid the risks associated with sea freight disruptions.
- **Rising Air Cargo Demand Due to Ocean Shipping Delays**
 - International air cargo volumes from India have surged as exporters shift from sea to air transport.
 - In Ahmedabad, international air cargo volumes rose by **33%** from April to October 2024 compared to the same period the previous year, showing a significant shift in logistics trends.
 - Sectors such as electronics, pharmaceuticals, and fast fashion are particularly affected, as they require time-sensitive shipments that cannot afford prolonged ocean transit times.
- **Shift in Logistics Strategy: From Ocean to Air Freight**
 - Due to continued uncertainties in ocean freight, businesses are modifying their logistics strategies and incorporating air freight into their supply chain planning.
 - Companies that previously relied heavily on sea transport are now diversifying their shipping methods, ensuring they have air freight options available in case of further maritime disruptions.
 - Analysts predict that the increased reliance on air cargo due to ocean shipping issues will continue, potentially reshaping long-term logistics trends.

Warehousing Infrastructure

India's air cargo traffic plays a crucial role in facilitating trade and economic growth, yet it faces significant challenges due to inadequate warehousing infrastructure. Many airports lack dedicated cargo handling facilities, leading to congestion, inefficiencies, and delays in shipment processing. The absence of modern storage solutions, including temperature-controlled warehouses for perishable goods, further hampers the efficiency of air cargo logistics. Additionally, limited multimodal connectivity between airports and other logistics hubs restricts the seamless movement of goods, increasing transportation costs and turnaround times. Despite government initiatives to improve infrastructure, the existing warehousing facilities remain insufficient to meet the growing demands of e-commerce, pharmaceuticals, and international trade.

Another major challenge is the high logistics cost associated with inefficient warehousing operations. India's logistics expenses account for nearly **13-14%** of GDP, significantly higher than the global average. This is largely due to outdated warehouse management systems, manual cargo handling processes, and cumbersome customs clearance procedures. Regulatory hurdles and complex documentation requirements further slowdown air cargo movement, increasing operational costs for businesses. Additionally, the sector faces a shortage of skilled professionals trained in cargo handling and digital logistics, limiting the potential for



automation and technology adoption. These inefficiencies make India's air cargo sector less competitive compared to global standards.

To overcome these challenges, India must prioritize investments in modernizing warehousing infrastructure, integrating digital technologies, and improving regulatory frameworks. The adoption of artificial intelligence, automated cargo handling systems, and real-time tracking can significantly enhance operational efficiency. Furthermore, strengthening multimodal connectivity and streamlining customs clearance processes can reduce delays and lower logistics costs. Encouraging private sector participation and foreign direct investment in airport warehousing can also accelerate infrastructure development. By addressing these challenges, India can enhance its air cargo capabilities, supporting its ambition to become a global trade and logistics hub.

Major Gaps in the Indian Warehousing Sector

The Indian warehousing sector is a critical backbone of the country's logistics and supply chain ecosystem, supporting manufacturing, retail, e-commerce, pharmaceuticals, and perishable goods sectors. Over the past decade, the sector has witnessed substantial growth due to rising domestic demand, expansion of organized retail, and the surge of e-commerce. However, despite these developments, the Indian warehousing industry still faces several structural and operational gaps that hinder efficiency, productivity, and global competitiveness. Compared to international standards, Indian warehouses are estimated to be **40–50%** less productive, primarily due to inefficiencies in technology, infrastructure, workforce capabilities, and supply chain integration. These deficiencies result in higher operational costs, longer processing times, frequent inventory discrepancies, and limited responsiveness to market demands. The following sections explore the major gaps in the sector in detail.

I. Technology Deficiency and Automation Gap

A significant shortcoming in Indian warehousing is the limited adoption of advanced technology and automation. In global markets, warehouses increasingly leverage robotic automation, AI-driven inventory management, predictive analytics, and IoT-enabled real-time tracking to optimize operations and improve efficiency. In contrast, many Indian warehouses continue to rely heavily on manual processes and outdated software systems. This technological lag results in several operational inefficiencies:

- **Longer Order Processing Times:** Manual handling and labour-intensive operations slow down the movement of goods, delaying order fulfilment and increasing turnaround times.
- **Higher Error Rates:** Reliance on manual inventory tracking and documentation increases the likelihood of discrepancies, stockouts, and overstocking, which can disrupt supply chains.
- **Limited Real-time Visibility:** The absence of IoT-enabled sensors and cloud-based warehouse management systems limits real-time monitoring of inventory levels, shipment status, and equipment utilization, reducing responsiveness to sudden demand fluctuations.



The lack of technological adoption not only hampers operational efficiency but also affects competitiveness, particularly in sectors requiring high-speed fulfillment, such as e-commerce and pharmaceuticals.

2. Poor Infrastructure and Inadequate Storage Facilities

Infrastructure challenges remain a major bottleneck in the Indian warehousing sector. Many warehouses operate in outdated facilities that are ill-equipped to handle modern logistics requirements. Specific gaps include:

- **Outdated Storage Systems:** A significant number of warehouses lack high-rise racking systems, automated storage and retrieval systems (ASRS), temperature-controlled storage, and specialized handling facilities for perishable or high-value goods. This limits storage density and handling efficiency.
- **Unorganized and Scattered Warehouses:** The sector is highly fragmented, with a large proportion of small-scale, unorganized warehouses operating in limited spaces and with inadequate standardization. This fragmentation reduces operational efficiency and makes consolidation of shipments difficult.
- **Inadequate Transportation Connectivity:** Poor road networks, limited proximity to industrial hubs, and weak last-mile connectivity delay the movement of goods to and from warehouses, increasing lead times and logistics costs.

These infrastructural challenges make it difficult for Indian warehouses to meet the speed, reliability, and scale of operations demanded by modern supply chains.

3. Regulatory and Policy Challenges

Despite reforms like the Goods and Services Tax (GST) and the National Logistics Policy, regulatory and policy hurdles continue to affect warehouse expansion and operational efficiency. Key challenges include:

- **Complex Land Acquisition Procedures:** Procuring land for modern warehouses involves navigating lengthy legal and bureaucratic procedures, including multiple approvals from local authorities.
- **High Compliance Costs:** Various licensing, safety, and environmental regulations increase operational costs, particularly for small and medium-sized warehouse operators.

These regulatory complexities discourage investment in state-of-the-art facilities and slow down the modernization of the sector.

4. Skilled Workforce Shortage

Efficient warehouse operations require specialized skills in areas such as inventory management, material handling, equipment operation, data analytics, and warehouse automation. However, India faces a persistent shortage of trained warehouse professionals due to:



- **Limited Training and Skill Development Programs:** Most workers receive minimal technical training, affecting accuracy in order fulfillment and the efficient handling of modern equipment.
- **High Dependence on Manual Labor:** The absence of skilled automation specialists prevents the adoption of advanced warehouse management systems, robotic handling, and AI-driven process optimization.

The workforce gap reduces operational efficiency and hinders the adoption of modern technologies, keeping Indian warehouses behind global benchmarks.

5. Inefficient Supply Chain Integration

A modern warehouse functions optimally only when integrated seamlessly with transport networks, suppliers, and retailers. Indian warehouses often face challenges in achieving this integration:

- **Poor Data Synchronization:** Lack of real-time data exchange across warehouses, transporters, and retail systems leads to stock mismatches, delayed shipments, and inefficiencies in order fulfillment.
- **Fragmented Logistics Ecosystem:** Small and medium-sized enterprises (SMEs) often struggle to access efficient warehousing solutions due to high costs, limited capacity, and lack of standardized services.

The fragmented and poorly integrated supply chain ecosystem limits visibility, responsiveness, and coordination, which are essential for just-in-time inventory management and efficient distribution.

6. Limited Adoption of Sustainable Practices

Environmental sustainability is gaining importance globally, but Indian warehouses lag in implementing green practices. This gap has both ecological and economic implications:

- **Low Usage of Renewable Energy:** Few warehouses utilize solar energy, energy-efficient lighting, or other renewable energy solutions, resulting in higher operational costs and environmental impact.
- **High Carbon Footprint:** Inefficient logistics, outdated equipment, and poor energy management contribute to excessive greenhouse gas emissions.
- **Waste Management Issues:** Limited recycling practices and poor disposal mechanisms increase environmental degradation and reduce compliance with emerging sustainability standards.

Bridging these gaps is essential for India to enhance its warehousing sector's productivity and competitiveness. By investing in technology, improving infrastructure, simplifying regulations, upskilling the workforce, and promoting sustainable practices, the country can develop a more efficient and globally aligned warehousing ecosystem. Addressing these challenges will not only reduce operational costs but also strengthen India's position as a global logistics hub.

Regulatory Hurdles

1. Land Acquisition and Zoning Issues

Land acquisition remains one of the most critical bottlenecks for warehouse infrastructure development in India. The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 (LARR Act) significantly slows down project timelines due to extensive procedural requirements. Under Section 4, a compulsory Social Impact Assessment (SIA) must be conducted for acquisitions exceeding **50 acres** in urban areas or **100 acres** in rural areas, a process that typically takes 6–9 months and involves public hearings, multiple field surveys, and impact evaluation reports. Additionally, Section 10 mandates that 80% landowner consent for private projects and **70%** consent for PPP projects must be obtained before acquisition can proceed. In practice, achieving this level of consensus is extremely difficult, resulting in prolonged negotiations, higher land prices, and frequent project delays.

State-level differences add to these hindrances:

- **Maharashtra Land Revenue Code (1966):** Under Section 42, converting agricultural land into industrial or warehousing use requires paying **50–75%** of the ready reckoner rate as a conversion fee. This significantly raises the upfront cost for developers and often makes large-scale warehousing projects financially unviable, especially near Mumbai, Pune, and Nagpur.
- **Haryana Development and Regulation of Urban Areas Act (1975):** As per Section 3, developers must secure NOCs from five or more state departments, including town planning, environment, fire services, and irrigation authorities. The lack of a unified approval mechanism results in repetitive inspections, document submissions, and cross-department delays, extending approval timelines by several months.
- **Karnataka Town and Country Planning Act (1961):** Section 14A requires additional zoning clearances before land can be used for warehousing or logistics purposes. This process involves detailed scrutiny of land utility, environmental suitability, and traffic impact studies. As a result, approvals often take **120–180 days**, pushing back construction schedules and creating uncertainty for logistics investors.

Overall, these multi-layered legal and procedural requirements increase project timelines, escalate land acquisition costs, and reduce investor confidence. The lack of uniformity across states also forces companies to navigate different regulatory environments for each warehousing project, hindering the creation of an integrated national warehousing network.

2. Multi-Layered Approval Framework

Warehouse development in India is burdened by an extensive and fragmented approval ecosystem. As per the Ease of Doing Business Report 2023, a typical warehouse project must comply with 47 separate approvals from municipal bodies, environmental authorities, fire departments, and state-level planning agencies. The



absence of a unified single-window clearance mechanism significantly slows project execution, increases administrative costs, and creates uncertainty for developers. These delays are further amplified by differing state-level procedures and the lack of digital integration among approving agencies.

Fire Safety Compliance: National Building Code 2016 (Part IV): Mandates

Under the National Building Code (NBC) 2016, warehouses must meet stringent fire safety norms to obtain a Fire NOC, which is a mandatory prerequisite for project commencement and occupancy.

- **Minimum 6m wide motorable access (Clause 3.4.2):** Warehouses must provide a 6-metre-wide all-weather motorable road around the building to ensure access for fire tenders. This often requires redesigning site layouts, acquiring additional land, or altering boundary limits, adding both time and cost to the development process.
- **Automatic sprinklers for storage >500m³ (Clause 5.2):** Storage volumes exceeding 500 cubic metres must be equipped with automated sprinkler systems, which require sophisticated hydraulic calculations, certified equipment, and periodic inspections. Installing these systems can delay construction by several weeks and substantially raise the project's capital expenditure
- **Fire NOC process takes 90-120 days (Source: BIS Notification No. CD-1/2023):** As per BIS Notification No. CD-1/2023, Fire NOCs typically require 90–120 days due to multiple site inspections, document submissions, and compliance checks. Any deviation from the NBC even minor design changes necessitates re-inspection, further delaying approvals.

Environmental Clearances: Environment Impact Assessment Notification (2006)

- **Category B2 projects need EC for >20,000m² built-up:** Warehouses with built-up areas exceeding 20,000 square metres fall under Category B2 and must obtain an Environmental Clearance (EC). This requires conducting baseline environmental studies, traffic impact assessments, and risk analyses, all of which extend project timelines.
- **Form I and IA applications require 6 months of processing:** Developers must submit Form I/IA with detailed project and environmental data. The approval cycle review, site inspection, public consultation (if applicable), and Expert Appraisal Committee evaluation generally takes 6 months, delaying construction commencement and affecting financing schedules.

Municipal Approvals: Model Building Bye-Laws 2016:

Municipal corporations impose an additional layer of procedural approvals that vary across states and cities, further complicating the development process.

- **Extra 2% of project cost as scrutiny fee:** Under the Model Building Bye-Laws 2016, developers must pay a scrutiny fee amounting to **2% of the project cost** for building plan approvals. For large warehouses spanning 5–10 lakh square feet, this fee becomes a significant financial burden.



- **45-day compulsory waiting period for objections:** Municipal bodies impose a mandatory 45-day public notice period, allowing local residents and stakeholders to raise objections regarding land use, traffic impact, or environmental concerns. Even a single objection can trigger re-evaluation, hearings, and additional documentation, pushing approval timelines well beyond the standard 45-day period.



Capex trend and other initiative to strengthen the Warehousing Infrastructure in India

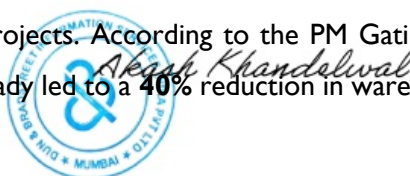
India's warehousing infrastructure is undergoing rapid transformation, driven by increasing capital expenditure (Capex), government policies, and technological advancements. The sector is a critical component of India's logistics ecosystem, influencing supply chain efficiency, trade competitiveness, and economic growth. To enhance warehousing capacity and optimize storage and distribution networks, India is implementing large-scale infrastructure projects, production-linked incentives (PLI) schemes, and policy-driven investments. These initiatives aim to position India as a global manufacturing and logistics hub by fostering innovation, improving multimodal connectivity, and promoting sustainable storage solutions.

The Indian government and private sector players have significantly increased investments in warehousing infrastructure to meet the growing demands of e-commerce, manufacturing, and cold chain logistics. Some key trends in Capex allocation include:

I. Large-Scale Investments in Multi-Modal Logistics Parks (MMLPs)

The Government of India's PM Gati Shakti National Master Plan, launched in 2021, has provided a significant impetus to the country's warehousing and logistics sector by promoting large-scale investments in Multi-Modal Logistics Parks (MMLPs). These parks are designed to integrate various modes of transportation, road, rail, and coastal shipping with modern warehousing facilities, aiming to enhance supply chain efficiency, reduce logistics costs, and improve last-mile connectivity across the country. By bringing together 16 ministries under a unified framework, the initiative ensures coordinated planning and development, minimizing redundancies and optimizing resource utilization across infrastructure projects.

- **Chennai MMLP:** A total of **35 MMLPs** have been proposed under this plan, with investments aimed at creating state-of-the-art warehousing, automated material handling systems, and dedicated freight corridors linking major industrial hubs and ports. For instance, the Chennai MMLP, with an **investment of INR 7,200 crore**, will feature **5 million** square feet of automated warehousing. This facility will be strategically linked to Chennai Port via dedicated rail corridors, enabling seamless movement of goods and reducing transit times significantly. Such large-scale, integrated logistics hubs are expected to serve as key nodes in the national supply chain network, supporting both domestic distribution and international exports.
- **GIS-Based Planning:** The planning and development of these MMLPs are increasingly being guided by GIS-based site selection and AI-powered analytics, which optimize warehouse locations, minimize duplication, and ensure efficient land use. This data-driven approach enhances operational efficiency, enabling logistics operators to plan routes, allocate storage, and manage inventory more effectively.
- **Investment Inflow:** In terms of investment scale, the PM Gati Shakti initiative has already mobilized over **INR 1.2 lakh crore** in infrastructure and logistics projects. According to the PM Gati Shakti Progress Report (Q1 2024), these developments have already led to a **40%** reduction in warehouse-



to-port transit times, demonstrating tangible improvements in supply chain speed and reliability. Beyond efficiency gains, these MMLPs are expected to generate employment, support industrial clusters, and attract private-sector participation, further boosting the growth and modernization of India's logistics and warehousing sector).

2. Production-Linked Incentive (PLI) Scheme Driving Investment

The PLI scheme is a transformative initiative to boost domestic manufacturing and strengthen warehousing infrastructure. With **INR 1.61 lakh** crore in approved investments and **INR 14 lakh** crore in projected production output, the scheme is driving significant upgrades in warehousing and supply chain capabilities:

- **Capacity Expansion:** The scheme has accelerated investment in automated storage and retrieval systems (ASRS), high-density racking, and temperature-controlled cold storage. These upgrades enable warehouses to handle high-value and perishable goods efficiently, reduce human errors, optimize space utilization, and improve turnaround times. This is particularly beneficial for sectors like electronics, pharmaceuticals, automobiles, and FMCG.
- **Export-Oriented Growth:** Warehouses developed under the PLI framework adhere to international quality, safety, and regulatory standards, enabling smoother export operations. Modernized facilities support seamless trade flows, faster customs clearance, and compliance with global certifications, making India a more attractive destination for international trade and supply chain investments.
- **Job Creation:** Over **11.5 lakh** new jobs have been generated through PLI-linked investments, expanding the logistics workforce. The scheme also promotes skill development in warehouse management, material handling, automation systems, and data-driven inventory management, enhancing employability and creating a technically skilled workforce to operate modern warehouses.
- **Operational Efficiency:** Modernization through the PLI scheme reduces order processing times, minimizes inventory discrepancies, and improves overall supply chain responsiveness. Enhanced warehouse systems enable predictive inventory planning, efficient first-mile and last-mile operations, and better integration with manufacturing and distribution networks.

3. Rise of Private Sector Investments in Grade-A Warehousing

Private sector participation in India's warehousing industry has grown significantly, particularly in Grade-A warehouses large, modern, technology-enabled facilities designed to meet global standards. These



investments are concentrated in high-demand regions such as Delhi-NCR, Mumbai, Bengaluru, and Chennai, driven by industrial growth, e-commerce expansion, and investor interest. Key highlights include:

- **Institutional Funding & REITs:** Warehousing-focused Real Estate Investment Trusts (REITs), global private equity firms, and infrastructure funds have invested billions of dollars into developing modern logistics parks and Grade-A warehouses. This infusion of institutional capital is not only expanding warehouse capacity but also improving professional management, standardization, and long-term sustainability of assets. REIT-backed projects ensure high-quality design, regulatory compliance, and modern safety features, attracting large corporate clients and multinational companies.
- **E-Commerce Boom:** Rapid growth in e-commerce and omnichannel retail has created an unprecedented demand for fulfillment centers, last-mile distribution hubs, and just-in-time inventory management solutions. This demand has driven **10–12%** annual growth in warehouse leasing, particularly in urban and suburban industrial corridors. Technology-enabled warehouses with real-time tracking, automated material handling, and integrated inventory management are becoming critical to supporting fast deliveries and high-volume order processing.
- **Sustainability Focus:** Private sector investors are increasingly incorporating green and sustainable practices into Grade-A warehouses. Initiatives include solar-powered storage, thermal energy storage (TES) backup systems, energy-efficient lighting, rainwater harvesting, and smart waste management. Such practices not only reduce operational costs but also align with global environmental standards, attract multinational clients with ESG (Environmental, Social, Governance) mandates, and support India's overall sustainability goals.

Key Initiatives to Strengthen Warehousing Infrastructure

I. Solar-Powered Cold Storage Systems with TES Backup

To address energy inefficiencies in cold chain logistics, the government has introduced new guidelines regulating solar-powered cold storage systems. These systems incorporate thermal energy storage (TES) backup, reducing reliance on conventional power grids and enhancing operational efficiency.

- **Impact:** The adoption of solar-powered cold storage systems with TES backup significantly lowers energy costs, which form a major portion of operational expenses in cold chain logistics. The TES backup ensures continuous temperature control even during periods of low solar generation or power outages, minimizing spoilage of temperature-sensitive products. These systems also enhance the effective storage capacity for perishable goods, benefiting sectors such as pharmaceuticals, vaccines, dairy, fruits, and vegetables, while ensuring compliance with regulatory and quality standards.



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- **Scalability:** These energy-efficient systems encourage private sector investments in renewable energy-powered warehousing infrastructure and can be replicated across industrial corridors, logistics parks, and standalone facilities. Integration with smart energy management and IoT-enabled monitoring allows predictive maintenance, optimizes performance, and further reduces operational costs. Overall, this approach supports sustainability goals, strengthens the cold chain ecosystem, and enhances India's competitiveness in domestic and export markets.

2. Logistics Vision 2030 – Attracting USD 50 Billion in FDI

The NITI Aayog's Logistics Vision 2030 outlines a comprehensive roadmap to transform India into a highly efficient, technology-driven, and globally competitive logistics hub. A central goal is to attract **USD 50 billion** in foreign direct investment (FDI) into warehousing, cold chain, and multimodal logistics. The initiative strengthens India's long-term infrastructure pipeline, improves ease of doing business, and aligns with global supply chain standards.

- **Policy Reforms:** Logistics Vision 2030 emphasizes simplifying land acquisition processes, reducing bureaucratic delays, and creating a friendlier regulatory environment for global investors. Policy changes include single-window clearances, rationalized logistics-related taxes, and incentives for setting up large, modern warehouses. These reforms aim to significantly cut compliance burdens and improve investor confidence, enabling faster project execution and larger inflows of foreign capital.
- **Technological Integration:** A key component of the vision is the adoption of advanced technologies such as AI-driven Warehouse Management Systems (WMS), IoT-enabled tracking, automated sorting, and digital twins for warehouse planning. These technologies enhance efficiency by improving real-time visibility, reducing operational errors, and optimizing inventory management. By encouraging tech-intensive warehousing, the policy positions India to meet the evolving expectations of global supply chains and high-value industries.
- **Global Competitiveness:** Through modern infrastructure, digital transformation, and supportive policies, Logistics Vision 2030 aims to position India as a leading destination for third-party logistics (3PL), fourth-party logistics (4PL), and contract warehousing services. Improved service quality, faster turnaround times, and compliance with international standards allow India to compete more effectively with global logistics hubs. This strengthens India's attractiveness for MNCs, export-oriented companies, and integrated logistics providers seeking efficient and scalable warehouse operations.

3. Digital & Smart Warehousing Development

The government is actively promoting the modernization of warehousing through the adoption of digital and smart technologies that significantly enhance operational accuracy, speed, and scalability. By encouraging



automation, data-driven decision-making, and interconnected warehouse ecosystems, India aims to build a future-ready logistics backbone capable of supporting high-volume, time-sensitive, and global supply chain operations.

- **Automated Guided Vehicles (AGVs) and Robotics:** AGVs, robotic arms, and automated picking systems are being introduced to streamline warehouse activities such as sorting, picking, packaging, and pallet movement. These technologies drastically reduce manual errors, increase throughput, and enable warehouses to operate at higher efficiency levels. Robotics also supports 24/7 operations, ensuring faster order fulfilment and minimizing dependence on manual labour during peak seasons.
- **Blockchain for Supply Chain Transparency:** Blockchain solutions are being integrated to enhance visibility, authenticity, and traceability across the supply chain. By creating secure, tamper-proof records of every transaction and movement of goods, blockchain helps reduce fraud, prevent counterfeiting, and improve regulatory compliance. This is especially valuable for industries such as pharmaceuticals, electronics, and food processing, where traceability is essential. The use of blockchain also strengthens trust among stakeholders, including exporters, importers, and logistics service providers.
- **5G & IoT Connectivity:** With the rollout of 5G and advanced IoT sensors, warehouses are transitioning into fully connected environments where every asset, package, and piece of equipment can be monitored in real time. High-speed connectivity supports continuous inventory tracking, automated replenishment systems, and predictive analytics that help identify and resolve disruptions before they occur. These technologies ensure better space utilization, reduced downtime, and improved decision-making, ultimately strengthening the resilience and responsiveness of the overall logistics network.

India's warehousing infrastructure is undergoing a structural transformation with high Capex allocations, government-driven policies, and technological innovations. The convergence of initiatives such as PM Gati Shakti, PLI schemes, solar-powered cold storage, and smart warehousing solutions is set to elevate India's logistics ecosystem to global standards. With sustained investment momentum, India is well-positioned to enhance supply chain resilience, boost exports, and emerge as a logistics powerhouse by 2030.



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Performance Standards

1. Structural & Safety Standards

- Structural building standards assure that warehouses can withstand mechanized handling and heavy loads. The National Building Code 2016 requires a minimum ceiling height of **9** meters to support modern stacking equipment such as high-reach forklifts and automated storage systems (BIS SP 7:2016, Part IV). For heavy-duty flooring, floors need to be able to support a **5-7.5** metric tons per square meter load capacity according to WDRA specifications, essential for the storage of industrial products or bulk commodities. Earthquake resistance standards conform to IS 1893:2016, necessitating certain structural reinforcements dependent upon seismic zones, with the most protection being required for Zone V.
- Fire safety measures are also stringent. Automatic smoke detection systems need to cover all spaces according to NBC Part IV requirements, with heat/smoke detectors at **9m** intervals. Hydrant systems, as per IS 3844:2019 standards, need to be installed every **30** meters with sufficient water pressure. These are supplemented by mandatory Fire NOCs from local authorities, which include inspections of emergency exits (at least two per warehouse), fire extinguisher locations (one per 500 sqm), and electrical safety compliance.

2. Cold Chain Standards

- Temperature regulation is critical for perishable items. FSSAI Regulation 4.1.5 allows only $\pm 2^{\circ}\text{C}$ deviation from specified temperatures, with the need for precision cooling systems and digital monitoring. Frozen foods require -18°C to -25°C conditions according to NCCD TS-102, with ongoing temperature recording. Pharmaceutical warehouses require even more stringent controls - the CDSCO requires $\pm 1^{\circ}\text{C}$ for vaccines with backup power for cooling equipment.
- Cold storage design specifications ensure proper air circulation. The mandated 0.5-1.0 m/s airflow velocity (IS 14812) prevents hot spots while minimizing product dehydration. Pharma facilities require dual redundant cooling systems - if the primary fails, the backup must activate within 90 seconds. All units need temperature mapping reports showing uniform distribution across storage zones.

3. Technology Compliance

- Automation grading divides facilities into capabilities. Grade-A warehouses employ Automated Storage/Retrieval Systems (ASRS) with less than 5% manual handling - including robotic pallet movers, conveyor systems, and automated guided vehicles. Grade-B facilities combine Warehouse Management Systems (WMS) with barcode scanners and semi-automated picking.
- Data reporting requirements harmonize synchronizations with taxation systems. Real-time GSTN portal updates monitor inventory movements, while E-way bill integration provides automatic documentation for dispatches. Cybersecurity standards (ISO 27001) require encrypted databases,



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biometric access controls, and penetration testing once a year - particularly essential for e-commerce warehouses dealing with customer data.

4. Environmental & Sustainability Standards

- Energy efficiency standards conform to climate objectives. The 1-star BEE rating (ECBC 2017) mandates LED lighting, insulated panels, and energy-efficient HVAC systems. Big warehouses (50,000+ sqft) have to get **15%** of the power from on-site solar as per SECI guidelines, usually through rooftop installations.
- Water management adheres to CPWD standards with no liquid discharge - treating and reusing all wastewaters. Rainwater collection systems require at least **50,000 litres of** capacity for every acre of roofing area. Handling of materials moves to electric fleets - the NITI Aayog 2025 target eliminates diesel-operated forklifts with a carbon-reduction of 8-10 tons per warehouse per year.

5. Labour & Safety Standards

- Work conditions laws strike a balance between production and well-being. The 48-hour work week maximum (OSH Code 2020) supports 8-hour shifts with 30-minute mandatory meal breaks. For material handlers, 15-minute breaks every 2 hours avoid musculoskeletal damage - a requirement from the Ergonomics Code (DGMS Circular 5/2021).
- Safety gear regulations prescribe ANSI-quality hard hats, high-stacking full-body harnesses, and NFPA-approved PPE kits. Biennial medical screening tests for hearing loss (due to conveyor noise) and respiratory problems. The 100-hour biennial training (NSDC Standards) involves equipment operation, first aid, and computer systems for inventory.



Threats and Challenges

India's logistics sector is undergoing a transformation driven by rising demand from manufacturing, e-commerce, and international trade. The government's infrastructure push, digitalization efforts, and policy interventions like the National Logistics Policy (NLP) and Gati Shakti are steps toward streamlining operations and reducing logistics costs. Despite these positive developments, the sector remains complex, with operational inefficiencies, cost pressures, and infrastructural constraints impacting its full potential. Air cargo logistics, in particular, has emerged as a crucial mode for time-sensitive and high-value shipments, yet it faces specific hurdles that distinguish it from surface logistics.

Key Threats and Challenges

- **Fuel Price Volatility**

Crude oil price movements have a direct and substantial impact on air logistics. Jet fuel accounts for a large portion of air cargo operating expenses, and fluctuations can sharply erode margins. For instance, the dramatic rise in fuel prices in 2022 following the Russia-Ukraine conflict significantly increased freight costs for air cargo operators, prompting them to introduce fuel surcharges or renegotiate rates. While some large players could partially absorb or pass on the cost, mid-sized firms like Skyways had to optimize routes and reduce non-essential legs to stay competitive.

- **Geopolitical and Global Trade Risks**

Disruptions like the COVID-19 pandemic, U.S.-China trade tensions, and the Suez Canal blockage in 2021 have highlighted how fragile global supply chains can be. These events had cascading effects on international cargo schedules, customs clearance times, and freight pricing, particularly for air shipments requiring just-in-time delivery.

- **Climate Change and Environmental Concerns**

The global shift toward sustainable logistics is placing pressure on air cargo operators to lower emissions and carbon footprints. European regulations, for example, are increasingly favouring low-emission freight alternatives, which could marginalize non-compliant air carriers. Companies like Lufthansa Cargo and FedEx have started integrating Sustainable Aviation Fuel (SAF), while Indian players are still in early stages, indicating a widening technology gap.

- **Cybersecurity Threats**

The logistics sector is increasingly reliant on digital tools for inventory management, cargo tracking, and customs documentation. However, this reliance exposes firms to data breaches and cyber-attacks. In 2021, global shipping giant CMA CGM suffered a ransomware attack that disrupted its booking systems. For Indian firms transitioning to digital platforms, a lack of robust cybersecurity measures makes them vulnerable to similar threats.



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- **Fragmented Industry Structure**

India's logistics space remains dominated by small, unorganized players lacking integrated systems. This fragmentation leads to inefficiencies in cargo handling, poor intermodal coordination, and limited scalability. For air cargo, which demands time-bound and precision-driven execution, working with multiple disconnected vendors increases risks of delay and service inconsistency.

- **Infrastructural Gaps**

While metro cities like Delhi and Mumbai have reasonably advanced cargo facilities, many secondary airports lack dedicated air freight terminals, cold-chain support, and night-handling capabilities. For example, a pharma shipment requiring temperature control may get delayed or damaged in Tier-2 airports due to lack of cold storage or skilled handling staff.

- **Regulatory Complexity**

Logistics players often deal with overlapping regulations from customs, aviation, and GST departments. Delays in customs clearances, frequent changes in documentation requirements, and inconsistent enforcement across states result in operational bottlenecks. For instance, exporters frequently report delays in DGFT approvals or scanning clearances, even when shipments are time-sensitive.

- **Limited Skilled Workforce**

Specialized logistics especially air freight involving hazardous or perishable goods requires trained manpower in cargo handling, packaging, documentation, and compliance. The industry currently suffers from a shortage of such skilled professionals, increasing reliance on on-the-job training and risking errors or shipment rejections.

- **Technology Adoption Gap**

Although many large logistics firms are adopting AI, IoT, and blockchain solutions to enhance efficiency, smaller and mid-tier companies often struggle with the cost and complexity of implementing these systems. As a result, companies like Skyways may find it difficult to offer end-to-end visibility or real-time tracking at the level of global competitors.

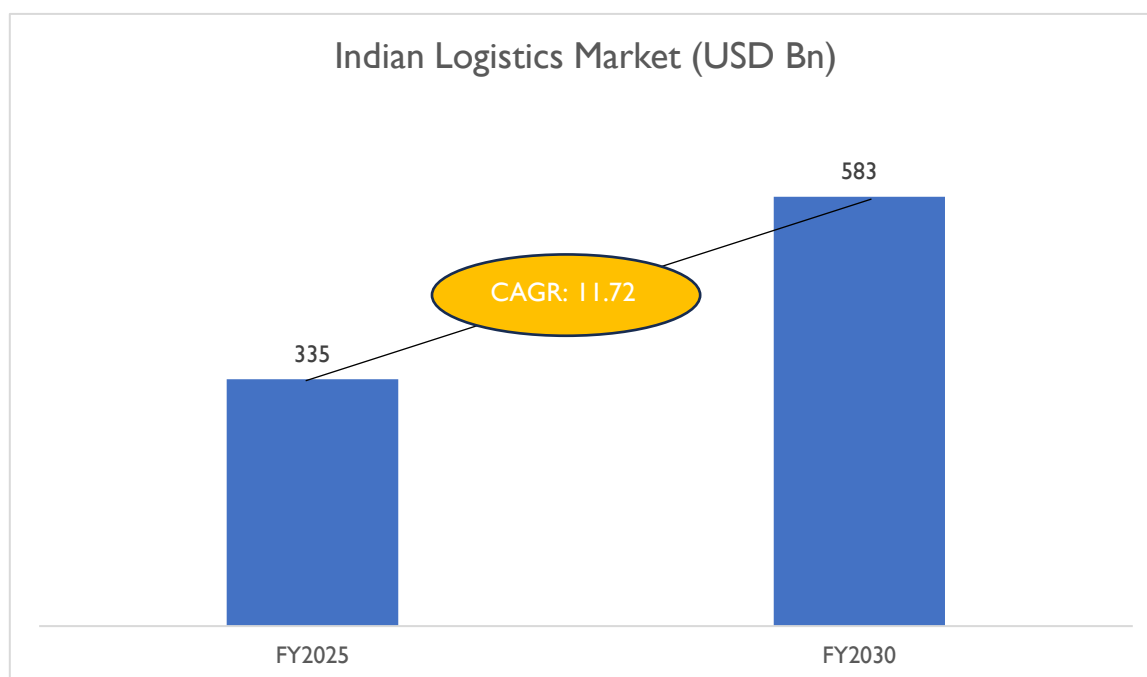


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Growth scenario

By nature, logistic demand corresponds to the healthy economic GDP growth, which facilitates demand for goods in the domestic as well as in the global market. Economic growth is expected to augment the freight movement. India has emerged as one of the fastest-growing countries globally, even amidst a challenging operating environment, driving the demand for logistics services. The necessity of having improved logistic services to have an efficient supply chain management, which fulfils the growing need of the economy, has made the government undertake several focused initiatives in terms of improving road infrastructure, power supply, policy reforms, etc. Increasing government spending on infrastructure is expected to support the growth of the logistics sector in the long run.

The Indian government's Union Budget aims to achieve inclusive and sustainable growth while adhering to the fiscal glide path. The growth in the coming years would be majorly backed by growing freight traffic, substantial government spending planned under the PM Gati Shakti scheme, the Comprehensive Logistics Action Plan, the National Logistics Policy, and other projects like Sagarmala and Bharatmala. Additional efforts like DFC, the e-way bill, introduction of lead index and technological advancement are expected to streamline the logistic operations, resulting in reduced cost and enhanced efficiency of the freight movement across the country.



Sources: Dun & Bradstreet Desk Research

In light of the above factors, the overall market size of the logistics industry in India is estimated to reach **USD 583 Bn by 2030**, registering a **CAGR of 11.72%** between 2025-2030.



Several factors are fuelling this market expansion:

1. Infrastructure Development

- **Bharatmala Project – Strengthening Road Connectivity:** The Bharatmala Pariyojana is significantly improving national highway networks by linking economic corridors, border roads, and rural-urban clusters. Projects like the Delhi–Mumbai Industrial Corridor (DMIC) have enhanced freight mobility between major industrial hubs, reducing travel time and improving long-haul trucking efficiency. This expanded connectivity helps manufacturers, exporters, and logistics companies move goods faster with fewer bottlenecks.
- **Sagarmala Project – Port-Led Development:** Sagarmala aims to modernize India’s ports and integrate them with Special Economic Zones (SEZs), coastal industrial zones, and inland waterways. This boosts cargo handling capacity and reduces turnaround time at ports. The improved coastal shipping routes and multimodal linkages created under Sagarmala are making maritime logistics more cost-effective and reliable for both domestic and international trade.
- **Dedicated Freight Corridors (DFCs) – Rail Logistics Transformation:** The Delhi–Mumbai and Amritsar–Kolkata Dedicated Freight Corridors are expanding India’s rail freight network, allowing high-speed, high-capacity movement of goods across long distances. These corridors reduce logistics costs by improving fuel efficiency, shortening transit times, and freeing up passenger rail lines. The DFCs are expected to handle nearly **70%** of freight traffic on these routes, supporting faster and safer movement of industrial cargo.

2. E-Commerce Expansion

- **Quick Commerce Growth:** India’s rapid-commerce sector has experienced exponential growth, with platforms delivering groceries and essentials within minutes. This trend has increased demand for dense micro-warehousing networks and optimized last-mile delivery systems. Quick commerce now accounts for over two-thirds of e-grocery orders, reflecting shifting consumer behavior toward instant delivery services across major cities.
- **International Company Example – Amazon’s Network Expansion:** Global e-commerce giants like Amazon have invested heavily in India’s logistics infrastructure by expanding fulfillment centers, same-day delivery hubs, and advanced sorting facilities. Amazon’s focus on automation, robotics, and predictive analytics in its warehouses has pushed industry standards higher and accelerated sector-wide digital adoption, strengthening the overall e-commerce logistics ecosystem.

3. Implementation of the Goods and Services Tax (GST)



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- **Streamlined Tax Structure & Faster Freight Movement:** GST replaced the earlier fragmented tax system, eliminating interstate checkpoints and reducing compliance burdens. This has cut transit delays significantly and improved truck turnaround time. As a result, the average distance travelled per truck per day increased from **225 km to nearly 300–325 km**. This improvement has lowered overall logistics costs, strengthened supply chain reliability, and made nationwide distribution far more efficient.

4. Technological Advancements

- **IoT-enabled Fleet Monitoring (Rivigo Example):** Rivigo has integrated IoT-based sensors across its fleet, enabling real-time monitoring of critical data such as fuel consumption, engine performance, location tracking, and temperature. This visibility helps reduce breakdowns, prevent delays, enhance driver productivity, and optimize route planning, ultimately improving overall fleet efficiency.
- **Blockchain Adoption for Supply Chain Transparency:** Companies are increasingly experimenting with blockchain technology to ensure safer, tamper-proof, and fully traceable documentation of goods across the supply chain. Blockchain enhances trust by securing records like invoices, shipment data, and quality certificates. This reduces fraud, eliminates documentation errors, and accelerates customs and compliance processes.

5. Growing Demand for Cold Chain Logistics

- **Expansion of Cold Storage Capacity:** India's cold storage infrastructure has grown to **39.37 million metric tons**, driven by rising demand from agriculture, food processing, pharmaceuticals, and dairy sectors. States like Uttar Pradesh and West Bengal lead this expansion due to their strong output of perishable products. This increased capacity supports better preservation, reduces food wastage, and strengthens agro-supply chains.
- **Vaccine Distribution Requirements & ULT Freezers:** The nationwide COVID-19 vaccination drive highlighted the importance of advanced cold chain systems, requiring over **2,000** ultra-low temperature (ULT) freezers for safe distribution. This created a shift toward more robust, temperature-controlled logistics networks capable of handling sensitive pharmaceuticals, biologics, and high-value medical supplies.

Future Trends and Opportunities in the Indian Logistics Market

The Indian logistics industry presents immense opportunities for growth and innovation in the coming years. Several key trends will shape the future of the sector:

1. Adoption of 3PL and 4PL Services



- Companies are increasingly outsourcing their logistics operations to third-party (3PL) and fourth-party (4PL) logistics providers to enhance efficiency and reduce costs.
- This shift allows businesses to focus on core competencies while leveraging professional logistics expertise.

2. Sustainable and Green Logistics

- With growing concerns about carbon emissions, companies are shifting toward electric vehicles (EVs), eco-friendly packaging, and energy-efficient warehouses.
- The Indian government is promoting green supply chain initiatives to achieve sustainability goals.

3. Expansion of Cold Chain Logistics

- Rising demand for perishable food, dairy products, and pharmaceuticals will drive investments in temperature-controlled warehouses and refrigerated transport.
- The Indian government is offering incentives and subsidies to support the expansion of cold storage facilities.

4. Smart Warehousing and Automation

- Adoption of AI-driven inventory management, automated storage systems, and robotics is enhancing warehouse productivity.
- Smart warehouses equipped with IoT sensors and predictive analytics improve real-time stock monitoring and reduce wastage.

5. Investment in Multi-Modal Transportation

- Integration of road, rail, air, and water transport will optimize freight movement and reduce transit times.
- The expansion of logistics parks and inland container depots (ICDs) will further facilitate seamless goods transportation.

The Indian logistics industry is undergoing rapid transformation, fuelled by technological advancements, policy reforms, and infrastructure investments. While the market is set to grow at a CAGR of **10.7%**, addressing high logistics costs, regulatory challenges, and infrastructure gaps remains crucial for sustainable development. With continued investments in digitalization, automation, green logistics, and multi-modal transport, India is well-positioned to emerge as a global logistics hub in the near future. Strengthening supply chains, streamlining operations, and fostering collaborations between government and private players will be key to unlocking the full potential of India's logistics sector.

Airport Network Expansion



India's aviation industry has undergone a remarkable transformation in the past decade, emerging as one of the fastest-growing air transport markets globally. The sector has seen a surge in domestic air travel, infrastructure expansion, and fleet growth. The rapid expansion of operational airports and an increase in air passengers underscore the industry's bright future.

India's aviation industry has witnessed rapid growth, with the number of operational airports more than doubling from **74 in 2014 to 162 in 2025**. The government's expansion plan aims to reach **350-400** airports by 2047, significantly improving regional connectivity.

By 2030, the number of airports is expected to cross **200**, driven by increasing air travel demand. By 2040, around **300** airports are projected, aligning with India's infrastructure growth. The long-term goal of **350-400** airports by 2047 highlights a strong government commitment to making India a global aviation hub. This expansion will enhance connectivity, boost economic development, and support India's growing air passenger market, which is set to become the third-largest globally by 2030.



Source: Government of India - PIB

Key factors affecting the Air Transport Industry:

➤ Growth in Air Passenger Traffic

Over the past decade, domestic air passenger traffic in India has more than doubled, reflecting the increasing affordability and accessibility of air travel. According to the International Air Transport Association (IATA), India is poised to become the third-largest air passenger market by 2030, surpassing China and the United States. This growth is fuelled by rising middle-class incomes, improved regional connectivity, and government initiatives such as the UDAN (Ude Desh Ka Aam Nagrik) scheme, which promotes affordable regional air travel.



➤ **Fleet Expansion**

To meet growing passenger demand, Indian airlines have significantly expanded their fleets. Major airlines like IndiGo, Air India, and Akasa Air have placed large aircraft orders, ensuring fleet modernization and increased capacity. The increase in fleet size will enhance connectivity, reduce congestion, and improve service efficiency across domestic and international routes.

➤ **Future Growth Prospects**

India's commitment to increasing its number of operational airports to **350-400 by 2047** suggests a strong focus on infrastructure development. The expansion of airports in tier-2 and tier-3 cities will play a crucial role in enhancing regional air connectivity, reducing dependency on metro hubs, and promoting economic growth in smaller cities.

Rising Passenger Demand

- India's domestic air passenger traffic is projected to grow at a CAGR of **8-10%** over the next decade.
- By 2030, India will become the third-largest air passenger market, overtaking the U.S. and China.
- Increased disposable incomes, business travel, and tourism will continue to fuel this demand.

➤ **Increase in Aircraft Fleet**

With rising passenger demand, Indian airlines are expected to double their fleet size by 2035. Fleet expansion will enable greater connectivity between domestic and international destinations, reinforcing India's position as a key global aviation hub. India's aviation industry is on a robust growth trajectory, driven by increasing airport infrastructure, rising passenger traffic, and fleet expansion. Government policies, economic growth, and technological advancements will further enhance connectivity and efficiency in the sector. With the projected increase in operational airports and air passenger traffic, India is set to become one of the world's leading aviation markets in the coming decades.

Factors and Trends Impacting the Growth of Air Transportation in India

1. Infrastructure Development

- The Bharatmala Project is enhancing road connectivity, with initiatives like the Delhi-Mumbai Industrial Corridor facilitating faster freight movement between industrial hubs.
- The Sagarmala Project focuses on port-led development, modernizing ports and integrating them with special economic zones to improve maritime logistics.
- The Dedicated Freight Corridors, such as the Delhi-Mumbai and Amritsar-Kolkata corridors, are expanding rail freight infrastructure, reducing transportation costs, and boosting trade efficiency.

2. E-Commerce Expansion



- Companies like Amazon and Flipkart increasingly use air cargo networks (e.g., Amazon Air in India) to ensure same-day or next-day delivery, directly increasing demand for aviation logistics.
- E-commerce shipments require speed and reliability, making air transport the preferred mode for intercity and international deliveries.

3. Implementation of the Goods and Services Tax (GST)

- The GST reform has streamlined tax structures, reduced logistics costs and increasing the average daily distance travelled by trucks from **225** kilometres to **300-325** kilometres, significantly enhancing efficiency in goods transportation.

4. Technological Advancements

- Rivigo, a logistics startup, has integrated IoT technology to enable real-time tracking of truck metrics like fuel levels and engine temperature, improving operational efficiency.
- Companies are increasingly exploring blockchain technology to enhance transparency and secure documentation in supply chains, ensuring trust and efficiency.

5. Growing Demand for Cold Chain Logistics

- India's cold storage capacity has expanded to **39.37 million** metric tons, with Uttar Pradesh and West Bengal leading due to their strong agricultural production.
- The COVID-19 vaccination drive required over **2,000** ultra-low temperature freezers, emphasizing the critical role of cold chain logistics in distributing temperature-sensitive vaccines.



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Industry SWOT Analysis

The Indian logistics industry is a cornerstone of the nation's economy, ensuring the seamless movement of goods across its vast and diverse landscape. A detailed SWOT analysis examining Strengths, Weaknesses, Opportunities, and Threats offer a comprehensive understanding of the industry's current state, with a particular emphasis on airways logistics.

Strengths:

- **Strategic Geographic Position:** India's central location in South Asia positions it as a crucial hub for international trade routes connecting the East and West. This strategic advantage facilitates efficient distribution channels for both domestic and international markets.
- **Robust Economic Growth:** India's consistent economic expansion drives demand for sophisticated logistics services. As industries grow and consumer markets expand, the need for timely and reliable transportation, especially via air freight, becomes increasingly critical.
- **Expanding Air Cargo Market:** The Indian air cargo market has demonstrated significant growth, reaching approximately **USD 13.44 billion** in **2024**. Projections indicate a compound annual growth rate (CAGR) of 5.80% between 2025 and 2034, potentially attaining a market value of **USD 23.62 billion** by **2034**. This growth is fuelled by factors such as increased export activities and advancements in airport infrastructure.
- **Technological Integration:** The adoption of advanced technologies, including automation, real-time tracking, and data analytics, has enhanced operational efficiency within the air cargo sector. These innovations lead to improved cargo handling, reduced turnaround times, and increased customer satisfaction.

Weaknesses:

- **Infrastructure Limitations:** Despite growth, many Indian airports face challenges like inadequate cargo handling facilities, limited warehousing space, and outdated equipment. These limitations can cause delays and increase operational costs.
- **Regulatory Challenges:** Complex customs procedures and bureaucratic hurdles often impede the swift movement of goods. Lengthy clearance processes can diminish the advantages of air freight's speed, affecting overall supply chain efficiency.
- **High Operational Costs:** Air freight is inherently more expensive than other transportation modes. Factors such as volatile fuel prices, high airport charges, and maintenance expenses contribute to elevated operational costs, which can deter price-sensitive customers.
- **Limited Domestic Connectivity:** While major metropolitan areas are well-served, regional connectivity remains underdeveloped. This gap restricts the reach of air cargo services, particularly affecting businesses in remote or underserved regions.



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Opportunities:

- **E-commerce Expansion:** The rapid rise of e-commerce in India has escalated demand for quick and reliable delivery services. Air cargo is uniquely positioned to meet these needs, offering expedited shipping that aligns with consumer expectations for speed.
- **Government Initiatives:** The Indian government has launched several initiatives aimed at bolstering the logistics sector. Policies focused on modernizing infrastructure, streamlining regulations, and promoting public-private partnerships are set to enhance the efficiency and capacity of air cargo operations.
- **Investment in Infrastructure:** Ongoing and planned investments in airport infrastructure, including the development of dedicated cargo terminals and integrated logistics parks, present opportunities to alleviate current bottlenecks and improve service delivery.
- **Regional Trade Agreements:** India's participation in regional trade agreements can open new markets and increase cargo volumes. Strengthening ties with neighbouring countries and trading blocs can lead to enhanced cross-border air cargo movement.

Threats:

- **Intense Competition:** The air cargo sector faces stiff competition from other modes of transport, such as road, rail, and sea, which may offer more cost-effective solutions for certain types of cargo. Additionally, international carriers operating in India add to the competitive pressure.
- **Environmental Concerns:** The environmental impact of air freight, particularly its carbon footprint, is under increasing scrutiny. Regulatory measures aimed at reducing emissions and promoting sustainability could impose additional costs and operational constraints on air cargo operators.
- **Global Economic Fluctuations:** The air cargo industry is vulnerable to global economic downturns, trade restrictions, and geopolitical tensions. Such factors can disrupt supply chains, reduce trade volumes, and impact profitability.
- **Pandemic and Health Crises:** Health crises, like the COVID-19 pandemic, have demonstrated the potential for severe disruptions in air cargo operations. Travel restrictions, reduced flight frequencies, and lab or supply shortages can significantly impact the industry's ability to function effectively during such events.



Competitive landscape

India's air cargo logistics sector forms a vital link in the broader supply chain, facilitating the rapid movement of time-sensitive and high-value goods across domestic and international markets. As the economy grows and global trade volumes increase, the demand for efficient air freight services continues to rise. Sectors such as pharmaceuticals, electronics, automotive components, and e-commerce are increasingly reliant on the speed and reach that air cargo offers.

The segment is also witnessing structural improvements driven by government support, including logistics infrastructure development, cargo-specific airport terminals, and digital initiatives aimed at enhancing transparency and turnaround time. While air cargo represents a smaller share compared to surface freight, its significance lies in handling urgent consignments and servicing regions with limited access.

Operational complexity, infrastructure limitations at key airports, and the need for specialized handling remain challenges in this space. However, advancements in cargo tracking technologies, growing adoption of multimodal logistics models, and the expansion of express logistics services are steadily reshaping the sector. Service providers are focusing on improving agility, optimizing capacity, and ensuring reliability to meet evolving client expectations.

Key players in the Indian air cargo logistics segment include a mix of specialized freight forwarders and diversified logistics providers. Companies like **Skyways Air Services Limited** focus primarily on-air freight and international cargo movement, offering tailored solutions for time-sensitive shipments. On the other hand, larger integrated logistics firms such as **Delhivery Ltd**, **Mahindra Logistics Ltd**, and **TVS Supply Chain Solutions Ltd** incorporate air cargo as part of a broader multimodal logistics portfolio, combining it with surface transport, warehousing, and last-mile delivery. These companies leverage scale, technology, and a pan-India presence to serve varied industry verticals. Together, they shape a competitive market where niche specialization and end-to-end service capabilities both find relevance depending on client needs.

Peer Profiling

Delhivery Limited

Established in 2011 and headquartered in Gurgaon, Haryana, Delhivery Limited is a leading logistics and supply chain services provider in India. The company has developed a wide-reaching pan-India presence, servicing over 18,700 pin codes with infrastructure comprising 24 automated sort centres, 94 gateways, and 2,880 direct delivery centres. As of August 2023, Delhivery had fulfilled over 2 billion orders. By March 2025, it achieved a milestone: for the first time, a full fiscal-year net profit, marking a strong turnaround in performance.



Delhivery's logistics capabilities are designed to cater to businesses of all sizes, from small and medium enterprises to large-scale e-commerce platforms. Its infrastructure is supported by robust automation and process engineering that facilitate high-speed, accurate order sorting, tracking, and delivery. The company has built strategic partnerships both domestically and globally, enabling seamless cross-border and multi-modal transportation options. The firm's operational footprint extends beyond metropolitan hubs, reaching Tier 2 and Tier 3 cities to provide inclusive logistics coverage. With significant investments in technology and process innovation, Delhivery has positioned itself not only as a service provider but also as an enabler of modern commerce. It continues to expand its infrastructure, including warehousing space and transport fleet, to support a growing and diverse customer base.

In April 2025, Delhivery initiated the acquisition of Ecom Express Limited, which was subsequently completed in June 2025 after receiving approval from the Competition Commission of India (CCI). The transaction, valued at approximately INR 1,407 crore on a fully diluted basis, represents a significant strategic consolidation in India's logistics sector. This acquisition strengthens Delhivery's express parcel and last-mile delivery capabilities, enhances network density across Tier 2 and Tier 3 cities, and enables improved operational efficiencies through the integration of infrastructure, technology platforms, and customer networks.

Product & Service Offerings:

Delhivery provides a wide range of logistics and supply chain services designed to meet the diverse needs of its customers across various sectors:

- **Express Parcel:** Time-bound delivery services for high-volume e-commerce and direct-to-consumer shipments, covering first mile, line haul, and last-mile logistics.
- **Part Truckload (PTL):** Freight solutions that allow multiple clients to share space on a single vehicle, offering cost efficiency and regular dispatches across key routes.
- **Full Truckload (FTL):** Dedicated truck services for businesses with bulk transportation needs, providing direct connectivity between manufacturing units, warehouses, and retail points.
- **Warehousing & Fulfilment:** A network of warehousing facilities offering inventory management, order processing, packaging, and fulfilment services. These are supported by technology platforms for inventory visibility and control.
- **Cross-border Services:** International shipping solutions including import/export clearance, freight forwarding, and last-mile delivery in select markets through global partnerships.
- **Retail Chains:** Enables intra-city and inter-city movement of inventory for organized retail outlets through structured warehousing and FTL services.
- **Supply Chain Software:** A proprietary logistics operating system enabling shipment tracking, network optimization, demand forecasting, and operational analytics.



- **Data Intelligence Solutions:** Technology-driven insights for customers to optimize their logistics planning, demand prediction, and cost structures.
- **Solutions for D2C and SMEs:** Customizable logistics services to support direct-to-consumer brands, small sellers, and businesses with flexible scale requirements.

Key Customer Segments Serviced:

Delhivery caters to a broad and diversified customer base across industries, offering tailored logistics solutions to meet varying supply chain needs:

- **E-commerce Marketplaces:** Serves some of India's largest online platforms by providing high-frequency express parcel and reverse logistics services across the country.
- **Direct-to-Consumer (D2C) Brands:** Supports digitally native and emerging consumer brands with scalable fulfilment and delivery services, enabling faster market reach and customer responsiveness.
- **Small and Medium Enterprises (SMEs):** Offers flexible warehousing, PTL, and last-mile solutions to SMEs across sectors, helping streamline logistics and improve delivery timelines.
- **Large Enterprises:** Works with established firms in automotive, retail, FMCG, and manufacturing sectors, providing end-to-end transportation and supply chain management solutions tailored to complex business needs.
- **Consumer Durables & Electronics:** Facilitates safe and timely movement of high-value and bulky goods, leveraging tracking systems and secure delivery protocols.
- **Retail Chains:** Enables intra-city and inter-city movement of inventory for organized retail outlets through structured warehousing and FTL services.

Key Strengths

- **Integrated Multi-Modal Logistics Network:** Delhivery's extensive infrastructure includes road, air, and cross-border logistics capabilities, allowing it to offer express parcel, part-truckload (PTL), full-truckload (FTL), and warehousing services. Its ability to manage multi-modal movement enhances network flexibility.
- **Proven Profitability & Scale:** In FY25, Delhivery reported total services revenue of Rs 8,932 Cr and posted a net profit of Rs 162 Cr, its first full-year PAT profitability, marking a significant turnaround from FY24. For the first half of FY26, Delhivery Limited reported revenue from operations of INR 4,853 crore and a PAT of INR 40.5 crore, reflecting continued scale-up of operations and improving financial performance during the period.
- **Proprietary Technology Platform:** The company's logistics operating system integrates warehousing, last-mile delivery, order management, and real-time visibility. This tech-led approach supports dynamic



routing, shipment optimization, and predictive analytics, improving service performance and cost efficiency.

- **Diverse Customer Base:** With over **30,000** active customers as of FY24, Delhivery supports a varied clientele ranging from new-age e-commerce startups to traditional enterprises, enabling it to mitigate demand concentration risk and maintain a balanced business portfolio.
- **Strong Emphasis on Automation and Scale:** The company continues to invest in automation across its sortation centers and delivery operations. These investments are aligned with cost management goals and operational scalability.
- **Strategic Partnerships and Expansions:** Delhivery has partnered with several international logistics providers and also engages with ONDC (Open Network for Digital Commerce), allowing it to explore collaborative opportunities in both B2B and B2C segments.

Mahindra Logistics Ltd

Mahindra Logistics Ltd (MLL), founded in **2008** and part of the Mahindra Group, is a leading integrated logistics and mobility solutions provider in India. The company offers end-to-end services spanning contract logistics, freight forwarding, last-mile delivery, express logistics, and enterprise mobility solutions. Serving a diversified portfolio of corporate clients across automotive, e-commerce, consumer goods, pharmaceuticals, and manufacturing sectors, MLL leverages technology-enabled operations, data analytics, and a strong partner network to deliver scalable, efficient, and sustainable logistics solutions across the country.

As of 2025, Mahindra Logistics commands an impressive operational footprint with 22.1 million+ sq. ft. of warehousing space spread across 1,100+ locations nationwide, supported by a dynamic workforce of over 30,000. Every month, the company executes 50,000+ full truck trips, manages 10,000+ TEUs of ocean freight, and delivers nearly 12 crore packages, reflecting its ability to seamlessly connect businesses and consumers across India. With such scale, MLL not only handles complex, multi-modal supply chains with efficiency but also demonstrates the agility and reliability that have made it a trusted partner for hundreds of corporate clients across industries.

On 21 April 2025, Mahindra Logistics Ltd (MLL) announced a leadership transition at the top, with Hemant Sikka named as MD & CEO designate effective 22 April 2025. He officially assumed the role on 5 May 2025, succeeding the previous MD & CEO, Ram Praveen Swaminathan, who stepped down from the position on 20 July 2025. This leadership change marks a new chapter in MLL's strategic growth and operational expansion.

Product Profile & Operational Infrastructure

Mahindra Logistics operates primarily through two business segments:



- **Supply Chain Management (SCM):**
 - Inbound and outbound logistics
 - Distribution and network design
 - Fulfilment and warehousing
 - Freight forwarding (domestic and international)
 - E-commerce logistics
 - Project cargo and express logistics
- **Enterprise Mobility Services (People Transport Solutions):**
 - Employee transportation solutions for corporates
 - Tech-enabled routing, scheduling, and fleet management
 - Electric vehicle fleet options to support sustainability

MLL also offers end-to-end logistics consulting and design through its integrated service platform.

Key Customer Segments Serviced

Mahindra Logistics serves a wide range of customer segments across various industries, providing logistics and mobility solutions tailored to specific needs. Key customer segments include:

- **Automotive Industry:** Serving original equipment manufacturers (OEMs) and component suppliers with transportation, warehousing, and distribution services.
- **E-Commerce:** Providing end-to-end logistics solutions to e-commerce platforms, including fulfilment, last-mile delivery, and reverse logistics.
- **Consumer Goods & Retail:** Offering supply chain management services for fast-moving consumer goods (FMCG) companies and retail networks.
- **Pharmaceuticals & Healthcare:** Ensuring efficient and safe transportation of pharmaceutical products, medical devices, and healthcare equipment.
- **Engineering & Manufacturing:** Delivering project logistics, specialized transport services, and warehouse management solutions for industrial sectors.

Key Strengths

Integrated Logistics Platform: Mahindra Logistics provides a range of services across the supply chain, including transportation, warehousing, and fulfilment. This integration helps streamline operations and improve efficiency for various industries.



Technology-Driven Operations: The company uses technologies like AI, IoT, and automation to optimize routes, manage inventory, and enhance tracking. These tools help improve operational performance and provide valuable insights for decision-making.

Pan-India (and Regionally Diverse) Footprint, including Tier-II / Tier-III & Northeast: Through recent expansion, leases and new warehouses (e.g. Telangana, Phaltan near Pune, Northeast; Guwahati, Agartala), MLL is broadening its reach beyond metros, tapping growth in industrial corridors and emerging regions. This geographic diversity helps resilience and access to newer markets.

Asset-Light Business Model: Operating with a non-asset-based model, Mahindra Logistics avoids heavy investments in infrastructure, relying on a network of partners and technology to offer scalable and flexible logistics solutions.

Sustainability Focus: Mahindra Logistics integrates electric vehicles into its fleet and focuses on green logistics practices, such as energy-efficient warehouses and optimized routing, to reduce the environmental impact of its operations.

Strong Parentage and Brand Equity: As part of the Mahindra Group, the company benefits from brand recognition and access to a wide network of resources, which strengthens its position in the logistics market.

Experienced Workforce and Operational Expertise: Mahindra Logistics invests in its workforce by providing ongoing training, ensuring that its team is equipped to handle the complexities of logistics and supply chain management.

Commitment to Green / ESG-Friendly Logistics: Through emissions analytics, sustainable warehouses, and integration of green energy / technologies, MLL positions itself as a forward-looking, environmentally conscious logistics partner — a strength given increasing regulatory and customer focus on ESG.

Analysis of key factors shaping competition in the air transportation sector

The Indian air transportation sector is a dynamic and evolving industry, influenced by various competitive factors and dominated by key players who shape its trajectory.

I. Market Structure and Dominant Players:

- **IndiGo Airlines:** Holding over **60%** of India's domestic market share, IndiGo has established itself as the largest carrier in the country. The airline is transitioning from its budget airline image by expanding internationally and introducing business-class seating, aiming to compete with global long-haul carriers.



- **Air India:** Under the ownership of the Tata Group, Air India is undergoing modernization efforts. The airline is in discussions with Boeing and Airbus for a substantial order of 30 to 40 widebody jets, aiming to elevate its modernization efforts and recapture market share lost to global rivals.

2. Fleet Expansion and Modernization:

Airlines are investing aggressively in expanding and modernizing their fleets to cater to rising passenger volumes, improve operational efficiency, and strengthen global competitiveness. Air India, for example, has initiated one of the largest fleet renewal programs in aviation history, with significant widebody jet orders proposed from Airbus and Boeing. This strategy is aimed at upgrading its long-haul capabilities, entering new international routes, and restoring its position in the global aviation market. In addition, IndiGo Airlines is pursuing an ambitious expansion roadmap starting August 2025, reflecting its long-term strategic focus on operational resilience and cost optimization. The airline is developing a state-of-the-art maintenance facility in Bengaluru, designed to support advanced fleet servicing, reduce turnaround times, and minimize dependence on external MRO partners. Alongside this, IndiGo plans to increase its owned aircraft share to **30–40% by 2030**, a major shift from its predominantly leased fleet model. This move is intended to strengthen balance-sheet stability, reduce lease liabilities, mitigate supply chain risks, and enhance overall operational discipline.

3. Infrastructure Development:

The Indian government has been investing heavily in airport infrastructure to support the rapidly growing aviation sector. This includes the construction of new airports under the UDAN regional connectivity scheme, along with major capacity expansion and modernization of existing airports to handle rising passenger and cargo volumes. To address the surge in air traffic, airport operators are set to invest a cumulative **INR 60,000 crore** in capital expenditure between FY 2025 and FY 2027, marking a **12%** increase compared to the **INR 53,000 crore** invested during FY 2022–24. This investment push is critical as the aviation industry is projected to add **65 million** passengers per year, requiring significant upgrades in terminal capacity, runway expansion, cargo handling zones, and airside infrastructure.

4. Regulatory Environment:

India's regulatory environment has been steadily evolving to support the expansion of the aviation sector. The government has introduced policies aimed at improving safety standards, enhancing passenger convenience, and promoting investment across airports and airlines. Reforms such as the DigiYatra initiative, single-window clearances for airport development projects, and the modernization of DGCA processes have reduced compliance complexities and encouraged industry growth. Additionally, reforms in slot allocation, bilateral agreements, ground handling, and air cargo policy frameworks are helping create a more competitive and efficient aviation market.



5. Pilot Shortage:

India's aviation industry is currently grappling with a severe pilot shortage, with an estimated **12–15%** deficit in trained, experienced pilots required to meet projected growth. Ironically, despite this shortage, **4,000–5,000** newly licensed pilots remain unemployed because airlines prefer candidates with substantial flying hours and real-world experience. The high cost of Type Rating training, limited simulator availability, and the long training cycles make it challenging for new pilots to enter commercial operations quickly. This mismatch between supply and demand places additional pressure on airlines, forcing them to depend on expatriate pilots for widebody operations or delay route expansion plans due to insufficient crew availability. The pilot shortage remains one of the biggest operational bottlenecks for India's fast-growing aviation sector.

6. Technological Advancements:

Airlines are increasingly leveraging advanced technologies to deliver a more seamless passenger experience and improve operational efficiency. Innovations such as mobile check-in, biometric boarding, AI-driven flight scheduling, and smart baggage tracking systems are becoming standard across major carriers. The adoption of in-flight Wi-Fi, digital payment systems, predictive maintenance tools, and real-time performance monitoring is helping airlines reduce delays and enhance customer satisfaction. Additionally, backend operations such as crew management, route optimization, and fuel efficiency monitoring are increasingly being automated using AI and big-data analytics. This growing digital transformation not only improves convenience for passengers but also boosts cost efficiency and operational reliability for airlines.

7. Sustainability Initiatives:

Growing concerns over climate change and global carbon emissions have compelled Indian airlines to adopt more sustainable practices. Carriers such as Air India, IndiGo, and Vistara are replacing older aircraft with newer, fuel-efficient models like the A320neo and Boeing 787 Dreamliner to reduce fuel consumption and emissions. Airlines are also exploring sustainable aviation fuels (SAF), participating in global carbon-offset programs, and implementing waste reduction measures onboard. Airports, too, are transitioning to renewable energy sources, green building standards, and electric ground support equipment. These sustainability initiatives are becoming essential as aviation's environmental footprint faces increasing scrutiny from regulators, international bodies, and environmentally conscious passengers.

8. Market Dominance by Foreign Carriers

Foreign carriers exert overwhelming dominance in India's international air cargo market, controlling nearly **94–95%** of total freight volumes. Their strong position is backed by extensive global networks, operational efficiencies, and access to a large fleet of dedicated freighter aircraft advantages that Indian



carriers currently lack. Airlines like Emirates SkyCargo, Qatar Airways Cargo, Lufthansa Cargo, FedEx, and UPS provide high-frequency services, faster connections, and superior global coverage, making them the preferred choice for exporters and freight forwarders. Meanwhile, Indian carriers rely mostly on belly cargo capacity, limiting their flexibility and scale. This imbalance influences market pricing, cargo routes, and service levels, making it difficult for local airlines to compete effectively. Until domestic carriers expand their widebody and freighter operations, foreign airlines will continue to dominate India's international cargo ecosystem.



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TVS Supply Chain Solutions Ltd

TVS Supply Chain Solutions Ltd (TVS SCS), **established in 2004** and headquartered in Chennai as part of the TVS Mobility Group, is a global leader in integrated supply-chain and logistics solutions. Operating across **26 countries**, the company delivers end-to-end services including inbound and outbound logistics, in-plant operations, warehousing, international freight forwarding, final-mile deliveries, and value-added supply-chain solutions. By combining technology-enabled platforms with a network-focused approach, TVS SCS helps businesses streamline operations, improve visibility, and manage complex global supply chains with agility and precision.

As of 2025, TVS SCS operates a vast global network, managing 19.9 million sq. ft. of warehouse and managed space across 26 countries, supported by a dedicated workforce of around 17,000 professionals. Every day, the company delivers seamless logistics and supply-chain services to over 6,200 customers worldwide, including 91 Fortune Global 500 clients, across industries like automotive, industrial, consumer goods, retail, healthcare, aerospace, defence, and technology. This scale, combined with integrated end-to-end solutions and technology-driven processes, allows TVS SCS to handle complex, multi-modal supply chains efficiently while offering highly customized services that meet the unique needs of each client.

Product & Service Offerings

TVS SCS delivers supply chain solutions under two key business segments:

- **Integrated Supply Chain Solutions (ISCS):** Includes inbound logistics, outbound distribution, warehousing, inventory planning, packaging, and in-factory logistics operations.
- **Network Solutions (NS):** Covers international freight forwarding (air, sea, road), express logistics, time-sensitive final mile deliveries, and aftermarket support including spare parts logistics and returns management.

The company also offers industry-specific solutions with a strong emphasis on technology integration to streamline operations and improve turnaround time.

Key Customer Segments Serviced

TVS SCS serves a wide spectrum of industries including:

- Automotive and Auto Components
- Engineering and Industrial Equipment
- Consumer Electronics and Appliances
- Retail and E-commerce
- Pharmaceuticals and Healthcare



- Rail & Related sectors
- Aerospace and Defence
- Utilities and Infrastructure

Key Strengths

- **Global Scale with Local Execution:** With presence in 25+ countries and nearly 20 million sq.ft warehouse capacity globally, TVS SCS blends global logistics standards with local understanding.
- **Multi-Sector Experience:** Provides supply chain solutions across varied industries, enabling adaptability and scalability. The cross-industry exposure supports process innovation and allows transfer of best practices across segments.
- **Technology-Driven Operations:** Leverages proprietary digital tools to improve visibility, enhance control, and drive data-informed logistics planning. These platforms also facilitate performance tracking, automated reporting, and predictive analytics.
- **Tailored Logistics Models:** Designs solutions that align with customer-specific operational challenges and regulatory environments. This customization helps improve turnaround time and ensures alignment with client objectives.
- **Integrated Capabilities:** Offers a seamless supply chain experience by combining warehousing, transportation, freight forwarding, and aftermarket services. Such integration simplifies vendor management and improves coordination across the supply chain.
- **Global Expansion and Facility Growth:** TVS SCS has recently expanded its presence in North America, including the inauguration of a 225,000 sq. ft. facility in Iowa in September 2025, reflecting ongoing investments in its global operations and capacity growth.
- **Focus on Efficiency:** Works toward optimizing logistics networks and operational processes to support cost-effective service delivery. Continuous improvement initiatives and lean practices help in minimizing waste and reducing delivery timelines.



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Financial Analysis

Particular	Unit	Skyways Air Services Limited				Delhivery Ltd				Mahindra Logistics Ltd				TVS Supply Chain Solutions Ltd			
		HI FY 2026	FY 2025	FY 2024	FY 2023	HI FY 2026	FY 2025	FY 2024	FY 2023	HI FY 2026	FY 2025	FY 2024	FY 2023	HI FY 2026	FY 2025	FY 2024	FY 2023
Revenue From Operations	INR in Crore	1,328.20	2,247.82	1,289.11	1484.12	4,853.32	8,931.90	8,141.54	7,225.30	3,309.86	6,104.83	5,505.97	5,128.29	5,254.94	9,995.72	9,199.98	10,235.37
EBITDA	INR in Crore	57.15	86.49	48.34	58.71	212.79	377.72	112.93	-438.03	161.25	284.04	231.84	256.97	445.58	696.48	635.74	678.42
EBITDA Margin	in %	4.30	3.85	3.75	3.96	4.38	4.23	1.39	-6.06	4.87	4.65	4.21	5.01	8.48	6.97	6.91	6.63
PAT	INR in Crore	25.46	48.14	34.49	37.90	40.55	162.11	-249.19	-1,007.78	-17.80	-30.00	-53.09	24.63	87.47	-9.64	-90.49	41.76
PAT Margin	in %	1.92	2.14	2.68	2.55	0.84	1.81	-3.06	-13.95	-0.54	-0.49	-0.96	0.48	1.66	-0.10	-0.98	0.41
Net Worth	INR in Crore	305.07	247.14	154.26	125.19	9,529.28	9,432.15	9,144.65	9,177.14	1,145.38	437.88	492.47	561.67	1,950.03	1,801.77	1,815.03	723.55
DE Ratio	Multiple	1.25	1.42	1.92	1.34	0.01	0.004	0.01	0.02	0.06	0.97	0.69	0.71	0.53	0.48	0.44	2.75
ROE	in %	5.54	19.52	22.37	31.35	0.43	1.75	-2.72	-13.32	-2.67	-7.71	-10.39	4.74	4.58	-0.75	-7.97	5.54
RoCE	in %	8.26	14.61	15.57	35.56	-1.18	-1.67	-6.63	-13.66	2.08	6.68	3.14	8.61	8.72	8.32	4.28	12.30
ROA	in %	1.94	3.64	4.36	8.45	0.32	1.34	-2.18	-8.99	-0.57	-1.16	-2.14	0.96	1.39	-0.17	-1.55	0.67

Note: For all companies consolidated balance sheet is considered.

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Company Profiling: Skyways Air Services Limited

Skyways Airservices is a professional aviation and logistics company specializing in air cargo transportation, freight forwarding, and related services. Established in 1983 by Chairman Mr. S.L. Sharma, the company has built a strong foundation in the aviation sector, offering reliable and efficient solutions for cargo movement across domestic and international routes. Headquartered in New Delhi, Skyways Airservices operates with a focus on streamlined logistics, ensuring timely and secure deliveries for a diverse range of industries.

With a well-established regional presence, the company serves key markets presence across India with 31 locations, including major cities such as Ahmedabad, Bangalore, Chennai, Cochin, Delhi, Hyderabad, Jaipur, Kolkata, Mumbai, Panipat, and Surat., enabling seamless connectivity between major trade hubs. Internationally, Skyways has expanded its operations to Germany, Vietnam, Bangladesh, Dubai, Hong Kong, and Thailand. Skyways Airservices collaborates with a network of carriers, ground handling agents, and logistics partners to provide customized transport solutions that cater to the specific needs of businesses. By leveraging industry expertise and technology-driven processes, the company continues to enhance its service offerings and expand its operational footprint in the aviation and logistics industry.

Product profile & operational infrastructure

Skyways Air Services Limited provides a range of logistics and freight solutions tailored to diverse industry needs directly or through its subsidiaries. The company's key services include:

1. Air Freight Services

- Direct and consolidated air cargo solutions
- Time-sensitive and high-value cargo handling
- Partnerships with major airlines for efficient routing
- Specialized handling of perishable, dangerous, and oversized goods

2. Ocean Freight Services

- Full Container Load (FCL) and Less than Container Load (LCL) shipping
- Multi-modal transport solutions (sea-air, sea-road)
- Global partnerships with shipping lines for optimized costs

3. Road Transportation Services

- Domestic and cross-border road freight solutions
- Full truckload (FTL) and less than truckload (LTL) options



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- Real-time tracking and end-to-end logistics support

4. Express & Courier Services

- Fast-track delivery for documents and parcels
- Domestic and international courier services
- Last-mile delivery solutions

5. Warehousing & Distribution

- Temperature-controlled and general warehousing
- Inventory management and order fulfilment
- Distribution network across key locations

6. Customs Clearance & Regulatory Compliance

- Import and export documentation support
- Customs brokerage and duty management
- Trade compliance advisory

7. Specialized Logistics Solutions

- Handling of hazardous materials and live animals
- Cold chain logistics for pharmaceuticals and perishables
- E-commerce and retail supply chain solutions

Operational Infrastructure:

1. Network & Regional Presence

- Headquartered in New Delhi, India
- Operational presence in 31 locations across India, including major cities such as Mumbai, Bangalore, Chennai, Hyderabad, Kolkata, and Ahmedabad
- International offices in Germany, Vietnam, Bangladesh, Dubai, Hong Kong, and Thailand

2. Logistics Hubs & Facilities

- Strategically located warehouses near major transport hubs
- Cold storage facilities for temperature-sensitive shipments



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- Automated handling systems for faster processing

3. Technology Integration

- Digital tracking and real-time shipment visibility
- AI-driven logistics optimization
- Integrated supply chain management systems

4. Certifications & Compliance

- IATA-accredited air freight services
- AEO (Authorized Economic Operator) certified
- ISO 9001:2015 for quality management in logistics

Key customer segments serviced:

Skyways Airservices provides logistics solutions to businesses across multiple industries, ensuring seamless supply chain management tailored to each sector's specific requirements. The key customer segments include:

1. Fast-Moving Consumer Goods (FMCG)

- Supports large FMCG brands such as Parle and Britannia in the efficient distribution of food products and packaged goods.
- Ensures timely deliveries to maintain stock availability in retail chains and supermarkets.
- Offers warehousing and last-mile delivery solutions to optimize supply chain efficiency.

2. Pharmaceuticals & Healthcare

- Provides cold chain logistics for temperature-sensitive medications and vaccines for companies like Morepen and IPCA.
- Ensures regulatory-compliant transportation of medical devices and pharmaceutical products.
- Offers express delivery services for urgent medical supplies, reducing lead times for critical healthcare shipments.

3. Automotive Industry

- Partners with leading automotive brands such as Tata Motors and Eicher to manage the transport of vehicle parts and assemblies.
- Supports just-in-time (JIT) supply chains, ensuring uninterrupted production lines.



- Provides multimodal transport options, including air, sea, and road freight, for the global automotive supply network.

4. Textiles & Apparel

- Works with major textile manufacturers such as Vardhman and Arvind to streamline fabric and garment transportation.
- Ensures smooth international trade logistics for raw materials and finished clothing.
- Offers warehousing and distribution services to manage seasonal inventory fluctuations.

5. Engineering Goods & Industrial Equipment

- Supports companies like Honeywell and JCB Power Systems in transporting heavy machinery, industrial tools, and engineering components.
- Provides secure logistics solutions for high-value industrial goods, ensuring safety during transit.
- Manages global freight forwarding for manufacturing units, ensuring smooth operations across supply chains.

Skyways Airservices continues to cater to a broad spectrum of industries by offering specialized logistics solutions, advanced tracking systems, and a robust global network to support businesses in managing their supply chain operations efficiently.

Financial Performance Snapshot on Consolidated Basis

Particular	Unit	Skyways Air Services Limited			
		HI Fiscal 2026	Fiscal 2025	Fiscal 2024	Fiscal 2023
Revenue From Operations	INR in Crore	1,328.20	2,247.82	1,289.11	1484.12
EBITDA	INR in Crore	57.15	86.49	48.34	58.71
EBITDA Margin	in %	4.30	3.85	3.75	3.96
PAT	INR in Crore	25.46	48.14	34.49	37.90
PAT Margin	in %	1.92	2.14	2.68	2.55
Net Worth	INR in Crore	305.07	247.14	154.26	125.19
Debt Equity Ratio	Multiple	1.25	1.42	1.92	1.34
Return on Equity (ROE)	in %	5.54	19.52	22.37	31.35
Return on capital	in %	8.26	14.61	15.57	35.56



employee (RoCE)					
Return on Asset (RoA)	in %	1.94	3.64	4.36	8.45

Particular	Formula
Revenue From Operations	Revenue from Operations
EBITDA	Profit After Tax + Tax + Depreciation Cost + Finance Cost – Other Income
EBITDA Margin	EBITDA / Revenue from Operations
PAT	Profit After Tax
PAT Margin	Profit After Tax / Revenue from Operations
Net Worth	Shareholder Equity
Debt Equity Ratio	(Short Term Borrowing + Long Term Borrowing) / Shareholder Equity
Return on Equity (ROE)	Profit after Tax / Shareholder Equity
Return on capital employee (RoCE)	(Profit Before Tax + Finance Cost) / (Shareholder Equity + Short Term Borrowing + Long Term Borrowing)
Return on Asset (RoA)	Profit After Tax / Total Asset



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